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AIR FORCE MATERIEL COMMAND

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FUNCTIONAL DEPOT MAINTENANCE COURSEWARE DEVELOPMENT AND MANAGEMENT

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This manual implements Department of the Air Force Policy Directive (DAFPD) 36-26, *Total Force Development and Management*, and further aligns with Department of the Air Force Manual (DAFMAN) 36-2689, *Training Program*, and Department of the Air Force Handbook (DAFH) 36-2675, *Information for Designers of Instructional Systems*. This manual presents Instructional System Design (ISD) methodology and activities as they apply to functional courseware development and management in Air Force Materiel Command (AFMC). This manual applies to military, civilian, and contractor personnel responsible for developing and managing AFMC functional depot maintenance training courseware. This manual does not apply to the Air National Guard, Air Force Reserve Command, or the United States Space Force. This publication may be supplemented at any level, but all supplements must be routed to the Office of Primary Responsibility (OPR) of this publication for coordination prior to certification and approval. Ensure that all records created as a result of processes prescribed in this publication are maintained in accordance with Air Force Instruction (AFI) 33-322, *Records Management and Information Governance Program,* and disposed of in accordance with the Air Force Records Disposition Schedule located in the Air Force Records Information Management System. Refer recommended changes and questions about this publication to the OPR using Department of the Air Force (DAF) Form 847, *Recommendation for Change of Product*; route DAF Forms 847 from the field through the appropriate functional chain of command. Submit request for waivers through the chain of command to the publication OPR for non-tiered compliance. The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the Department of the Air Force.

SUMMARY OF CHANGES

Minor changes to this manual include 1) revised the command courseware signature pages to expedite turn time; 2) revised the courseware review/revision gated process used for command courseware; 3) removed redundant and unnecessary ISD Process Flow Chart; 4) added the process of localizing command courseware for specific complex needs; and 5) made administrative changes.

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Chapter 1

INTRODUCTION

1.1. Purpose. In order for participants in AFMC Courseware Development and Management Process to work efficiently and effectively together, procedures must be well documented. Such documentation is particularly valuable when applying a complex and dynamic methodology such as Instructional System Design (ISD). ISD is flexible and systematic. It is a tool to get the right training for the problem, not a lockstep linear process. This manual promotes standardization of courseware development and management activities across AFMC Depot Maintenance Organizations and reduces time needed to train new participants entering the process. It lists typical activities used for the full spectrum of courseware issues and allows a courseware program/project manager to select only those activities appropriate for a particular situation.

1.2. Application. Use of the processes in this manual for performed activities is mandatory for development of formal, functional training provided to AFMC Depot Maintenance personnel. It is recommended for development of education and employee development instruction. It applies to military, civilian and contractor personnel performing one or more activities in the AFMC Courseware Development and Management Process for depot maintenance functional training.

1.2.1. Instruction Defined. Instruction is information provided in a structured manner for education, employee development, or training purposes.

1.2.2. Formal Training Defined. Formal training is instruction based on established learning objectives that enables delivery of training to be accurate, complete and consistent information time after time. Formal training courses are assigned course numbers to document student completions in employee records.

1.2.3. Informal Training. Informal training does not have defined learning objectives. It provides basic information but not always the same way or at a similar level of detail. It is tailored specifically to the situation and audience. Coaching, tutoring and informal on-the-job training that is based entirely on variables of each situation (e.g., actual workload, prior employee experience, trainer background etc.) are considered forms of informal training.

1.2.4. Functional Training Defined. Functional training is instruction that helps employees learn how to perform specific work processes so they can efficiently and economically accomplish their mission. It differs from education and employee development instruction which focus on learning for growth of an individual rather than immediate support of a mission.

1.2.5. Soft Skills Defined. Soft skills address feelings, emotions and attitudes, which are part of the affective domain of learning. Instruction on teambuilding, change management and customer service are examples of soft skills training. Soft skills courses are considered functional training for purposes of this manual if they are implemented to support performance in the workplace.

1.2.6. Education Defined. Education are opportunities that helps learners acquire new knowledge disciplines, skill sets, and viewpoints that are not based on performance requirements of the current job. It prepares learners for non-specific future requirements and its effectiveness cannot be fully evaluated until the learner is on a job that requires performance related to educational learning objectives.

1.2.7. Employee Development Defined. Employee development is opportunities which focus on growth of an individual rather than performance of a job. While it may indirectly help the mission by increasing promotion potential of employees or improving their perspective or morale, employee development instruction does not directly relate to specific job tasks.

1.2.8. Instructional System Defined. An instructional system is an organized combination of procedures, techniques, and resources (e.g., instructors, students, courseware, equipment, tools, facilities, etc.) that are used to instruct specified learning objectives.

1.2.9. Courseware Defined. Courseware is a term commonly used in the Department of Defense (DoD) for course control documents, instructional materials, evaluation materials, and supporting documentation for an instructional system. Examples of courseware items are plans of instruction, training simulation programs, instructional presentations, student exercises, written tests, source code and authoring files, correspondence, decision documentation, etc.

1.3. Scope. This manual does not address methods and resources needed to deliver instruction after courseware validation. This manual covers:

1.3.1. Planning of courseware projects;

1.3.2. Analysis of instructional requirements;

1.3.3. Design of instruction;

1.3.4. Development of courseware;

1.3.5. Implementation of instruction;

1.3.6. Evaluation/Finalization of instruction;

1.3.7. Formal Re-Evaluation (Periodic Review) of instruction;

1.3.8. Management of courseware.

1.4. How to Use This Manual. Depending on the situation, users may skip directly to the relevant phase and activity, selecting the sections of this manual that best support their specific roles.

1.4.1. Content of the Manual. This manual contains a group of activities typically performed during functional courseware development and management.

1.4.1.1. Methods and Formats. When an activity can be accomplished in many effective ways, or when the outcome of an activity is likely to be impacted by rapidly changing technology, this manual does not specify methods or formats for accomplishing the outcome.

1.4.1.2. Reference to Additional Guidance. The *AFMC Depot Maintenance Training Resource* Site and the *AFMC ISD Resource* Site, both on SharePoint, provide courseware forms, examples, and valuable working aids. It does not restrict experienced developers from incorporating creative ideas, quality improvements, and emerging technology into their processes and products. To add new content to either site, contact Headquarters Air Force Materiel Command / Workforce Development and Training (HQ AFMC/A4PT). Assistance may also be provided by a Complex Courseware Program Manager (CPM).

1.4.2. Manual Structure.

1.4.2.1. Overview of AFMC Process. **Chapter 2** contains an overview of the AFMC ISD model and AFMC Courseware Development and Management Process.

1.4.2.2. Courseware Activity Chapters. Chapters **3-10** describes various activities that typically take place in phases and functions of the courseware process. Each activity includes these categories of information:

1.4.2.2.1. Purpose. Explains the reason an activity is performed.

1.4.2.2.2. Activity Outcomes. Defines typical deliverables and decisions that result from the activity.

1.4.2.2.3. Additional Guidance. Lists working aids (procedural guidance, examples, templates, worksheets, quality checklists, etc.) and references found in the *AFMC ISD Resource* Site or other sources.

1.4.2.2.4. Evaluation Criteria. Identifies elements and factors that should be considered when deciding if an activity has been accomplished well enough to move to the next process step.

1.4.2.3. AFMC ISD Release Points Checklist.

1.4.2.3.1. Courseware references.

1.4.2.3.2. Acronyms and terms associated with courseware.

1.4.2.3.3. Guidance on structure and content of courseware master files.

1.5. Roles and Responsibilities. While anyone in any organization may be assigned an activity in the Courseware Development and Management Process, responsibilities of such individuals can be categorized into a few basic roles.

1.5.1. Complex Maintenance Training Manager. An Air Force, or equivalent, ISD Theory course can help prepare a training manager for duties relating to processing of training gaps, courseware/training/non-training issue worksheets, search for existing content, manage courseware documentation, etc. Official responsibilities are contained in AFI 36-2650_AFMCSUP, *Maintenance Training*. As part of his/her responsibility, a Complex Maintenance Training Manager will:

1.5.1.1. Recommend courseware be developed only when it is determined (1) there is a valid instructional need; (2) the proposed instruction is an effective, cost-efficient solution; and (3) existing material cannot meet a defined requirement.

1.5.1.2. Ensure a Training Needs Assessment (TNA) and recommend courseware development only when training gaps are clearly identified, and when: (1) there is a valid instructional need; (2) the proposed training offers an effective and cost-efficient solution; and (3) existing materials are insufficient to meet the defined requirements.

1.5.1.3. Initiate development of training in support of upcoming workload, changes in work processes, acquisition of new systems, etc., as early as possible to provide adequate lead time and resources for developing needed instruction.

1.5.1.4. Base decisions on careful analysis of data collected from the customer and other expert sources, and not on assumptions or incomplete information. **Note:** Education

officers and employee development managers may perform the same role as training managers for education and employee development courseware.

1.5.2. Complex CPM. A Complex CPM is responsible for applying ISD methodology and AFMC Courseware Development and Management Processes for a functional area. The primary role is to serve as a liaison between their area of responsibility and higher Education and Training (E&T) offices on matters related to courseware. Official responsibilities are contained in AFI 36-2650_AFMCSUP. An Air Force, or equivalent, ISD Theory course will prepare a CPM to effectively address these issues. As part of his/her responsibility, a Complex CPM will:

1.5.2.1. Conduct or assign periodic evaluations of instructional courseware under your responsibility, and remove any content that is not aligned with the mission or job requirements.

1.5.2.2. Assign course numbers In Accordance With (IAW) AFI 36-2650_AFMCSUP.

1.5.2.3. Oversee electronic and physical courseware master libraries.

1.5.2.4. Implement and tracks official changes to courseware master files and processes courseware Interim Change Memos (as needed).

1.5.2.5. Distribute courseware to authorized individuals for review or implementation.

1.5.2.6. Activate archives, deactivates/reactivates courseware upon request from authorized individuals.

1.5.2.7. Coordinate courseware signature page process as outlined in AFI 36-2650_AFMCSUP.

1.5.2.8. Coordinate Subject Matter Expert (SME) support for command and local courseware development and periodic review projects.

1.5.3. Courseware Project Manager. A Courseware Project Manager is responsible for monitoring progress of assigned courseware projects, assisting courseware development team in removing obstacles that threaten a timeline or project budget, and reporting project status to appropriate training offices and customer representatives. Courseware Project Managers oversee tasks in each ISD phase and function, where their qualifications enable them to identify and prioritize instructional requirements, and to intercede as necessary to ensure that they are met. An Air Force or equivalent ISD Theory course will prepare a courseware project manager to perform these and other critical functions. As part of his/her responsibility, a Courseware Project Manager will apply/identify:

1.5.3.1. The principles of learning, systems engineering, Project Management Body of Knowledge (PMBOK) and quality improvement that are the foundation of ISD.

1.5.3.2. Various instructional design models, methods, techniques, media and computer technologies are used in a blended learning approach to instruction.

1.5.3.3. Processes within Air Force and AFMC education, training, and employee development programs.

1.5.3.4. Reason for each activity in the courseware process.

1.5.3.5. The risks of not performing a particular activity in the courseware process.

1.5.3.6. Formulas and adjustment factors for computing assignment priority and rough order of magnitude of resources needed to accomplish courseware projects.

1.5.4. Courseware Developer. A Courseware Developer is responsible for designing and producing instruction that is relevant, engaging, effective, and delivered in a cost-efficient, low-risk manner. The developer's role spans the Planning, Analysis, Design, Development, and Implementation phases of the instructional process. Completion of an Air Force or AFMC ISD Theory course equips developers with the foundational skills needed to perform these tasks successfully. As part of their responsibilities, Courseware Developers must demonstrate proficiency in the following areas:

1.5.4.1. Principles of learning, systems engineering, Project Management Body of Knowledge (PMBOK) and quality improvement which are the foundations of ISD.

1.5.4.2. Various instructional design models, methods, and techniques are used to develop instruction.

1.5.4.3. Courseware Developer should attend a formal Air Force ISD Theory course, or equivalent, training prior to assignment to a development position.

1.5.4.4. Various methods, techniques, media and computer technologies are used in a blended learning approach to instruction.

1.5.4.5. Purpose and value of each activity is part of the AFMC courseware process.

1.5.4.6. Ensures adequate planning activities are accomplished before a project enters a later phase in a Courseware Development and Management Process.

1.5.4.7. Sets reasonable suspense dates for courseware activities, realizing that development of instructional systems can be a relatively slow and complex process.

1.5.5. Instructor or Trainer. The instructor or trainer serves as an expert on effectiveness of instructional and evaluation activities/materials in meeting the intent of the course. An Air Force, or equivalent, ISD Theory course and Principles of Instruction (POI) will help to prepare the instructor. The following are responsibilities of an instructor or trainer:

1.5.5.1. Assists in course or learning aid planning.

1.5.5.2. Provides input on course flow, design and content of instructional and evaluation materials.

1.5.5.3. Tests and provides feedback on effectiveness of presentation and learning activities.

1.5.5.4. Attends train-the-trainer events (as needed) to become qualified to teach new courses.

1.5.5.5. Assists in formal Re-Evaluation (periodic review) of courses.

1.5.5.6. Initiates Interim Changes and out-of-cycle reviews of courseware (as needed).

1.5.6. Subject Matter Expert (SME). The SME has a high level of knowledge and skill in a particular topic, task, system, or process. The primary role of a SME is to verify accuracy and completeness of course content. As part of his/her responsibility, SMEs will:

1.5.6.1. Assist in development of course or learning aid.

1.5.6.2. Advise on specific tasks, performance steps and decision points to be covered, along with conditions and standards for acceptable performance.

1.5.6.3. Contribute to content identification of instructional and evaluation materials.

1.5.6.4. Verify accuracy and completeness of information in instructional and evaluation materials.

1.5.6.5. Assist in formal Re-Evaluation (periodic review) of courseware related to their area of expertise to verify currency and accuracy of materials.

1.5.6.6. Participate in technical reviews, small group tryouts, and operational tryouts of courseware to advise on content and provide input on ways to add relevance to instruction (real-world examples, lessons learned, etc.).

1.5.6.7. Alert instructors or appropriate training office about changes to work processes, directives, or systems when such changes may impact instruction.

1.5.7. Maintenance Supervisor. The maintenance supervisor is ultimately responsible for training, qualification, and/or certification of assigned personnel. Official responsibilities are contained in AFI 36-2650_AFMCSUP. The maintenance supervisor:

1.5.7.1. Works with the appropriate Group Training Manager to identify work center training requirements based on regulatory guidance and work center tasks.

1.5.7.2. Supports Group Commander in providing SMEs and students as requested in support of courseware development and management activities.

1.5.7.3. Provides graduate assessment survey feedback as necessary to evaluate the relevancy and effectiveness of courses and learning aids.

1.5.8. ISD Evaluator. ISD Evaluators at each installation are functional experts on the application of ISD methodology and AFMC Courseware Development and Management Process. Only experienced Courseware Developers or Complex Courseware Program/Project Managers with the ability to interpret courseware policy for their installation should be assigned this role. The ISD Evaluator reviews courseware activities performed by others to verify evaluation criteria in this manual have been met and the project complies with ISD process intent. Attending an approved ISD course will prepare selected personnel to perform this function.

Chapter 2

OVERVIEW OF AFMC INSTRUCTIONAL DEVELOPMENT PROCESS

2.1. Instructional System Development Defined. ISD is a practical and flexible approach for developing and delivering instruction that promotes transfer of learning from an instructional setting to the workplace. It adapts a systems engineering process to provide a means for sound decision making and integrates art and technology with scientific principles of psychology, sociology, andragogy, and anthropology to ensure quality of instruction. ISD produces relevant, effective, and economical instructional systems.

2.2. Benefits of ISD. Although the ISD method may look cumbersome and costly, it has proven itself to be both efficient and inexpensive in the long run. ISD requires a systematic collection and careful analysis of data and encourages objectivity. ISD focuses on specific requirements and expectations of the customer to promote relevancy. Use of the ISD process results in learning objectives are based on job performance requirements and instruction is effective and economical to sustain.

2.3. History of ISD. ISD is not new. It has its roots in the 1950s. A partnership between the Army and Florida State University (FSU) to research better ways to develop instruction for adult learners advanced the concept. In the 1970s, the DoD commissioned faculty members from FSU to formally define a process for development and delivery of instruction for all DoD services. Those efforts resulted in the original ISD model.

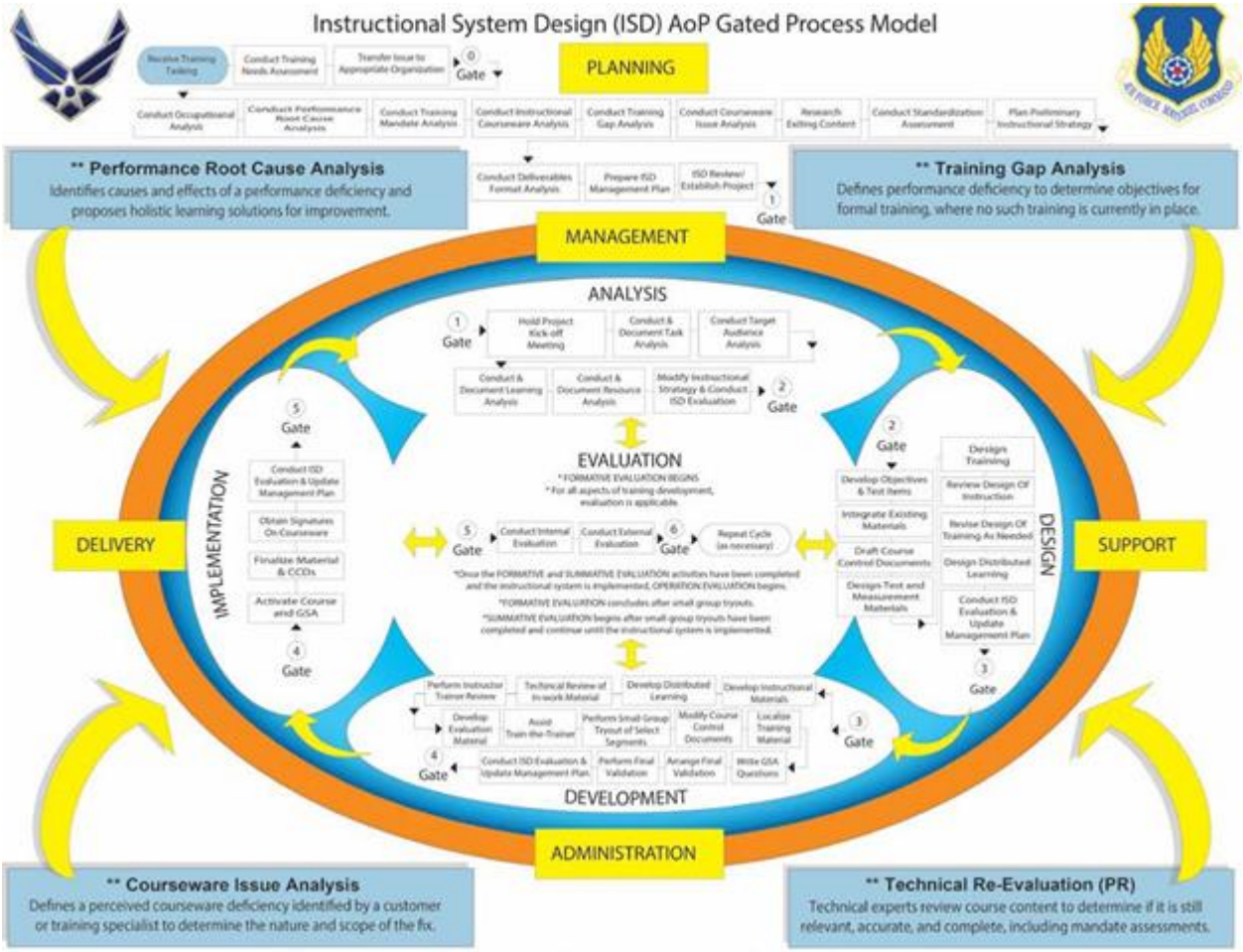
2.4. ISD Model Evolution. Over time, advances in instructional technology added more conditions and variables to the original ISD model. A systems engineering process was adapted to control and manage an increasingly complex process. Next, the ISD process was made flexible and interactive with the introduction of phases that could be entered at any point depending on the specific situation. This Analysis, Design, Development, Implementation, Evaluation (ADDIE) model is the basis of nearly all instructional design methodologies in use today. Additionally, advances in learning theory and educational technology caused major changes in some ISD variables. Continual evaluation and quality improvement are embedded into all phases and functions of ISD process to facilitate sound decision-making based on research and analysis.

2.4.1. Upon its stand-up in 2012, AFSC adopted “Art of the Possible” as its business model. The AoP business model uses a gated process philosophy to create process visibility and allow pro-active management of the items moving through AFSC processes. The gated Art of the Possible (AoP) philosophy integrates with the Instructional Systems Development (ISD) process to enhance training delivery by systematically identifying and resolving constraints at each phase. For more information refer to Air Force Sustainment Center Handbook (AFSCH) 60-101, *Art of the Possible Handbook*, and *AFMC ISD Resource* site for guidance on calibrating the gated machines.

2.5. ISD Process Variations. Nearly all businesses, academic institutions, and government agencies that use the ISD process modify it somewhat to meet their specific needs. Some variations go by different names such as: Systems Approach to Training (SAT), Instructional Development (ID), or Instructional Systems Design & Development (ISDD). However, the systems engineering process and basic ADDIE model are still at the core of all of these customized methodologies.

2.6. Current Department of Defense (DoD) ISD. Below is an illustration of a DoD ISD model configured for Air Force and AFMC ISD Art of the Possible (AoP) processes. Military Handbook (MIL-HDBK)-29612-2A, *Instructional Systems Development/Systems Approach to Training and Education*, and DAFH 36-2675, provide additional information on the ISD model and methodology. This manual includes a brief overview to provide context for later chapters.

Figure 2.1. AFMC DoD ISD Model Example.



2.7. Quality Improvement (QI) Infrastructure. Phases and functions of the ISD model are embedded in, and held together by, an overarching QI process. This QI infrastructure permeates every activity and outcome of the ISD process. It continually generates refinements to an instructional system based on the evaluation of activity outcomes against defined performance requirements. If a product or service resulting from an activity does not meet specified standards, then quality controls in each phase of the ISD process enable management to quickly redirect the project to an appropriate step or activity so that needed remedies or requested improvements are accomplished with little threat to budget or schedule.

2.7.1. ISD System Functions. Management, support, administration, and delivery are basic functions that sustain QI infrastructure and core phases of the ISD process. Each of these functions involves numerous responsibilities and activities that ensure instructional systems are developed, delivered, and maintained in an effective manner.

2.7.1.1. Management. This function directs, monitors, and controls all activities associated with the instructional system throughout its life cycle. Various participants (e.g., supervisors, instructors, courseware developers, etc.) can perform management roles. Some basic management activities are:

2.7.1.1.1. Planning for instructional system activities.

2.7.1.1.2. Arranging resources such as skills, facilities, equipment, funding, etc., needed to accomplish planned activities in support of the instructional system.

2.7.1.1.3. Reporting on status of instructional system elements and activities.

2.7.1.2. Support. This function involves long-range and day-to-day tasks that are performed in order to implement and maintain an instructional system throughout its life cycle. Some basic support activities are:

2.7.1.2.1. Providing funding, facilities, manpower authorizations, and services in support of instructional activities.

2.7.1.2.2. Supplying equipment, skilled personnel, courseware, and instructional aids.

2.7.1.2.3. Maintaining facilities, equipment, skilled personnel, courseware, and instructional aids.

2.7.1.3. Administration. This function conducts day-to-day operations of the instructional system throughout its life cycle. Some basic administration activities are:

2.7.1.3.1. Preparing documentation such as instructional materials, course control documents, plans, and reports.

2.7.1.3.2. Maintaining records such as courseware master files, rosters, test scores, student completions, budgets, and contracts.

2.7.1.3.3. Providing staff support such as processing personnel actions and maintaining employee development programs.

2.7.1.3.4. Scheduling resources such as personnel, equipment, and facilities.

2.7.1.3.5. Implementing and maintaining automated information management systems and other tools to support instructional system activities.

2.7.1.3.6. Administering funds and contracts.

2.7.1.4. Delivery. This function provides instruction to students throughout its life cycle. Some basic delivery activities are:

2.7.1.4.1. Providing an infrastructure for distance learning including an on-line course management system, equipment, and facilities.

2.7.1.4.2. Supplying skilled personnel to instruct courses.

2.7.1.4.3. Maintaining current, effective, and appealing courseware.

2.7.1.5. Evaluation. This function is the bridge between QI infrastructure and system phases of the ISD process. It is a continual measurement of activity outcomes against defined performance requirements and is part of every step in the ISD process. The

evaluation function gathers feedback to assess instructional system and student performance. Some basic evaluation activities include:

2.7.1.5.1. Formative evaluations include internal reviews, individual tryouts, small group tryouts, and ISD validations conducted during Analysis, Design and Development Phases.

2.7.1.5.2. Summative evaluations include operational tryouts during Implementation Phase.

2.7.1.5.3. Operational evaluations include periodic instructor/trainer review, subject matter expert/program manager review, graduate assessment surveys, and other types of reviews in the instructional system after implementation.

2.8. AFMC Functional Courseware Development and Management Process. AFMC employs DoD ISD model as its standardized process for courseware development and management. The AFMC process includes Planning, Analysis, Design, Development, Implementation, Evaluation/Finalization and Periodic Review of course material, as well as Courseware Management, Acquisition and Contract Oversight Functions, as defined below. DAFH 36-2675 provides additional information on the ISD model:

2.8.1. Planning. Analyze performance deficiencies, define training requirements, decide on instructional strategies, identifies activities and deliverables of a proposed courseware project, and estimates resources needed to accomplish it. **Note:** DAFH 36-2675, Chapter 2, provides additional Air Force related information and guidance for completing the Planning function of the ISD process.

2.8.2. Analysis Phase. Further analyze requirements to determine precisely what should be included in instruction, types of learning involved, proficiency level of learning needed for the target audience, and any prerequisite or follow-on instruction required. **Note:** DAFH 36-2675, Chapter 3, provides additional Air Force related information and guidance for completing the Analysis Phase of the ISD process.

2.8.3. Design Phase. Create a blueprint of the course or learning aid that shows what instruction, evaluation, and media will be used in each module to meet defined requirements. **Note:** DAFH 36-2675, Chapter 4, provides additional Air Force related information and guidance for completing the Design Phase of the ISD process.

2.8.4. Development Phase. Develop, test, and revise instructional and evaluation materials and prepare the instructional system for full operational tryout in the field. **Note:** DAFH 36-2675, Chapter 5, provides additional Air Force related information and guidance for completing the Development Phase of the ISD process.

2.8.5. Implementation Phase. Conduct evaluation of an instructional system under field conditions, make final changes, posts official courseware master file in the appropriate installation functional courseware library, distribute courseware to authorized users, and personalize lesson plans so that a course or learning aid is fully operational and ready for use in the field. **Note:** DAFH 36-2675, Chapter 6, provides additional Air Force related information and guidance for completing the Implementation Phase of the ISD process.

2.8.6. Evaluation/Finalization of course material. Consists of operational, internal and external evaluations conducted throughout the ISD process and is the process for closing out

the project and distributing the courseware for instructional delivery. **Note:** DAFH 36-2675, Chapter 7, provides additional Air Force related information and guidance for completing the Evaluation Phase of the ISD process.

2.8.7. Periodic Review of course material. Periodically review courseware to determine if courseware is still needed, identify if changes are needed and either revalidate instruction, initiate a revision project, or archive the courseware.

2.8.7.1. Periodic reviews also apply to standard maintenance of course material as a result of minor changes to processes. Examples include data refreshes and reference updates.

2.9. ISD Process. ISD is by no means a rigid or linear process, and individual projects include only those activities needed to ensure relevant, effective and economical instructional systems.

2.10. Applying AFMC Courseware Process. The AFMC Courseware and Development Process for functional training may at first appear intimidating, restrictive, and too time-consuming to be practical in the real world. Actually, it is quite flexible and serves more as a problem-solving and decision-making model than a mandatory list of activities.

2.10.1. Choosing Activities. The courseware project manager or other instructional design expert decides which activities will likely be needed in each phase, depending on constraints and requirements of the particular project.

2.10.2. Mandatory Steps and Procedures. Mandatory steps and procedures are clearly identified in activity chapters. **Note:** Courseware developers are free to incorporate emerging technology and their own creative ideas, lessons learned, and quality improvements into the process and its products.

2.10.3. Entering and Re-Entering Process. Issues and projects enter the courseware process in the first step of Planning. After the second step of Planning, (**Note:** Refer to **Chapter 10**) a courseware project manager and courseware development team enter or re-enter a process after deciding on the next appropriate step for the situation.

2.10.4. Adapting Process. The steps in the AFMC courseware process are intended to remind participants of activities involved in developing and managing instructional systems in a variety of situations and are not supposed to be an obstacle course of rules that must be followed. Issues often arise that require quick and ingenious solutions that a courseware project manager and courseware developer may need to bypass, modify, switch the sequence, or add new steps of their own in response. If constraints of a particular situation prevent them from following the standard courseware process, they determine ways to mitigate risks, document their decisions, and implement the plan.

2.10.5. Flexible Sequence of Activities. While flow charts illustrate a courseware process in a linear manner, steps and phases should not be thought of as consecutive in nature. For example, one activity does not have to be completed before the next one is started and steps do not always have to be accomplished in a recommended sequence. Each project has its own natural flow of activities depending on each situation. As long as required activities are completed, and variation in sequence does not waste resources or add unacceptable risk, flexibility is encouraged.

2.10.6. Making Decisions Based on Data. The ISD approach is successful because decisions are based on careful analysis of data collected from customers and other expert sources rather

than on assumptions. Skipping the Planning and Analysis steps out of convenience, or making decisions based on assumptions or incomplete data threatens the success of a project.

2.10.7. Evaluation and Quality Control. Evaluation and quality control measures have been built into the courseware process to ensure that instruction is relevant, economical, and effective. If an activity outcome does not meet evaluation criteria or does not provide enough information to make a sound decision, the ISD Evaluator or courseware project manager should not hesitate to request rework or to re-insert a project into a remedial activity to mitigate risk and ensure quality.

2.10.8. Documenting Decisions. Decisions to skip a step, modify requirements for an activity outcome, or redirect a project are documented for audit. Supporting documents/arguments should be provided in ISD Evaluator review notes.

2.10.9. Experience Counts. The less experience that participants have with developing and managing courseware, the closer they should follow the AFMC process in this manual. Not fully understanding scientific principles and theories underlying ISD and the reason for each activity in the courseware process has led many novices in a wrong direction, resulting in wasted resources and unnecessary, ineffective, or inefficient instruction.

2.11. Adapting AFMC Courseware Process to Education and Employee Development. The AFMC Functional Courseware and Development Process can be adapted easily to development and management of education and employee development instruction. Whenever an activity specifies “training,” simply substitute “education” or “employee development” as appropriate, and guidance will readily apply.

2.11.1. Planning Activities. Because functional training is the most complex type of instruction, some activities in Planning may not be necessary when developing education and employee development courseware.

2.11.2. Academic Accreditation. Accreditation guidance for developing education courseware for formal schools is not covered in this manual, but may be obtained from the academic institution where instruction will be delivered.

Chapter 3

PLANNING

3.1. Planning Defined. This is the foundation of a courseware process. During the Planning process, SMEs, functional area supervisors, subject area program managers, and automated system Officers of Primary Responsibility (OPRs) work together with courseware specialists to establish training concepts, decide on training strategies, analyze performance deficiencies, define training requirements, identify activities and deliverables of the proposed courseware project, and estimate resources needed to accomplish it. For additional information and guidance relating to the Planning process of the Air Force ISD process, refer to DAFH 36-2675.

3.2. Reason for Separate Planning Process. Some courseware development is contracted out to professional instructional designers. Many of the activities which normally fall under the Analysis Phase of the ISD model need to be accomplished before writing contract requirement documents. A second reason for establishing a separate Planning process is to emphasize the importance of these often overlooked activities. Keep in mind that data collected and decisions made during Planning are later validated and adjusted (as needed) during the Analysis Phase, based on changes in the situation and deeper analyses of student demographics and content to be trained.

3.3. Receive Potential Training Issue. A customer or training specialist identifies a perceived training issue and submits it to the appropriate Maintenance Training Office level for action. This is a required step in the AFMC courseware process.

3.3.1. Purpose. This activity represents the start of a courseware process regardless of what other activity (e.g., courseware review, occupational training analysis, performance root cause analysis, etc.) triggers it.

3.3.2. Activity Outcome. Outcome is an issue submitted to the appropriate training office for action.

3.3.2.1. Identify and obtain project SMEs.

3.3.3. Additional Guidance. Sample tasking worksheets are at the *AFMC Depot Maintenance Training Resource* Site.

3.3.4. Evaluation Criteria. Is documentation of a training issue sufficient to determine:

3.3.4.1. Nature of perceived issue?

3.3.4.2. Priority of assignment?

3.3.4.3. Appropriate organization to handle tasking request?

3.3.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

3.3.5.1. Nature of perceived issues?

3.3.5.2. Priority of assignment?

3.3.5.3. Appropriate organization to handle tasking request?

3.3.6. Follow-on: Is the decision to send this issue to another organization for action without accomplishing additional activities supported by regulation, policy, or sound logic?

3.4. Training Needs Assessment. In this activity, a training function decides how to act on a perceived training issue submitted by the customer or training specialist. This is a required step in the AFMC courseware process.

3.4.1. Purpose. This activity allows a training function to decide if a training request is best resolved through training and where to enter the courseware process based on how much is known about the training issue. It enables experienced participants to solve a problem efficiently without compromising effectiveness or relevance of the outcome.

3.4.2. Activity Outcomes:

3.4.2.1. Training Needs Assessment documentation.

3.4.2.2. Decision that either an issue is outside the scope of a courseware process, or that the courseware process will be used to address an education, training or employee development issue.

3.4.2.3. Decision on which organization should handle the issue at this point.

3.4.2.4. Decision on where to go next in the courseware process.

3.4.3. Additional Guidance. See *AFMC Depot Maintenance Training Resource* Site

3.4.4. Evaluation Criteria.

3.4.4.1. Is there enough information in the issue description to justify a determination that courseware is or is not a direct part of the solution to this problem?

3.4.4.2. Is a decision to send this issue to another organization for action without accomplishing additional activities supported by regulation, policy, or sound logic?

3.4.4.3. Is there existing training that will fulfill this requirement?

3.5. Transfer Issue to Appropriate Organization. In this activity, a training specialist forwards the issue to a more appropriate organization to handle remaining project activities.

3.5.1. Purpose. This activity allows a training function to forward an issue to an organization that is more responsible for subject matter or scope of the work involved in it.

3.5.2. Activity Outcome. Outcome is a message with attached issue description that is submitted to a more appropriate organization.

3.5.3. Additional Guidance. Sample tasking worksheets are at the *AFMC Depot Maintenance Training Resource* Site.

3.5.4. Evaluation Criteria. Does justification in the message explain:

3.5.4.1. Why an approached training function is not the most appropriate for tasking?

3.5.4.2. Why this falls into the jurisdiction of another organization?

3.6. Conduct an Occupational Analysis. In this activity, a training office identifies training needed to perform work of an occupation from entry level through advanced journeyman. It is performed in support of Civilian Career Field Education and Training Plan, Civilian Training Plan, Career Development Plan, Occupational Training Template development, etc.

3.6.1. Purpose. This activity consolidates several other courseware activities to identify functional training needed for an occupation. Occupational training analysis helps supervisors

and training specialists develop individual and occupational training plans for employees, and assists courseware developers by identifying training requirements and target populations.

3.6.2. Activity Outcomes:

3.6.2.1. Occupational Training Analysis Report.

3.6.2.2. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

3.6.2.3. Decision on where to go next in the courseware process.

3.6.3. Additional Guidance. Sample tasking worksheets are in the *AFMC Depot Maintenance Training Resource* Site.

3.6.4. Evaluation Criteria.

3.6.4.1. Does an Occupational Training Analysis Report include, at a minimum:

3.6.4.1.1. Knowledge/Task Analysis Report that meets criteria for that analysis activity?

3.6.4.1.2. Training Requirements Matrix that meets criteria for that activity?

3.6.4.1.3. Instructional System Analysis Report that meets criteria for that activity?

3.6.4.1.4. Training Standardization Assessment that meets criteria for that activity?

3.6.4.1.5. Completed tasking worksheets for any action items defined during an occupational training analysis?

3.6.4.2. Did data come from, or was it approved by, those in authority to define training requirements for the subject area?

3.6.4.3. Are any issues/action items that emerged during this activity documented sufficiently to determine:

3.6.4.3.1. Nature of perceived issues?

3.6.4.3.2. Priority of assignment?

3.6.4.3.3. Appropriate organization to handle tasking request?

3.6.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

3.6.5.1. Nature of perceived issues?

3.6.5.2. Priority of assignment?

3.6.5.3. Appropriate organization to handle tasking request?

3.6.6. Is a decision to send this issue to another organization for action without accomplishing additional activities supported by regulation, policy, or sound logic?

3.7. Conduct Performance Root Cause Analysis. In this activity, a training office or customer organization examines a performance deficiency, analyzes underlying causes, and determines a solution. Include occupational and training requirements analysis.

3.7.1. Purpose. This activity identifies causes and effects of a performance deficiency and proposes holistic solutions for improving performance which may or may not include a training intervention. Performance root cause analysis considers not only a lack of skills or knowledge, but other factors that might contribute to the performance deficiency.

3.7.2. Activity Outcomes:

3.7.2.1. Performance Root Cause Analysis.

3.7.2.2. Decision on where to go next in the courseware process.

3.7.2.3. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

3.7.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

3.7.4. Evaluation Criteria.

3.7.4.1. Were appropriate personnel involved in collection and analysis of data?

3.7.4.2. Was adequate data collected to support the analyses and recommendations?

3.7.4.3. Were appropriate analysis methods and statistical process control techniques used to evaluate and illustrate data?

3.7.4.4. Does the problem statement focus on what a deficiency is and not on why it exists? (Watch for “lack of” and “no” statements as they imply solutions.)

3.7.4.5. Does each cause statement clearly define a cause and effect of a performance deficiency so that anyone who becomes involved from this point forward can understand the reason for a proposed corrective actions?

3.7.4.6. Are plans included to evaluate the effectiveness of the intervention(s) after they have been implemented?

3.7.4.7. Is a proposed solution logical, feasible, and cost-effective? (**Note**: The average cost of one-hour training may exceed 1 million dollars.)

3.7.4.8. For each root cause identified:

3.7.4.8.1. Would a performance deficiency have occurred had the cause not been present?

3.7.4.8.2. Will correction or elimination of the cause prevent recurrence of a like condition?

3.7.4.9. Does a Performance Root Cause Analysis Report clearly:

3.7.4.9.1. Define performance deficiency?

3.7.4.9.2. Describe facts and root causes?

3.7.4.9.3. Identify pros and cons of possible corrective actions?

3.7.4.9.4. Provide recommendations on how to proceed?

3.7.4.9.5. Identify initial tasks for instruction if training is part of a performance deficiency’s root cause.

3.7.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

3.7.5.1. Nature of perceived issues?

3.7.5.2. Priority of assignment?

3.7.5.3. Appropriate organization to handle tasking request?

3.8. Conduct Training Mandate Analysis. In this activity, a responsible office or customer organization documents training mandated and recommended (from initial awareness through refresher) for each target population in support of a particular work process, program, system, or subject area.

3.8.1. Purpose. This activity assists developers by identifying training mandates and target populations.

3.8.2. Activity Outcomes:

3.8.2.1. Training Mandate Analysis.

3.8.2.2. Possible tasking worksheets on training issues, courseware issues, training gaps, or non- training issues submitted to appropriate organization for action.

3.8.2.3. Decision on where to go next in the courseware process.

3.8.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

3.8.4. Evaluation Criteria.

3.8.4.1. Does a Training request include, at a minimum:

3.8.4.1.1. Numbers and titles of directives, training plans, etc., which prescribe training requirement?

3.8.4.1.2. List of all existing or proposed instruction (including training gaps)?

3.8.4.1.3. Priority of assignment?

3.8.4.1.4. Appropriate organization to handle tasking request?

3.8.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

3.8.5.1. Nature of perceived issues?

3.8.5.2. Priority of assignment?

3.8.5.3. Appropriate organization to handle tasking request?

3.8.5.4. Is a decision to send this courseware deficiency issue to another organization for action without accomplishing additional activities supported by regulation, policy, or sound logic?

3.9. Conduct Functional Courseware Analysis. In this activity, a courseware project manager or courseware developer analyzes courses and learning aids within available resources for a particular work process, program, system, or subject area to meet an identified training requirement. It is highly recommended that a Training Mandate Analysis be accomplished before analyzing the full instructional system.

3.9.1. Purpose. This activity determines the availability/compatibility of existing training for a particular work process, program, system, or subject area. It is performed in order to identify gaps and duplication, and to decide if existing courses and learning aids within a system are effective in meeting the defined training requirements (from initial awareness through refresher) for all target populations.

3.9.2. Activity Outcomes:

3.9.2.1. Instructional Courseware Analysis.

3.9.2.2. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

3.9.2.3. Decision on where to go next in the courseware process.

3.9.3. Additional Guidance. AFMC Form 96, *Existing Courseware Evaluation Checklist,* should be used to document research. See the *AFMC Depot Maintenance Training Resource* Site

3.9.4. Evaluation Criteria.

3.9.4.1. Does the Instructional Courseware Analysis Report adequately describe the following:

3.9.4.1.1. Courses and learning aids that exist or are in work for the instructional system?

3.9.4.1.2. Any gaps in instruction that appear to exist, in terms of tasks that need to be taught and proficiency level of learning required?

3.9.4.1.3. Any duplication of instruction that appears to exist?

3.9.4.1.4. Any unnecessary instruction that appears to exist?

3.9.4.1.5. Any courseware deficiencies that appear to exist, in terms of tasks that need to be taught and proficiency level of learning required?

3.9.4.2. Did data come from an adequate mix of novices, SMEs, training managers, subject area program managers, system OPRs, instructors, trainers, courseware developers, and other customer representatives to ensure accurate and complete information on which to base findings and recommendations?

3.9.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

3.9.5.1. Nature of perceived issues?

3.9.5.2. Priority of assignment?

3.9.5.3. Appropriate organization to handle tasking request?

3.10. Conduct Training Gap Analysis. In this activity, a courseware project manager or courseware developer works with a customer to better define and analyze a perceived gap in training and determine the nature and scope of a fix. This is accomplished by using various surveys to determine if an issue stems from a process, experience, attitudes, satisfaction, climate, management practices, equipment, facilities, budget, or a combination.

3.10.1. Purpose. This activity analyzes a perceived training gap or need identified by a customer or training specialist so that a solution can be planned.

3.10.2. Activity Outcomes:

3.10.2.1. Training Gap Analysis Report.

3.10.2.2. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

3.10.2.3. Decision on whether currently tasked training function should continue to handle this project for now.

3.10.2.4. Decision on where to go next in the courseware process.

3.10.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

3.10.4. Evaluation Criteria.

3.10.4.1. Does a Training Gap Analysis Report include findings, recommendations, and supporting data sufficient to define:

3.10.4.1.1. Why a training gap or need exists?

3.10.4.1.2. How employees are currently obtaining needed knowledge/skills/attitudes?

3.10.4.1.3. The nature and scope needed instruction, in terms of tasks that need to be taught and the proficiency level of learning required?

3.10.4.1.4. Who needs the proposed instruction?

3.10.4.1.5. The impact of a training gap?

3.10.4.1.6. Benefits of various proposed solutions?

3.10.4.2. Did data come from an adequate mix of novices, SMEs, subject area program managers, system OPRs, instructors, trainers, courseware developers, training managers, etc., to ensure accurate and complete information on which to base findings and recommendations?

3.10.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

3.10.5.1. Nature of perceived issues?

3.10.5.2. Priority of assignment?

3.10.5.3. Appropriate organization to handle tasking request?

3.10.5.4. Is a decision to send this training gap issue to another organization for action without accomplishing additional activities supported by regulation, policy, or sound logic?

3.11. Conduct Courseware Issue Analysis. In this activity, a courseware project manager or courseware developer works with a customer to better define and analyze a perceived existing courseware deficiency and to determine the nature and scope of the fix.

3.11.1. Purpose. This activity analyzes a perceived courseware deficiency identified by a customer or training specialist so that a solution can be planned.

3.11.2. Activity Outcomes:

3.11.2.1. Courseware Issue Worksheet.

3.11.2.1.1. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

3.11.2.1.2. Decision on whether currently tasked training function should continue to handle this project for now.

3.11.2.1.3. Decision on where to go next in the courseware process.

3.11.2.1.4. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

3.11.3. Evaluation Criteria.

3.11.4. Does a Courseware Issue Analysis include findings, recommendations, and supporting data sufficient to define:

3.11.4.1. Why a courseware deficiency exists?

3.11.4.1.1. How employees are currently obtaining needed knowledge/skills/attitudes?

3.11.4.1.2. Nature and scope of needed instruction, in terms of tasks that need to be taught and proficiency level of learning required?

3.11.4.1.3. Who needs the proposed instruction?

3.11.4.1.4. The impact of a courseware deficiency?

3.11.4.1.5. Benefits of various proposed solutions?

3.11.4.1.6. Did data come from an adequate mix of novices, SMEs, subject area program managers, system OPRs, instructors, trainers, courseware developers, training managers, etc., to ensure accurate and complete information on which to base findings and recommendations?

3.11.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

3.11.5.1. Nature of perceived issues?

3.11.5.2. Priority of assignment?

3.11.5.3. Appropriate organization to handle tasking request?

3.11.5.4. Is a decision to send this courseware deficiency issue to another organization for action without accomplishing additional activities supported by regulation, policy, or sound logic?

3.12. Research and Evaluate Existing Content. In this activity, a courseware specialist searches for available materials (e.g., directives, technical data, working aids, courseware, vendor instruction, etc.), from courses identified that might meet all or part of a defined requirement and evaluates its suitability. A decision is made as to what can be easily adopted and what remaining courseware development (if any) is needed.

3.12.1. Purpose. This activity saves resources by adopting as much existing documentation, courseware, and vendor instruction as is practical.

3.12.2. Activity Outcomes:

3.12.2.1. AFMC Form 96.

3.12.2.2. Existing Content Suitability.

3.12.2.3. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

3.12.2.4. Decision on where to go next in the courseware process.

3.12.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

3.12.4. Evaluation Criteria.

3.12.4.1. Does the AFMC Form 96 include:

3.12.4.1.1. Specific sites and sources searched for existing materials? At a minimum, check with all the Air Logistics Complexes (ALC), Training Scheduling System (TSS), AF myLearning, Defense Acquisition University (DAU) and Air Force Institute of Technology (AFIT) to ensure that they do not have similar courseware.

3.12.4.1.2. Key words used to search for existing materials?

3.12.4.1.3. List of potentially applicable directives, technical data, working aids, courseware, vendor instruction, etc., found?

3.12.4.1.4. Description of suitability for each available item in meeting at least part of the requirements of a proposed course or learning aid?

3.12.4.1.5. Is material copyrighted, proprietary, or classified?

3.12.4.1.6. Is material current?

3.12.4.1.7. Estimate of how much rework on materials or adjustment of a proposed instructional approach will be necessary if a found item or available instruction is incorporated or adopted?

3.12.4.1.8. Has AFMC Form 96 been signed by the Complex Maintenance Training Manager and the Courseware Program Manager?

3.12.4.2. Were logical key words describing a proposed instructional content used when searching?

3.12.4.3. Were decisions on suitability of existing materials approved by lead SMEs, subject area program managers, system OPRs, training managers, instructors, trainers, courseware developers, etc.?

3.12.4.4. Were a satisfactory number of sites and sources in government, private sector and academia searched for existing materials, to include such AFMC-approved functional training management systems as the Courseware Management Database (CMD)?

3.12.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

3.12.5.1. Nature of perceived issues?

3.12.5.2. Priority of assignment?

3.12.5.3. Appropriate organization to handle tasking request?

3.13. Conduct Training Standardization Assessment. In this activity, a training specialist determines if proposed training development or revision can be expanded to include other organizations, installations, or agencies.

3.13.1. Purpose. This activity promotes consistency of instruction across organizations and saves resources by preventing duplicate courseware.

3.13.2. Activity Outcomes:

3.13.2.1. Training Standardization Assessment with documented decision on whether existing or proposed training can be expanded to include other organizations, installations, or agencies.

3.13.2.2. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

3.13.2.3. Decision on whether currently tasked training function should continue to handle this project for now.

3.13.2.4. Decision on where to go next in the courseware process.

3.13.3. Additional Guidance. See AFI 36-2650_AFMCSUP.

3.13.4. Evaluation Criteria.

3.13.4.1. Is a decision on whether or not to standardize training supported by regulation, policy, or sound logic?

3.13.4.2. Does the decision documentation include findings, recommendations, and supporting data sufficient to justify whether or not to standardize training?

3.13.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

3.13.5.1. Nature of perceived issues?

3.13.5.2. Priority of assignment?

3.13.5.3. Appropriate organization to handle tasking request?

3.14. Plan Preliminary Instructional Strategy. In this activity, a courseware specialist analyzes data from previous Planning Process activities and tentatively selects the most suitable approach for developing and delivering new or significantly revised courseware. This Instructional Strategy not only serves as a project plan, but also considers how to apply accepted learning theory to the situation.

3.14.1. Purpose. This activity is critical for determining rough order of magnitude of resources needed to complete a new development or major revision project, and to implement courseware in the field.

3.14.2. Activity Outcomes:

3.14.2.1. Preliminary Instructional Strategy Report.

3.14.2.2. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

3.14.2.3. Decision on where to go next in the courseware process.

3.14.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

3.14.3.1. References on theories and principles of learning and various models and methodologies for instructional design.

3.14.3.2. References on selecting appropriate instructional methods/tools and media based on the situation.

3.14.3.3. Instructional Strategy guidance and examples.

3.14.4. Evaluation Criteria.

3.14.4.1. Does a Preliminary Instructional Strategy Report identify and justify the following aspects of a preliminary instructional plan:

3.14.4.1.1. Proposed delivery method(s)?

3.14.4.1.2. Proposed media?

3.14.4.1.3. Initial tasks and knowledge topics?

3.14.4.1.4. Initial proficiency levels of learning?

3.14.4.1.5. Planned instructional and evaluation methods?

3.14.4.1.6. Planned instructor and student material?

3.14.4.1.7. Planned target population?

3.14.4.1.8. Anticipated length of instruction?

3.14.4.1.9. Anticipated location of instruction?

3.14.4.1.10. Facilities, equipment, tools, and materials that will be needed to deliver instruction?

3.14.4.1.11. Skills (instructors, trainers, computer support, etc.) that will be needed to deliver instruction?

3.14.4.1.12. Computer software, automated systems, training regions, licenses, network access, Learning Management Systems (LMSs), courseware databases, content libraries, etc., needed to deliver instruction?

3.14.4.1.13. Security clearances, controlled area access, etc., required by developers, students, and instructors to develop and deliver instruction?

3.14.4.2. Are elements described in enough detail to define courseware activities and a rough order of resources needed to accomplish the project?

3.14.4.3. Does justification for proposed delivery method(s) and media address task frequency, task criticality, task learning difficulty, task performance difficulty, and incidence of poor task performance?

3.14.4.4. Are proposed tasks, knowledge topics, instructional methods, evaluation methods, target population, and length of instruction based on adequate data from appropriate experts, and is the reasoning documented in a Preliminary Instructional Strategy Report?

3.14.4.5. Do planned instructional and evaluation materials support a blended learning approach to instruction?

3.14.4.6. Does a Preliminary Instructional Strategy Report adequately identify resources needed to deliver this instruction once it is developed?

3.14.4.7. Did the analyst involve an adequate mix of SMEs, training managers, subject area program managers, system OPRs, instructors, trainers, courseware developers, and other customer representatives to ensure accurate and complete information on which to base findings and recommendations?

3.14.4.8. Did the owning/instructing organization(s) acknowledge responsibility for preparing instructors/trainers; arranging for necessary facilities, tools, materials, and equipment; and delivering and sustaining this training?

3.14.4.9. Has a preliminary Instructional Strategy been approved by lead SMEs, appropriate subject area program managers, system OPRs, instructors, trainers, courseware developers, training managers, etc.?

3.14.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

3.14.5.1. Nature of perceived issues?

3.14.5.2. Priority of assignment?

3.14.5.3. Appropriate organization to handle tasking request?

3.15. Conduct Preliminary Reusable Learning Object (RLO) Assessment (as available). In this activity, a courseware project manager or courseware developer determines reusability of courseware components. If a particular slide, video clip, photograph, chart, or other piece of information has high potential for being useful in future instruction or management activities, it can be coded with metadata and stored in an electronic repository for rapid search, retrieval, and re-use.

3.15.1. Purpose. This activity estimates the number of RLOs (also referred to as Reusable Knowledge Objects [RKO]) that will likely be of value in later performance support activities (e.g., instruction, quality, process improvement, etc.) so that the labor involved in coding objects with metadata tags can be calculated. A learning object that can be re-used saves time and effort of creating a new item, promotes consistency in information provided to the workforce, and reinforces retention of knowledge.

3.15.2. Activity Outcomes:

3.15.2.1. Preliminary RLO assessment documentation.

3.15.2.2. Possible tasking worksheets on training issues, courseware issues, training gaps, or non- training issues submitted to appropriate organization for action.

3.15.2.3. Decision on where to go next in the courseware process.

3.15.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

3.15.4. Evaluation Criteria.

3.15.4.1. Does preliminary RLO assessment documentation include:

3.15.4.1.1. Why content of this instruction does or does not lend itself to reusability.

3.15.4.1.2. List of objects within courseware that will likely have value as RLOs.

3.15.4.1.3. Speculation on how objects might be re-used.

3.15.4.2. Is a preliminary RLO assessment based on recommendations from SMEs, subject area program managers, system OPRs, instructors, trainers, courseware developers, training managers, information technology experts, and others who would have a good idea about transferability of the content?

3.15.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

3.15.5.1. Nature of perceived issues?

3.15.5.2. Priority of assignment?

3.15.5.3. Appropriate organization to handle tasking request?

3.15.5.4. Is a decision to skip any follow-on steps in the Planning Phase an acceptable risk to the relevance and effectiveness of the outcome, and if so, was it documented in the ISD Evaluation Board review notes?

3.16. Conduct Preliminary Sharable Content Object Assessment (as available). A Sharable Content Object (SCO) is a self-contained package of knowledge objects (e.g., graphics, text, etc.) in an E-Learning Environment or learning aid that can be tracked electronically. In this activity, a courseware project manager or courseware developer determines if level of SCO packaging needed to support transportability of instruction to other courses, and to track student progress, test scores, and completions for self-paced, E-Learning instruction.

3.16.1. Purpose. This activity predicts the level of SCO granularity needed so that labor involved in coding objects with metadata tags and launch assets can be roughly estimated.

3.16.2. Activity Outcomes:

3.16.2.1. Preliminary SCO assessment documentation.

3.16.2.2. Possible tasking worksheets on training issues, courseware issues, training gaps, or non- training issues submitted to appropriate organization for action.

3.16.2.3. Decision on where to go next in the courseware process.

3.16.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

3.16.4. Evaluation Criteria.

3.16.4.1. Does preliminary SCO assessment documentation include:

3.16.4.1.1. Recommended level of SCO granularity for a course?

3.16.4.1.2. Why content of this course or learning aid requires a recommended level of SCOs, either for transportability of instruction to other courses, or for tracking student progress, test scores, and completions?

3.16.4.1.3. An estimated number of SCOs to be programmed?

3.16.4.2. Is a recommended level of SCO granularity at as high a level as possible in light of anticipated transportability and student tracking requirements? **Note:** SCO programming is expensive and should be kept to a minimum.

3.16.4.3. Is a preliminary SCO assessment based on recommendations from SMEs, subject area program managers, system OPRs, instructors, trainers, courseware developers, training managers, information technology experts, and others who would have a good idea about transportability of content and if there is a need to track student progress during the course?

3.16.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

3.16.5.1. Nature of perceived issues?

3.16.5.2. Priority of assignment?

3.16.5.3. Appropriate organization to handle tasking request?

3.16.5.4. Is a decision to skip any follow-on steps in the Planning Phase an acceptable risk to the relevance and effectiveness of the outcome, and if so, was it documented in the ISD Evaluation Board review notes?

3.17. Conduct Deliverables Format Analysis. In this activity, a courseware project manager or courseware developer decides if any deliverables need to be in a particular electronic format (e.g., an Adobe application, a Microsoft Office application, etc.) and/or computer technology for easy maintenance and reusability.

3.17.1. Purpose. This activity ensures deliverables will be in a government-approved format so that the government has organic capability to deliver and maintain courseware without buying additional software applications, licenses, training, etc. This activity also promotes blended learning by identifying deliverables that need to be in a format that is well-suited to posting on the Web for reference, or for transporting intact to other courseware.

3.17.2. Activity Outcomes:

3.17.2.1. Deliverables format analysis documentation.

3.17.2.2. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

3.17.2.3. Decision on where to go next in the courseware process.

3.17.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

3.17.4. Evaluation Criteria.

3.17.4.1. Does deliverables format analysis documentation include:

3.17.4.1.1. List of deliverables that need to have a particular electronic format and/or computer technology?

3.17.4.1.2. Reason that these particular deliverables need to be in a specified format?

3.17.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

3.17.5.1. Nature of perceived issues?

3.17.5.2. Priority of assignment?

3.17.5.3. Appropriate organization to handle tasking request?

3.17.5.4. Is a deliverables format analysis based on recommendations from instructors, trainers, courseware developers, information technology experts and others who would have a good idea about functionality and sustainability of various computer applications?

3.18. Prepare ISD Management Plan (ISDMP)/Project Definition Summary (PDS). In this activity, an experienced courseware project manager or courseware developer defines activities, deliverables and resources needed to accomplish a courseware project.

3.18.1. Purpose. This activity prepares a courseware project for assignment or outsourcing and allows organizations to plan for manpower/funding needed to accomplish required tasks.

3.18.2. Activity Outcomes:

3.18.2.1. ISDMP/PDS document.

3.18.2.2. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

3.18.2.3. Decision on whether currently tasked training function should continue to handle this project for now.

3.18.2.4. Decision on where to go next in the courseware process.

3.18.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

3.18.3.1. ISDMP/PDS guidance, sample template, instructions, quality checklist, and examples.

3.18.3.2. Courseware estimate formulas, guidance, worksheets, quality checklist, and examples.

3.18.4. Evaluation Criteria.

3.18.4.1. Does the ISDMP/PDS have a Project Description section that (as appropriate for the situation) includes:

3.18.4.1.1. Reason for a project (e.g., to revise existing courseware, to fill a training gap, etc.)?

3.18.4.1.2. Area of project applicability (e.g., USAF, AFMC, Robins AFB, etc.)?

3.18.4.1.3. Primary delivery method (e.g., instructor-led; self-paced, Web-based; computer-aided instruction; etc., outlined in AFI 36-2650_AFMCSUP) of courseware?

3.18.4.1.4. Target population (e.g., AFMC equipment specialists, Air Force spray painters, C-130 electricians, etc.) of instruction?

3.18.4.1.5. Highest proficiency level of learning (e.g., B/2b, C/3c, etc. as outlined in AFI 36-2650_AFMCSUP) required for instruction?

3.18.4.1.6. Estimated length of proposed course or learning aid in terms of hours of instruction?

3.18.4.1.7. Initial tasks and knowledge topics that are planned to be covered?

3.18.4.1.8. Course presentations, student handouts, reference guides and other support documents needed?

3.18.4.1.9. Whether specific types of evaluation (e.g., written test, proficiency evaluation, etc.) are required?

3.18.4.2. Does the ISDMP/PDS have a Basic Courseware Activities and Outcomes section that (as appropriate for the situation)?

3.18.4.2.1. Describes each courseware activity (e.g., conduct knowledge/task analysis, develop learning objectives, etc.) required to accomplish this project?

3.18.4.2.2. Specifies the outcome documentation associated with each courseware activity?

3.18.4.2.3. Identifies evaluation standards (e.g., directive, this manual, an ISD Evaluation checklist, etc.) for each activity?

3.18.4.3. Does the ISDMP/PDS have an Estimate of Resources Needed section that (as appropriate for the situation):

3.18.4.3.1. Estimates hours needed to accomplish tasks for project in a worst case scenario (i.e., inexperienced organic courseware developer performing work)?

3.18.4.3.2. Explains basic formula and adjustment factors used to determine a Rough Order of Magnitude (ROM) for resources needed for a proposed courseware project?

3.18.4.4. Does the ISDMP/PDS have attachments that include (as applicable) any Planning Process activities documentation such as Training Requirements Matrix, Training Standardization Assessment, Training Gap Analysis Report, etc.?

3.18.4.5. Is a Project Description section formatted so that it can be easily inserted into the introductory portion of the Description of Services section of a Performance Work Statement (PWS)?

3.18.4.6. Is a Basic Courseware Activities and Outcomes section formatted so that it can be easily inserted into the Basic Services sub-section of a PWS?

3.18.4.7. Does the ISDMP/PDS appear to include all of activities, deliverables, standards and other details needed for a potential service provider to prepare a realistic bid for this project?

3.18.4.8. Did an experienced courseware program/project manager oversee formulas and adjustment factors used to determine a ROM of resources needed to accomplish the project?

3.18.4.9. Did information in the ISDMP/PDS come from results of earlier planning activities and not from assumptions and incomplete data?

3.18.4.10. Is a Pre-Kickoff Meeting agenda drafted that identifies needed attendees, list of attendee roles and responsibilities, and an outline of meeting topics?

3.18.4.11. Have CPMs or Training Managers (TMs) been notified to provide names of SMEs for Kickoff Meeting?

3.18.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

3.18.5.1. Nature of perceived issues?

3.18.5.2. Priority of assignment?

3.18.5.3. Appropriate organization to handle tasking request?

3.19. Conduct ISD Review of Planning Activities. In this activity, at least one ISD Evaluator reviews activities performed during Planning Process to determine if they meet the intent of the AFMC Courseware Development and Management Process and its underlying principles of systems engineering, instructional design, and quality improvement, prior to ISD Gate release into Analysis. **Note:** This is a required step in the Planning Process.

3.19.1. Purpose. This activity ensures that AFMC Courseware Development and Management Process is applied so that instruction has a high probability of being relevant, effective and economical to sustain.

3.19.2. Activity Outcomes:

3.19.2.1. Completion of the applicable section of AFMC ISD Gates Release Points Checklist. The current version of the checklist can be found on the *AFMC ISD Resource* Site. This checklist is used for Command Courses but may also be used for Local Course development. It provides a decision that the appropriate activities in each gate have been accomplished to the standards of this manual, or that corrective actions are needed before the project goes into the next gate.

3.19.2.2. Decision on whether currently tasked training function should continue to handle this project for now.

3.19.2.3. Decision that appropriate activities in the Planning Process have been accomplished to the standards of this manual, or corrective actions are needed before a project is funded/assigned.

3.19.2.4. Update ISDMP/PDS (as needed).

3.19.2.5. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

3.19.2.6. Decision on where to go next in the courseware process.

3.19.3. Additional Guidance. See the *AFMC ISD Resource* Site.

3.19.4. Evaluation Criteria.

3.19.4.1. Do the ISD Evaluator notes for Planning Process include:

3.19.4.1.1. Basic information about project, date of ISD review, and participants?

3.19.4.1.2. Reasons Planning Process activities were skipped?

3.19.4.1.3. Observations about compliance/non-compliance of Planning Process activities with standards set forth in this manual and *AFMC ISD Resource* Site?

3.19.4.1.4. Explanations of waivers and adjustments of standards for Planning Process activities, (as applicable)?

3.19.4.1.5. Corrective actions which need to be accomplished with suspense dates, (as applicable)?

3.19.4.1.6. Planning Process review decision?

3.19.4.2. Is a decision to send this courseware project to another organization for action without accomplishing additional activities supported by regulation, policy, or sound logic?

3.19.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

3.19.5.1. Nature of perceived issues?

3.19.5.2. Priority of assignment?

3.19.5.3. Appropriate organization to handle tasking request?

3.20. Request/Approve Authorization to Proceed with Project. In this activity, a project manager/courseware manager/CPM/program manager obtains approval from appropriate management channels to proceed with a courseware development or major revision project (i.e., change in learning objectives or proficiency levels).

3.20.1. Purpose. This activity verifies that a proposed project does not duplicate existing courseware or another planned/in work project. It also ensures that training is standardized to the greatest extent practical.

3.20.2. Activity Outcomes:

3.20.2.1. Request for authorization of courseware development or revision Project.

3.20.2.2. Decision to approve or disapprove request to proceed with a courseware development or revision Project.

3.20.2.3. Decision on where to go next in the courseware process.

3.20.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

3.20.4. Evaluation Criteria.

3.20.4.1. Did request for authorization of a courseware development or revision project include an ISDMP/PDS with applicable Planning Process documentation and supporting documentation attached?

3.20.4.2. Is there adequate information to determine if proposed project duplicates any existing courseware or planned/in work project?

3.21. Request/Assign Course Number(s). In this activity, Courseware Program Managers request the assignment of a 15 digit course number for new developments IAW AFI 36-2650_AFMCSUP.

3.21.1. Purpose. This activity allows HQ AFMC/A4PT to assign course numbers to courses and learning aids so they can be managed easily, and completions can automatically be documented in employee training records.

3.21.2. Activity Outcomes:

3.21.2.1. AFMC Form 853, *AFMC Course Chart*, and AFMC Form 96, *Existing Courseware Evaluation Checklist* (when applicable), are drafted to request course number(s) from the Complex Courseware Program Manager (CPM). The CPM will forward the request to HQ AFMC/A4PT.

3.21.2.2. Assignment of course number(s) by HQ AFMC/A4PT Courseware Manager.

3.21.2.3. Course number(s) and basic information entered in appropriate command-approved functional training management system, such as TSS, as directed in AFI 36-2650_AFMCSUP.

3.21.2.4. Decision on where to go next in the courseware process.

3.21.3. Additional Guidance. See the *AFMC ISD Resource* Site.

3.21.4. Evaluation Criteria.

3.21.4.1. Has course number been requested for an approved project?

3.21.4.2. Does the AFMC course number comply with AFI 36-2650_AFMCSUP as appropriate?

3.21.4.3. Was a course number and basic information about the course or learning aid entered in appropriate command-approved functional automated information system, such as CMD?

3.22. Create proposed project utilizing local tracking method.

3.22.1. Purpose. This activity is to ensure development project details are captured in the local tracking method for management to review progress of all development efforts.

3.22.2. Activity Outcomes:

3.22.2.1. Project created with details properly loaded into the local tracking method.

3.22.2.2. Responsible course manager is identified.

3.22.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

3.22.4. Evaluation Criteria.

3.22.4.1. Are project details available for review by all necessary individuals?

3.23. Set Project Assignment/Funding Priority. In this activity, a responsible management level determines a priority for the project as compared to other projects that require funding or assignment.

3.23.1. Purpose. This activity evaluates emerging issues against those that are already prioritized so that the appropriate manager can determine which are most important to the customer and need to be worked first.

3.23.2. Activity Outcomes:

3.23.2.1. Project Assignment/Priority Worksheet in CMD Project.

3.23.2.2. Possible tasking documentation on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

3.23.2.3. Decision on where to go next in the courseware process.

3.23.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

3.23.4. Evaluation Criteria.

3.23.4.1. Does assignment/funding priority assigned to this activity or project appear to be justified in light of activities/projects with lower and higher priorities?

3.23.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

3.23.5.1. Nature of perceived issues?

3.23.5.2. Priority of assignment?

3.23.5.3. Appropriate organization to handle tasking request?

3.24. Assign Project to Development Team. In this activity, a responsible office (lead developer office and CPM) decides if a project should be assigned to an organic courseware development team, assigned to a level-of-effort contractor development team already in place, or outsourced to a contractor development team.

3.24.1. Purpose. This activity helps organizations to determine the best approach for accomplishing priority courseware projects in light of their resource constraints.

3.24.2. Activity Outcomes:

3.24.2.1. Assignment of project to an organic or level-of-effort contractor development team, or decision to outsource project to a contractor development team.

3.24.2.2. Decision on where to go next in the courseware process.

3.24.2.3. Creation of Working Folder and Depot Maintenance Training Library (DMTL) Working Folder for Command Courses.

3.24.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site and the *AFMC ISD Resource* Site.

3.24.4. Evaluation Criteria. Did appropriate office have authority to assign selected individual/team this courseware project?

3.24.5. Were appropriate team members notified for kickoff meeting coordination?

3.24.6. Follow-on: Is a decision to skip processes in Planning an acceptable risk to the relevance and effectiveness of the outcome, and if so, was it documented in the ISD Evaluator review notes?

Chapter 4

ANALYSIS PHASE

4.1. Analysis Phase Defined. Once adequate planning has been accomplished, requirements are further analyzed to determine precisely what should be included in instruction, types of learning involved, proficiency level of learning needed for the target audience, and any prerequisite or follow-on instruction required. For additional information and guidance relating to the Analysis Phase of the Air Force ISD process, refer to DAFH 36-2675.

4.2. Hold a Project Kickoff Meeting. In this activity, a courseware project manager hosts a kickoff meeting with primary participants in the project to verify roles and responsibilities and review proposed project plan.

4.2.1. Purpose. This activity clarifies for all key participants what role each of them will have, including planned project activities and timelines. This reduces misunderstandings and allows the government to emphasize the importance of SME, subject area program manager, system OPR, instructor/trainer, Complex CPM, ISD Evaluator, customer training manager, etc., involvement in the Project.

4.2.2. Activity Outcomes:

4.2.2.1. Courseware project Kickoff Meeting.

4.2.2.2. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

4.2.2.3. Decision on where to go next in the courseware process.

4.2.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

4.2.4. Evaluation Criteria.

4.2.4.1. Did meeting include lead SME(s), lead subject area program manager(s), system OPR, primary instructor/trainer, Complex CPM, assigned ISD Evaluator, lead customer training manager, etc., as appropriate to kick off the project?

4.2.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

4.2.5.1. Nature of perceived issues?

4.2.5.2. Priority of assignment?

4.2.5.3. Appropriate organization to handle tasking request?

4.3. Conduct a Cognitive Knowledge/Task Analysis. In this activity, a courseware developer identifies precisely those tasks and subtasks which require instruction, conditions under which they are performed, performance standards that must be achieved in the workplace, and critical decisions and mental processes that separate an expert from a novice.

4.3.1. Purpose. This activity ensures that instruction will be relevant and efficient.

4.3.2. Activity Outcomes:

4.3.2.1. A draft Course Training Standard (CTS) is derived from a Knowledge/Task Analysis Report that includes a set of knowledge and task statements used to develop learning objectives for the course and to sequence instruction. (These topics and tasks may have been originally identified during the Planning Process and need to be further analyzed and validated during this phase.)

4.3.2.2. A set of critical decision point statements that are later incorporated into instructional materials as points emphasized and lessons learned. (These will be further analyzed during Design Phase when instructional materials are developed.)

4.3.2.3. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

4.3.2.4. Decision on where to go next in the courseware process.

4.3.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

4.3.3.1. Knowledge/Task Analysis guidance.

4.3.3.2. Knowledge/Task Analysis Report quality checklist and examples.

4.3.4. Evaluation Criteria.

4.3.4.1. Does report include tasks performed in a work center pertaining to the performance deficiency, even if some will not be included in the instruction?

4.3.4.2. Does report identify critical decision points in a work process/tasks/subtasks and describe cognitive strategies used by experts to decide what to do next?

4.3.4.3. Does report identify tasks and critical decision points that will be taught?

4.3.4.4. For any tasks that will be included in the instruction, does a report identify workplace conditions and standards?

4.3.4.5. Has a report been approved by SMEs, subject area program managers, system OPRs, instructors, trainers, etc., for topics that should be included in the instruction?

4.3.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

4.3.5.1. Nature of perceived issues?

4.3.5.2. Priority of assignment?

4.3.5.3. Appropriate organization to handle tasking request?

4.3.6. Follow-on: Is a decision to skip any follow-on steps in the Analysis Phase an acceptable risk to relevance and effectiveness of outcomes, and if so, was it documented in the ISD Evaluator review notes?

4.4. Conduct a Target Audience Analysis. In this activity, a courseware developer analyzes characteristics of the target audience.

4.4.1. Purpose. This activity helps a courseware developer decide what types of instructional methods, media, examples, exercises, and evaluation techniques will be most effective in gaining and holding students' attention.

4.4.2. Activity Outcomes:

4.4.2.1. Target Audience Analysis Report.

4.4.2.2. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

4.4.2.3. Decision on where to go next in the courseware process.

4.4.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

4.4.3.1. Target audience analysis guidance and examples.

4.4.3.2. Target Audience Report quality checklist, and samples.

4.4.4. Evaluation Criteria.

4.4.4.1. Does Target Audience Analysis Report include:

4.4.4.1.1. Age range of majority of students?

4.4.4.1.2. Education level of majority of students?

4.4.4.1.3. Work center locations of majority of students?

4.4.4.1.4. Experience in topics to be covered by this training?

4.4.4.1.5. Student attitudes toward topics to be covered by this training?

4.4.4.1.6. Estimated size of target population?

4.4.4.2. Has a Target Audience Analysis Report been accepted by lead SMEs, subject area program managers, system OPRs, instructors, trainers, etc., for types of learning needed?

4.4.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

4.4.5.1. Nature of perceived issues?

4.4.5.2. Priority of assignment?

4.4.5.3. Appropriate organization to handle tasking request?

4.5. Conduct a Learning Analysis. In this activity, a courseware developer analyzes tasks to be taught and demographics of a target audience in order to describe different types of learning involved, proficiency levels of learning needed, and any student prerequisites that will be required.

4.5.1. Purpose. This activity ensures instruction will be effective in resolving a performance deficiency.

4.5.2. Activity Outcomes:

4.5.2.1. Learning Analysis Report.

4.5.2.2. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

4.5.2.3. Decision on where to go next in the courseware process.

4.5.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

4.5.4. Evaluation Criteria.

4.5.4.1. Does Learning Analysis Report include:

4.5.4.1.1. A description of the target population demographic?

4.5.4.1.2. A description of learning needed by target audience?

4.5.4.1.3. Proficiency levels of learning required based on Proficiency Code Key (reference AFI 36-2650_AFMCSUP).

4.5.4.1.4. Identification of any course prerequisites and/or follow-ons?

4.5.4.2. Has a Learning Analysis Report been approved by instructors, trainers, etc.?

4.5.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

4.5.5.1. Nature of perceived issues?

4.5.5.2. Priority of assignment?

4.5.5.3. Appropriate organization to handle tasking request?

4.6. Modify Instructional Strategy (as needed). In this activity, a courseware developer validates data from previous Planning Process activities, integrates information from Analysis Phase activities, and modifies the plan for developing and delivering new or significantly revised courseware.

4.6.1. Purpose. This activity updates a Preliminary Instructional Strategy created during Planning Process to include more detail about types of learning and events of instruction planned for a course. It also identifies if changes to a courseware project will require any modification to a contract. Finally, this activity provides the lead time necessary to purchase tools and equipment, arrange for instructors and facilities, and obtain other resources needed to deliver the instruction.

4.6.2. Activity Outcomes:

4.6.2.1. Instructional Strategy Report approved by customer.

4.6.2.2. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

4.6.2.3. Decision on where to go next in the courseware process.

4.6.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

4.6.4. Evaluation Criteria.

4.6.4.1. Does the Instructional Strategy Report identify and justify the following aspects of the instructional plan:

4.6.4.1.1. Proposed delivery method(s)?

4.6.4.1.2. Proposed media?

4.6.4.1.3. Tasks and knowledge topics to be trained?

4.6.4.1.4. Proficiency levels of learning to be attained?

4.6.4.1.5. Planned instructional and evaluation methods?

4.6.4.1.6. Planned instructor and student materials?

4.6.4.1.7. Planned target population?

4.6.4.1.8. Anticipated length of instruction?

4.6.4.1.9. Anticipated location of the instruction?

4.6.4.1.10. Facilities, equipment, tools and materials that will be needed to deliver the instruction?

4.6.4.1.11. Skills that will be needed to deliver the instruction? Determine what it will cost to get the instructor/trainer/facilitator trained and how long this will take.

4.6.4.1.12. Computer software, automated systems, training regions, licenses, network access, Learning Management System (LMS), etc., that will be needed to deliver instruction?

4.6.4.1.13. Security clearances, controlled area access, etc., required by students and instructors to deliver instruction?

4.6.4.2. Does the Instructional Strategy Report explain what has changed from the Preliminary Instructional Strategy, and why?

4.6.4.3. Are any changes to the Instructional Strategy explained in enough detail to determine what project activities need to be adjusted, and what contract modifications will be required (as applicable)?

4.6.4.4. Does the justification for delivery method(s) and media address task frequency, task criticality, task learning difficulty, task performance difficulty, and incidence of poor task performance?

4.6.4.5. Are the selected tasks, knowledge topics, instructional methods, evaluation methods, target population and length of instruction based on adequate data from appropriate experts, and is the reasoning documented in the Preliminary Instructional Strategy Report?

4.6.4.6. Does the Instructional Strategy support a blended learning approach to instruction?

4.6.4.7. Did the owning/instructing organization(s) acknowledge responsibility for preparing instructors/trainers/facilitators; arranging for necessary facilities, tools, materials and equipment; and delivering and sustaining this training?

4.6.4.8. Has the Instructional Strategy been approved by owning/managing organization lead SMEs, subject area program managers, system OPRs, instructors, trainers, courseware developers, training managers, etc.?

4.6.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

4.6.5.1. Nature of perceived issues?

4.6.5.2. Priority of assignment?

4.6.5.3. Appropriate organization to handle tasking request?

4.7. Conduct ISD Evaluation of Analysis Phase Activities. In this activity, an ISD Evaluator reviews activities performed during the Analysis Phase to determine if they meet the intent of the

AFMC Courseware Development and Management Process and its underlying principles of systems engineering, instructional design, and quality improvement. **Note:** This is a required step in the Analysis Phase.

4.7.1. Purpose. This activity ensures AFMC Courseware Development and Management Process is applied to ensure instruction has a high probability of being relevant, effective and economical to sustain.

4.7.2. Activity Outcomes:

4.7.2.1. Completion of the applicable section of AFMC ISD Gates Release Points Checklist. The current version of the checklist can be found on the *AFMC ISD Resource* Site. This checklist is used for Command Courses but may also be used for Local Course development. It provides a decision that the appropriate activities in the each gate have been accomplished to the standards of this manual, or that corrective actions are needed before the project goes into the next gate.

4.7.2.2. Decision that appropriate activities in the Analysis Phase have been accomplished to the standards of this manual, or that corrective actions are needed before the project goes into Design Phase.

4.7.2.3. Update ISDMP/PDS (as needed).

4.7.2.4. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

4.7.2.5. Decision on where to go next in the courseware process.

4.7.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site and the *AFMC ISD Resource* Site.

4.7.3.1. ISD Evaluation review guidance, sample instructions, sample worksheet templates, quality checklists and examples.

4.7.4. Evaluation Criteria

4.7.4.1. ISD Evaluation Notes for Analysis Phase include:

4.7.4.1.1. Basic information about project, date of ISD review, and participants?

4.7.4.1.2. Reasons typical Analysis Phase activities were skipped, (as applicable)?

4.7.4.1.3. Explanations of waivers and adjustments of standards for Analysis Phase activities, (as applicable)?

4.7.4.1.4. Documentation and customer approval of changes to Preliminary Instructional Strategy?

4.7.4.1.5. Observations about compliance/non-compliance of Analysis Phase activities with standards set forth in this manual and *AFMC ISD Resource* Site?

4.7.4.1.6. Corrective actions to be accomplished with suspense dates, (as applicable)?

4.7.4.1.7. Analysis Phase review decision?

4.7.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

4.7.5.1. Nature of perceived issue?

4.7.5.2. Priority of assignment?

4.7.5.3. Appropriate organization to handle tasking request?

Chapter 5

DESIGN PHASE

5.1. Design Phase Defined. Instructional design is similar to architectural design. A courseware developer uses the results of Planning and Analysis Phase activities to create a blueprint for the course or learning aid that shows what instruction, evaluation, and media will be used in each module to meet defined objectives. For additional information and guidance relating to the Design Phase of the Air Force ISD process, refer to DAFH 36-2675.

5.2. Develop/Write Learning Objectives. In this activity, a courseware developer writes learning objectives for the course or learning aid.

5.2.1. Purpose. This activity allows the customer to review planned topics, tasks, conditions of performance, and standards of performance that will be required of the student for course completion before any resources are invested in developing instructional and evaluation materials.

5.2.2. Activity Outcomes:

5.2.2.1. Learning objectives for a course or learning aid, usually in the form of a draft Plan Of Instruction (POI).

5.2.2.2. Modified Instructional Strategy with customer approval (as needed).

5.2.2.3. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

5.2.2.4. Decision on where to go next in the courseware process.

5.2.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

5.2.4. Evaluation Criteria.

5.2.4.1. Does each learning objective contain:

5.2.4.1.1. One behavior demonstrating knowledge of the material or ability to complete a task?

5.2.4.1.2. A condition of performance that is closely related to actual conditions in a work environment?

5.2.4.1.3. A standard of behavior that is criterion-referenced and closely related to actual standards required in the work environment?

5.2.4.2. Does knowledge and tasks included in learning objectives coincide with task statements approved by the customer during knowledge/task analysis in the Analysis Phase?

5.2.4.3. Are learning objectives appropriate for proficiency levels defined during Planning and Analysis Phases?

5.2.4.4. Do objectives use types of learning appropriate for the situation?

5.2.4.5. Did changes to initially planned topics, tasks and proficiency levels occur, and if so, have these changes been approved by the customer and SME?

5.2.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

5.2.5.1. Nature of perceived issues?

5.2.5.2. Priority of assignment?

5.2.5.3. Appropriate organization to handle tasking request?

5.3. Plan Integration of Existing Materials. In this activity, a courseware developer decides how to incorporate any available content (e.g., directives, technical data, working aids, courseware, vendor instruction, etc.) that meets part of defined instructional requirement into the course or learning aid being developed.

5.3.1. Purpose. This activity saves resources by adopting as much existing material as is practical.

5.3.2. Activity Outcomes:

5.3.2.1. Existing Material Integration Plan.

5.3.2.2. Modified Instructional Strategy with customer approval, (as needed).

5.3.2.3. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

5.3.2.4. Decision on where to go next in the courseware process.

5.3.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

5.3.4. Evaluation Criteria.

5.3.4.1. Does an existing material integration plan documentation include:

5.3.4.1.1. How existing directives, technical data, working aids, courseware, vendor instruction, etc. will be incorporated into design of instruction?

5.3.4.1.2. Why excluded items are not suitable for inclusion in the instruction?

5.3.4.2. Did this activity impact the Instructional Strategy and, if so, have these changes been documented, justified and approved by the customer(s) and SME(s)?

5.3.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

5.3.5.1. Nature of perceived issues?

5.3.5.2. Priority of assignment?

5.3.5.3. Appropriate organization to handle tasking request?

5.4. Prepare Draft Course Control Documents.

5.4.1. Purpose. To begin developing course control documents to facilitate course development. These documents will be modified as a project progresses.

5.4.2. Activity Outcomes:

5.4.2.1. Draft of the Course Control Documents.

5.4.2.1.1. Learning objectives.

5.4.2.1.2. Course training standard.

5.4.2.1.3. Create/update AFMC Form 853.

5.4.2.1.4. Plan of instruction (as needed).

5.4.2.1.5. Outline (as needed).

5.4.2.1.6. Storyboard (as needed).

5.4.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site and the *AFMC ISD Resource* Site.

5.4.4. Evaluation Criteria.

5.4.4.1. Is an initial draft of the course control documents complete?

5.5. Design Test/Measurement Materials. In this activity, a courseware developer prepares the first draft of materials used to evaluate student progress and attainment of learning objectives.

5.5.1. Purpose. This activity allows the customer and instructor/trainer to review planned test/measurement materials before resources are invested in developing instructional materials.

5.5.2. Activity Outcomes:

5.5.2.1. Draft evaluation materials; such as pre-test bypass tests, review questions, class exercises/projects, quizzes, post-tests, proficiency evaluations, product/process checklists, etc.

5.5.2.2. Draft guidance for instructor/trainer on how to administer evaluation tools.

5.5.2.3. Modified Instructional Strategy with customer approval, (as needed).

5.5.2.4. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

5.5.2.5. Decision on where to go next in the courseware process.

5.5.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

5.5.4. Evaluation Criteria.

5.5.4.1. Do evaluations provide the customer with a high level of confidence that students have attained the proficiency level of learning specified in learning objectives?

5.5.4.2. Are media and delivery method(s) of student evaluations appropriate for instruction?

5.5.4.3. Is content in student evaluations directly related to topics, tasks, conditions and standards specified in learning objectives?

5.5.4.4. Do evaluation activities simulate conditions and standards of the work environment as closely as possible?

5.5.4.5. Is content of evaluation materials accurate and complete?

5.5.4.6. Have instructional concepts been applied when selecting student evaluation activities and creating test/measurement materials?

5.5.4.7. Is there enough evaluation guidance for a technically qualified substitute instructor/trainer to take over a course and administer exercises, projects and tests consistent with methods used by a lead instructor/trainer?

5.5.4.8. Are evaluations, where applicable, correctable to 100% so that students will be given immediate feedback and correct answers before completion of a course?

5.5.4.9. For pass/fail tests, are there at least two questions/scenarios for each learning objective tested or a question pool in Distributive Learning (DL) courseware to allow for remedial instruction and prevent test compromise situations?

5.5.4.10. Is content and minimum passing score for pass/fail tests consistent with training mandates (as applicable)?

5.5.4.10.1. General guidelines require a minimum passing score of 70% unless otherwise indicated through other published guidance.

5.5.4.11. Did this activity impact the Instructional Strategy and, if so, have these changes been documented, justified and approved by the customer(s) and SME(s)?

5.5.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

5.5.5.1. Nature of perceived issues?

5.5.5.2. Priority of assignment?

5.5.5.3. Appropriate organization to handle tasking request?

5.6. Design Training. In this activity, a courseware developer finalizes selection of media, instructional methods/tools, and student evaluation methods/tools used and creates a detailed outline for the course or learning aid.

5.6.1. Purpose. This activity prepares a blueprint of instruction that shows learning objectives, and types and sequence of instructional and student evaluation activities that will take place, so that the design of the course can be reviewed before extensive resources are invested in producing presentation, generic lesson plan, student handouts, etc.

5.6.2. Activity Outcomes:

5.6.2.1. A draft POI or detailed design outline that includes:

5.6.2.1.1. Instructional strategies and evaluation activities for the course or learning aid.

5.6.2.1.2. Identification of media used for each instructional and student evaluation activity.

5.6.2.1.3. A clear description of facilities, equipment, tools and materials used to deliver training.

5.6.2.1.4. An overview of content included in the instruction.

5.6.2.2. Modified Instructional Strategy with customer approval (as needed).

5.6.2.3. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

5.6.2.4. Decision on where to go next in the courseware process.

5.6.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

5.6.4. Evaluation Criteria.

5.6.4.1. Does the design outline for each module contain:

5.6.4.1.1. One or more criterion-referenced learning objectives?

5.6.4.1.2. Presentations, demonstrations, simulations and exhibits used to support each learning objective?

5.6.4.1.3. Student projects, exercises, review sessions, quizzes, tests and proficiency evaluations used to support each learning objective?

5.6.4.1.4. A clear description of facilities, equipment, tools and materials used to deliver training?

5.6.4.1.5. An overview of content supporting each learning objective?

5.6.4.1.6. Media used for each instructional and student evaluation activity?

5.6.4.2. Does the design outline:

5.6.4.2.1. Show sequence in which instruction will be delivered to students?

5.6.4.2.2. Confirm a selected delivery method is most appropriate for instruction?

5.6.4.2.3. Propose instructional and student evaluation events appropriate to support learning objectives?

5.6.4.2.4. Explain how relevant examples and non-examples in the form of illustrations, diagrams, demonstrations, scenarios, case studies, etc., will be used to apply instructional concepts?

5.6.4.2.5. Support tasks, conditions and standards of the work environment as closely as possible?

5.6.4.2.6. Describe any pre-assessment or bypass tests used to determine which modules, if any, a student is required to take?

5.6.4.3. Did this activity impact the Instructional Strategy and, if so, have these changes been documented, justified and approved by the customer(s) and SME(s)?

5.6.4.4. Was an appropriate mix of SMEs, subject area program managers, system OPRs, instructors, trainers, courseware developers, training managers, etc., involved in the design of this course or learning aid?

5.6.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

5.6.5.1. Nature of perceived issues?

5.6.5.2. Priority of assignment?

5.6.5.3. Appropriate organization to handle tasking request?

5.7. Review Design of Instruction. In this activity SMEs, subject area program managers, system OPRs, instructors, courseware developers, information technology experts, computer technology experts, etc., review overall design of the training and evaluation materials before instructional materials are prepared for review in Development Phase.

5.7.1. Purpose. This activity ensures the effectiveness of the instructional design before developing content or creating instructional aids.

5.7.2. Activity Outcomes:

5.7.2.1. Documented input from reviewers on design of the course or learning aid (Course Changes Log).

5.7.2.2. Documentation on changes made to topics, tasks, delivery methods, evaluation materials and flow of instruction based on input from experts.

5.7.2.3. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

5.7.2.4. Decision on where to go next in the courseware process.

5.7.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

5.7.4. Evaluation Criteria.

5.7.4.1. Did appropriate SMEs, training managers, subject area program managers, system OPRs, instructors, Computer-Based Instruction (CBI) experts, computer technology experts, etc., review draft design of the course?

5.7.4.2. Do proposed learning objectives meet the following evaluation criteria?

5.7.4.2.1. Does each learning objective contain:

5.7.4.2.1.1. One behavior (knowledge or task)?

5.7.4.2.1.2. A condition of performance closely related to actual conditions in the work environment?

5.7.4.2.2. A standard of behavior that is criterion-referenced and closely related to actual standards required in the work environment?

5.7.4.2.3. Do knowledge and tasks included in learning objectives coincide directly with the task statements approved by the customer during knowledge/task analysis in the Analysis Phase?

5.7.4.2.4. Are proficiency levels for learning objectives appropriate for instructional requirements defined during the Planning and Analysis Phases?

5.7.4.2.5. Do objectives use types of learning appropriate for the tasks?

5.7.4.2.6. Did changes to initially planned topics, tasks and proficiency levels occur, and if so, have these changes been approved by the customer and SME?

5.7.4.3. Do proposed student evaluation materials and activities meet the following evaluation criteria?

5.7.4.3.1. Does the amount and types of evaluation provide the customer with a high level of confidence that students have attained the proficiency level of learning specified in the learning objectives?

5.7.4.3.2. Are media and delivery method(s) of student evaluations appropriate for situation?

5.7.4.3.3. Is content of student evaluations directly related to topics, tasks, conditions and standards specified in learning objectives?

5.7.4.3.4. Do evaluation activities simulate conditions and standards of the work environment as closely as possible?

5.7.4.3.5. Is content of evaluation materials accurate and complete?

5.7.4.3.6. Have instructional concepts been applied when selecting student evaluation activities and creating test/measurement materials?

5.7.4.3.7. Is there enough evaluation guidance for a technically qualified substitute instructor/trainer to take over a course and administer exercises, projects and tests consistent with methods used by a lead instructor/trainer?

5.7.4.3.8. Are all applicable evaluations correctable to 100% so that students will be given immediate feedback and correct answers before completion of the course?

5.7.4.3.9. For pass/fail tests, are there at least two questions/scenarios for each learning objective tested or a question pool to allow for remedial instruction and prevent test compromise situations?

5.7.4.3.10. Is content and minimum passing score for pass/fail tests consistent with training mandates?

5.7.4.3.11. Did this activity impact the Instructional Strategy and, if so, have these changes been documented, justified and approved by the customer(s) and SME(s)?

5.7.4.4. Does the proposed design of the course meet the following evaluation criteria?

5.7.4.4.1. Does the design outline for each module contain:

5.7.4.4.1.1. One or more criterion-referenced learning objectives?

5.7.4.4.1.2. Presentations, demonstrations, simulations and exhibits used to support each learning objective?

5.7.4.4.1.3. Student projects, exercises, review sessions, quizzes, tests and proficiency evaluations used to support each learning objective?

5.7.4.4.1.4. A clear description of facilities, equipment, tools and materials used to deliver training?

5.7.4.4.1.5. An overview of content supporting each learning objective?

5.7.4.4.1.6. Media used for each instructional and student evaluation activity?

5.7.4.4.2. Does the design outline:

5.7.4.4.2.1. Show the sequence in which instruction will be delivered to students?

5.7.4.4.2.2. Confirm that selected delivery method is appropriate for the situation?

5.7.4.4.2.3. Propose instructional and student evaluation events appropriate for situation and support learning objectives?

5.7.4.4.2.4. Explain how relevant examples and non-examples in the form of illustrations, diagrams, demonstrations, scenarios, case studies, etc., used to apply instructional concepts?

5.7.4.4.2.5. Support tasks, conditions and standards of the work environment as closely as possible?

5.7.4.4.2.6. Describe any pre-assessment or bypass tests used to determine which modules, if any, a student will be required to take?

5.7.4.4.3. Did this activity impact the Instructional Strategy and, if so, have these changes been documented, justified and approved by the customer(s) and SME(s)?

5.7.4.4.4. Was an appropriate mix of SMEs, subject area program managers, system OPRs, instructors, trainers, courseware developers, training managers, etc., involved in the design of course or learning aid?

5.7.4.5. Did a courseware developer document input from reviewers and how recommended input will be accomplished?

5.7.4.6. Did a courseware developer document why any recommended changes will not be accomplished?

5.7.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

5.7.5.1. Nature of perceived issues?

5.7.5.2. Priority of assignment?

5.7.5.3. Appropriate organization to handle tasking request?

5.8. Revise Design of Training (as needed). In this activity, a courseware developer makes recommended changes to the design of a course or learning aid.

5.8.1. Purpose. This activity saves resources by accomplishing as much rework as possible before instructional and evaluation materials are created in Development Phase. **Note:** Only minor corrections to appearance and functionality of a course are permitted after prototype module receives final approval in the Design Phase.

5.8.2. Activity Outcomes:

5.8.2.1. Changes made to basic content, delivery methods, and flow of instruction based on input from experts on Course Changes Log.

5.8.2.2. Changes made to course control documents, (as needed).

5.8.2.3. Modified Instructional Strategy with customer approval (as needed).

5.8.2.4. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

5.8.2.5. Decision on where to go next in the courseware process.

5.8.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

5.8.4. Evaluation Criteria.

5.8.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

5.8.5.1. Nature of perceived issues?

5.8.5.2. Priority of assignment?

5.8.5.3. Appropriate organization to handle tasking request?

5.8.6. Follow-on: Has a courseware developer arranged for a follow-on review of revised materials (as applicable)?

5.9. Design Distributed Learning (DL). In this activity, a courseware developer finalizes selection of layout, media, instructional methods/tools, and student evaluation methods/tools used, and creates a detailed outline/storyboard of a computer-based course or learning aid and an initial functioning DL module that is a representation of a completed course.

5.9.1. Purpose. This activity prepares a blueprint of instruction that shows learning objectives, types and sequence of instructional and student evaluation activities that will take place, so that the design of a course can be reviewed before extensive resources are invested in producing detailed content storyboards.

5.9.2. Activity Outcomes:

5.9.2.1. A storyboard/outline or checklist that identifies Distance Learning (DL) instructional design characteristics.

5.9.2.2. Learning development methods employed (job sequence, psychological sequence, logical order sequence, branching scenario, etc.).

5.9.2.3. Identify Section 508 Compliance requirements for the target audience. **https://www.section508.gov/manage**

5.9.2.4. Identify LMS Sharable Content Object Reference Model (SCORM) level requirements for DL development.

5.9.2.5. Finalize SCO assessment.

5.9.2.5.1. Determine SCO level based on LMS SCORM version.

5.9.2.5.2. Determine number of SCOs needed for pre-assessment, review, scenarios, etc.

5.9.2.5.3. Identify SCO calls in storyboard/outline/checklist.

5.9.2.5.4. Determine if a contract requires modification to fund development of additional SCOs.

5.9.2.6. Determine coding methods or AF approved software for SCORM publishing to an LMS used in course development.

5.9.2.7. Determine instructional and evaluation method activities for the course or learning aid.

5.9.2.8. Identification of the media used for each instructional and student evaluation activity.

5.9.2.9. A clear description of screen design and student interactivity features.

5.9.2.10. An overview of content included in the instruction.

5.9.2.11. Modified Instructional Strategy with customer approval (as needed).

5.9.2.12. Decision on where to go next in the courseware process.

5.9.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

5.9.4. Evaluation Criteria.

5.9.4.1. Does the design outline for each module contain:

5.9.4.1.1. One or more criterion-referenced learning objectives?

5.9.4.1.2. Presentations, demonstrations, simulations and exhibits used to support each learning objective?

5.9.4.1.3. Student projects, exercises, review sessions, quizzes, tests and proficiency evaluations used to support each learning objective?

5.9.4.1.4. A clear description of screen design and student interactivity features?

5.9.4.1.5. An overview of content supporting each learning objective?

5.9.4.1.6. Media used for each instructional and student evaluation activity?

5.9.4.2. Does the design outline:

5.9.4.2.1. Show the sequence in which instruction will be delivered to students?

5.9.4.2.2. Confirm selected delivery method is appropriate for situation?

5.9.4.2.3. Does SCO assessment include a clear description of location in course or learning aid?

5.9.4.2.4. Is the final SCO assessment based on decisions made in preliminary SCO assessment and follow-on input from those who would have a good idea about the transportability of content and need to track progress of students during course?

5.9.4.2.5. Is final level of SCO granularity at as high a level as possible in light of anticipated transportability and student tracking requirements?

5.9.4.2.6. Has any decision to significantly increase the amount of SCOs been justified adequately to support a modification to a contract?

5.9.4.2.7. Propose instructional and student evaluation events that are appropriate for the situation and support learning objectives?

5.9.4.2.8. Explain how relevant examples and non-examples in the form of illustrations, diagrams, demonstrations, scenarios, case studies, etc., will be used to apply instructional concepts?

5.9.4.2.9. Support tasks, conditions, and standards of the work environment as much as is feasible?

5.9.4.2.10. Describe any pre-assessment or bypass tests used to determine which modules, if any, a student will be required to take if SCORM version of LMS allows?

5.9.4.3. Did this activity impact the Instructional Strategy and, if so, have these changes been documented, justified and approved by the customer(s) and SME(s)?

5.9.4.4. Was an appropriate mix of SMEs, subject area program managers, system OPRs, instructors, trainers, courseware developers, training managers, etc., involved in the design of this course or learning aid?

5.9.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

5.9.5.1. Nature of perceived issues?

5.9.5.2. Priority of assignment?

5.9.5.3. Appropriate organization to handle tasking request?

5.10. Review Design of Distributed Learning Training. In this activity, SMEs, subject area program managers, system OPRs, instructors, courseware developers, information technology experts, etc., review the overall design of distributed learning training and draft evaluation materials before content is written for an entire course. Only minor changes to design of a course or learning aid are permitted during this activity.

5.10.1. Purpose. This activity verifies effectiveness of the design before distributed learning training content is written and put in storyboard format.

5.10.2. Activity Outcomes:

5.10.2.1. Input from reviewers on content and design of a course or learning aid on Course Changes Log.

5.10.2.2. Documentation on changes made to topics, tasks, delivery methods, and flow of instruction based on input from experts.

5.10.2.3. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

5.10.2.4. Decision on where to go next in the courseware process.

5.10.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

5.10.4. Evaluation Criteria.

5.10.4.1. Did appropriate SMEs, training managers, subject area program managers, system OPRs, instructors, etc., review the draft design of the course?

5.10.4.2. Do proposed learning objectives meet the following evaluation criteria?

5.10.4.2.1. Does each learning objective contain:

5.10.4.2.1.1. One behavior (knowledge or task)?

5.10.4.2.1.2. A condition of performance closely related to actual conditions in the work environment?

5.10.4.2.1.3. A standard of behavior that is criterion-referenced and closely related

to actual standards required in work environment?

5.10.4.2.2. Do knowledge and tasks included in learning objectives coincide directly with task statements approved by the customer during knowledge/task analysis in the Analysis Phase?

5.10.4.2.3. Are proficiency levels for learning objectives appropriate for instructional requirements defined during the Planning and Analysis Phases?

5.10.4.2.4. Do objectives use types of learning appropriate for the situation?

5.10.4.2.5. Did changes to initially planned topics, tasks and proficiency levels occur, and if so, have these changes been approved by the customer and SME?

5.10.4.3. Do proposed student evaluation materials and activities meet the following evaluation criteria?

5.10.4.3.1. Do the amount and types of evaluation provide the customer with a high level of confidence that students have attained the proficiency level of learning specified in learning objectives?

5.10.4.3.2. Does media and delivery method(s) of student evaluations appear to be appropriate for the situation?

5.10.4.3.3. Is content of student evaluations directly related to topics, tasks, conditions and standards specified in the learning objectives?

5.10.4.3.4. Do evaluation activities emulate conditions and standards of the work environment as closely as possible?

5.10.4.3.5. Is content of evaluation materials accurate and complete?

5.10.4.3.6. Have instructional concepts been applied when selecting student evaluation activities and creating test/measurement materials?

5.10.4.3.7. Is there enough evaluation guidance for a technically qualified substitute instructor/trainer to take over a course and administer exercises, projects and tests consistent with methods used by a lead instructor/trainer?

5.10.4.3.8. Are applicable evaluations correctable to 100% so that students will be given immediate feedback and correct answers before completion of a course?

5.10.4.3.9. For pass/fail tests, are there at least two questions/scenarios for each learning objective tested or a test pool to allow for remedial instruction and prevent test compromise situations?

5.10.4.3.10. Is content and minimum passing score for pass/fail tests consistent with training mandates?

5.10.4.3.11. Did this activity impact the Instructional Strategy and, if so, have these changes been documented, justified and approved by the customer(s) and SME(s)?

5.10.4.4. Does the proposed design of a course meet the following evaluation criteria?

5.10.4.4.1. Does design outline for each module contain:

5.10.4.4.1.1. One or more criterion-referenced learning objectives?

5.10.4.4.1.2. Presentations, demonstrations, simulations and exhibits used to support each learning objective?

5.10.4.4.1.3. Student projects, exercises, review sessions, quizzes, tests and proficiency evaluations used to support each learning objective?

5.10.4.4.1.4. A clear description of screen design and student interactivity features?

5.10.4.4.1.5. An overview of content that will support each learning objective?

5.10.4.4.1.6. Media used for each instructional and student evaluation activity?

5.10.4.4.2. Does the design outline:

5.10.4.4.2.1. Show the sequence in which instruction is delivered to students?

5.10.4.4.2.2. Confirm selected delivery method is appropriate for the situation?

5.10.4.4.2.3. Proposed instructional and student evaluation events are appropriate for the situation and support learning objectives?

5.10.4.4.2.4. Explain how relevant examples and non-examples in illustrations, diagrams, demonstrations, scenarios, case studies, etc., will be used to apply instructional concepts?

5.10.4.4.2.5. Support tasks, conditions and standards of the work environment as closely as possible?

5.10.4.4.2.6. Describe any pre-assessment or bypass tests used to determine which modules, if any, a student is required to take?

5.10.4.4.3. Did this activity impact the Instructional Strategy and, if so, have these changes been documented, justified and approved by the customer(s) and SME(s)?

5.10.4.4.4. Was an appropriate mix of SMEs, subject area program managers, system OPRs, instructors, trainers, courseware developers, training managers, etc., involved in the design of this course or learning aid?

5.10.4.5. Did a courseware developer document input from reviewers, and how recommended input will be accomplished?

5.10.4.6. Did a courseware developer document why any recommended changes will not be accomplished?

5.10.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

5.10.5.1. Nature of perceived issues?

5.10.5.2. Priority of assignment?

5.10.5.3. Appropriate organization to handle tasking request?

5.11. Revise Design and Content of Distributed Learning. In this activity, a courseware developer makes recommended changes to proposed content, sequence, instructional activities and student evaluation activities of a distributed learning course or learning aid.

5.11.1. Purpose. This activity accomplishes as much rework as possible before a computer-based prototype module is produced. The earlier in the process that problems are discovered and corrections are made, the less expensive and time-consuming they are.

5.11.2. Activity Outcomes:

5.11.2.1. Changes made to content, delivery methods, and flow of instruction based on input from experts on Course Changes Log.

5.11.2.2. Changes made to AFMC Form 853.

5.11.2.3. Modified Instructional Strategy with customer approval (as needed).

5.11.2.4. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

5.11.2.5. Decision on where to go next in the courseware process.

5.11.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

5.11.4. Evaluation Criteria. N/A.

5.11.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

5.11.5.1. Nature of perceived issues?

5.11.5.2. Priority of assignment?

5.11.5.3. Appropriate organization to handle tasking request?

5.11.6. Follow-on: Has a courseware developer arranged for a follow-on review of revised materials?

5.12. Develop Initial Segment of Distributed Learning. In this activity, a courseware development team creates a representative segment of actual training that shows how screens will look, how content will appear, how navigational buttons will operate, how a narrator will sound, how interactivity will work, etc.

5.12.1. Purpose. This activity reduces the risk of complete rework by producing at least one representative module of DL for review by typical students and experts before additional modules are produced. Computer-material production is expensive, so identifying problems and testing changes or corrections using an initial segment of the actual course or learning aid saves time and money.

5.12.2. Activity Outcomes:

5.12.2.1. Segment of one or more modules for review by experts and typical students.

5.12.2.2. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

5.12.2.3. Modified Instructional Strategy with customer approval (as needed).

5.12.2.4. Decision on where to go next in the courseware process.

5.12.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

5.12.4. Evaluation Criteria.

5.12.4.1. Does the initial DL segment provided for review contain one or more criterion-referenced learning objectives?

5.12.4.2. Does the initial segment contain:

5.12.4.2.1. Content as it will appear in the actual course or learning aid?

5.12.4.2.2. Student projects, pre-assessment tests, exercises, review questions, quizzes, tests and proficiency evaluations that appear and operate as in an actual course or learning aid?

5.12.4.2.3. Navigation buttons that appear and operate as in an actual course or learning aid?

5.12.4.2.4. Screens as they will appear in an actual course or learning aid?

5.12.4.2.5. Narration as it will sound in an actual course or learning aid?

5.12.4.2.6. Student interactivity that appears and operates as it will in an actual course or learning aid?

5.12.4.3. Are screen design and navigational features of a segment engaging and easy to use?

5.12.4.4. Do learning objectives meet the following evaluation criteria?

5.12.4.4.1. Does each learning objective contain:

5.12.4.4.1.1. One behavior (knowledge or task)?

5.12.4.4.1.2. A condition of performance that is closely related to actual conditions in the work environment?

5.12.4.4.1.3. A standard of behavior that is criterion-referenced and closely related to actual standards required in the work environment?

5.12.4.4.2. Do knowledge and tasks included in learning objectives coincide directly with task statements approved by the customer during knowledge/task analysis in the Analysis Phase?

5.12.4.4.3. Are proficiency levels for learning objectives appropriate for instructional requirements defined during Planning and Analysis Phases?

5.12.4.4.4. Do objectives use types of learning appropriate for the situation?

5.12.4.4.5. Did changes to initially planned topics, tasks and proficiency levels occur, and if so, have these changes been approved by the customer and SME?

5.12.4.5. Do student evaluation materials and activities meet the following evaluation criteria?

5.12.4.5.1. Does the amount and types of evaluation provide the customer with a high level of confidence that students have attained the proficiency level of learning specified in learning objectives?

5.12.4.5.2. Does media and delivery method(s) of student evaluations appear to be appropriate for the situation?

5.12.4.5.3. Is content of student evaluations directly related to topics, tasks, conditions and standards specified in learning objectives?

5.12.4.5.4. Do evaluation activities emulate conditions and standards of the work environment as closely as possible?

5.12.4.5.5. Is content of evaluation materials accurate and complete?

5.12.4.5.6. Have widely-accepted instructional concepts been applied when selecting student evaluation activities and creating test/measurement materials?

5.12.4.5.7. Is there enough evaluation guidance for a technically qualified substitute instructor/trainer to take over a course and administer exercises, projects and tests consistent with methods used by a lead instructor/trainer?

5.12.4.5.8. Are applicable evaluations correctable to 100% so that students will be given immediate feedback and correct answers before completion of the course?

5.12.4.5.9. For pass/fail tests, are there at least two questions/scenarios for each learning objective tested or a question pool to allow for remedial instruction and prevent test compromise situations?

5.12.4.5.10. Is content and minimum passing score for pass/fail tests consistent with training mandates?

5.12.4.5.11. Did this activity impact the Instructional Strategy and, if so, have these changes been documented, justified and approved by the customer(s) and SME(s)?

5.12.4.6. Does design of the course meet the following evaluation criteria?

5.12.4.6.1. Does design outline for each module contain:

5.12.4.6.1.1. One or more criterion-referenced learning objectives?

5.12.4.6.1.2. Presentations, demonstrations, simulations and exhibits used to support each learning objective?

5.12.4.6.1.3. Student projects, exercises, review sessions, quizzes, tests and proficiency evaluations used to support each learning objective?

5.12.4.6.1.4. A clear description of screen design and student interactivity features?

5.12.4.6.1.5. An overview of content supporting each learning objective?

5.12.4.6.1.6. Media used for each instructional and student evaluation activity?

5.12.4.6.2. Does the design outline:

5.12.4.6.2.1. Show sequence in which instruction is delivered to students?

5.12.4.6.2.2. Confirm selected delivery method is appropriate for situation?

5.12.4.6.2.3. Propose instructional and student evaluation events appropriate for the situation and support the learning objectives?

5.12.4.6.2.4. Explain how relevant examples and non-examples in the form of illustrations, diagrams, demonstrations, scenarios, case studies, etc., apply

instructional concepts?

5.12.4.6.2.5. Support tasks, conditions and standards of the work environment as closely as possible?

5.12.4.6.2.6. Describe any pre-assessment or bypass tests used to determine which modules, if any, a student is required to take?

5.12.4.6.3. Did this activity impact the Instructional Strategy and, if so, have these changes been documented, justified and approved by the customer(s) and SME(s)?

5.12.4.6.4. Was an appropriate mix of SMEs, subject area program managers, system OPRs, instructors, trainers, courseware developers, training managers, etc., involved in the design of this course or learning aid?

5.12.4.7. Do instructional materials and activities meet the following evaluation criteria?

5.12.4.7.1. Are storyboards and supporting documentation detailed enough for reviewers to get a clear picture of precisely what content will be presented, how it will be instructed, and how students will be evaluated for attainment of each learning objective?

5.12.4.7.2. Do storyboards and supporting documentation describe any pre-assessment or bypass tests used to determine which modules, if any, a student is required to take?

5.12.4.7.3. Do storyboards for each module contain:

5.12.4.7.3.1. One or more criterion-referenced learning objectives?

5.12.4.7.3.2. Presentations, demonstrations, and simulations used to support each learning objective?

5.12.4.7.3.3. Relevant examples and non-examples in the form of illustrations, diagrams, demonstrations, scenarios, case studies, etc., used to support each learning objective?

5.12.4.7.3.4. Student projects, exercises, review sessions, quizzes, tests and proficiency evaluations used to support each learning objective?

5.12.4.7.3.5. A clear description of screen design and student interactivity features?

5.12.4.7.3.6. All content supporting each learning objective?

5.12.4.7.3.7. Media used for each instructional and student evaluation activity?

5.12.4.7.3.8. Sequence in which instruction will be delivered to the students?

5.12.4.7.4. Does content of course or learning aid adequately address Gagné’s Events of Instruction?

5.12.4.7.5. Are there sufficient instructional materials and activities to support learning objectives?

5.12.4.7.6. Have accepted instructional concepts been applied when creating examples, non-examples, illustrations, diagrams, demonstrations, scenarios, case studies, etc.?

5.12.4.7.7. Is the sequence of content and learning activities effective?

5.12.4.7.8. Do media and delivery method(s) of instructional materials appear to be appropriate?

5.12.4.7.9. Is content of instructional materials directly related to topics, tasks, conditions and standards specified in the learning objectives?

5.12.4.7.10. Are examples, non-examples, illustrations, diagrams, demonstrations, scenarios, case studies, and other instructional materials and activities relevant to the target audience?

5.12.4.7.11. Is content of instructional materials accurate and complete?

5.12.4.7.12. Is there enough navigational guidance for a student to easily take the course and any bypass pre-test that may be available?

5.12.4.7.13. Did this activity impact the Instructional Strategy and, if so, have these changes been documented, justified and approved by the customer(s) and SME(s)?

5.12.4.7.14. Was an appropriate mix of SMEs, subject area program managers, system OPRs, instructors, trainers, courseware developers, training managers, etc., involved in the design of this course or learning aid?

5.12.4.8. Did this activity impact the Instructional Strategy and, if so, have these changes been documented, justified and approved by the customer(s) and SME(s)?

5.12.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

5.12.5.1. Nature of perceived issues?

5.12.5.2. Priority of assignment?

5.12.5.3. Appropriate organization to handle tasking request?

5.13. Review Initial Segment of Distributed Learning. In this activity, SMEs, subject area program managers, system OPRs, training managers, DL experts, students from the target population, etc., review a representative segment of actual instruction to see how screens will look, content will appear, navigational buttons will operate, narrator will sound, interactivity will work, etc.

5.13.1. Purpose. This activity allows experts to visualize the appearance of the final product and test functionality of instruction before additional modules of a course or learning aid is produced. Computer-based material production is expensive, so review of this initial segment, in combination with earlier reviews serve as initial validation of a self-paced course or learning aid before remaining modules are developed and reviewed prior to deployment in the field.

5.13.1.1. Only minor corrections to content, appearance and functionality of a distributed learning course or learning aid are permitted after closeout of Activity in Design Phase.

5.13.2. Activity Outcomes:

5.13.2.1. Segment Acceptance Document.

5.13.2.2. Input from reviewers on content, appearance and functionality of the prototype module.

5.13.2.3. Documentation on changes made to content, appearance, navigation, and interactivity of instruction based on input from experts.

5.13.2.4. Decision on where to go next in the courseware process.

5.13.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

5.13.4. Evaluation Criteria.

5.13.4.1. Did appropriate SMEs, subject area program managers, system OPRs, training managers, DL experts, students from target population, etc., review the DL prototype?

5.13.4.2. Did a review team verify the initial segment provided for review contains:

5.13.4.2.1. One or more criterion-referenced learning objectives?

5.13.4.2.2. Content as it will appear in the actual course or learning aid?

5.13.4.2.3. Student projects, pre-assessment tests, exercises, review questions, quizzes, tests and proficiency evaluations that appear and operate as in the actual course or learning aid?

5.13.4.2.4. Navigation buttons that appear and operate as in the actual course or learning aid?

5.13.4.2.5. Screens as they will appear in the actual course or learning aid?

5.13.4.2.6. Narration as it will sound in the actual course or learning aid?

5.13.4.2.7. Student interactivity that appears and operates as it will in the actual course or learning aid?

5.13.4.3. Are screen design and navigational features of the segment engaging and easy to use?

5.13.4.4. Do learning objectives meet the following evaluation criteria?

5.13.4.4.1. Does each learning objective contain:

5.13.4.4.1.1. One behavior (knowledge or task)?

5.13.4.4.1.2. A condition of performance that is closely related to actual conditions in the work environment?

5.13.4.4.1.3. A standard of behavior that is criterion-referenced and closely related to actual standards required in the work environment?

5.13.4.4.2. Do knowledge and tasks included in learning objectives coincide directly with task statements approved by the customer during knowledge/task analysis in the Analysis Phase?

5.13.4.4.3. Are proficiency levels for learning objectives appropriate for the instructional requirements defined during the Planning and Analysis Phases?

5.13.4.4.4. Do objectives use types of learning appropriate for the situation?

5.13.4.4.5. Did changes to initially planned topics, tasks and proficiency levels occur, and if so, have these changes been approved by the customer and SME?

5.13.4.5. Do student evaluation materials and activities meet the following evaluation criteria?

5.13.4.5.1. Does the amount and types of evaluation provide the customer with a high level of confidence that students have attained the proficiency level of learning specified in the learning objectives?

5.13.4.5.2. Do media and delivery method(s) of student evaluations appear to be appropriate for the situation?

5.13.4.5.3. Is content of student evaluations directly related to topics, tasks, conditions and standards specified in the learning objectives?

5.13.4.5.4. Do evaluation activities emulate the conditions and standards of the work environment as closely as possible?

5.13.4.5.5. Is content of evaluation materials accurate and complete?

5.13.4.5.6. Have widely-accepted instructional concepts been applied when selecting student evaluation activities and creating test/measurement materials?

5.13.4.5.7. Are evaluations correctable to 100% so that students will be given immediate feedback and correct answers before completion of the course?

5.13.4.5.8. For pass/fail tests, are there random question pools for each learning objective tested to allow for remedial instruction and prevent test compromise situations?

5.13.4.5.9. Is content and minimum passing score for pass/fail tests consistent with training mandates?

5.13.4.5.10. Did this activity impact the Instructional Strategy and, if so, have these changes been documented, justified and approved by the customer(s) and SME(s)?

5.13.4.6. Does design of the course meet the following evaluation criteria?

5.13.4.6.1. Does design outline for each module contain:

5.13.4.6.1.1. One or more criterion-referenced learning objectives?

5.13.4.6.1.2. Presentations, demonstrations, simulations and exhibits used to support each learning objective?

5.13.4.6.1.3. Student projects, exercises, review sessions, quizzes, tests, and proficiency evaluations used to support each learning objective?

5.13.4.6.1.4. A clear description of screen design and student interactivity features?

5.13.4.6.1.5. An overview of content supporting each learning objective?

5.13.4.6.1.6. Media used for each instructional and student evaluation activity?

5.13.4.6.2. Does design outline:

5.13.4.6.2.1. Show the sequence in which the instruction will be delivered to students?

5.13.4.6.2.2. Confirm selected delivery method is appropriate for the situation?

5.13.4.6.2.3. Propose instructional and student evaluation events that are appropriate for the situation and support learning objectives?

5.13.4.6.2.4. Explain how relevant examples and non-examples in the form of illustrations, diagrams, demonstrations, scenarios, case studies, etc., will be used to apply instructional concepts?

5.13.4.6.2.5. Support tasks, conditions and standards of the work environment as closely as possible?

5.13.4.6.2.6. Describe any pre-assessment or bypass tests used to determine which modules, if any, a student is required to take?

5.13.4.6.3. Did this activity impact the Instructional Strategy and, if so, have these changes been documented, justified and approved by the customer(s) and SME(s)?

5.13.4.6.4. Was an appropriate mix of SMEs, subject area program managers, system OPRs, instructors, trainers, courseware developers, training managers, etc., involved in the design of this course or learning aid?

5.13.4.7. Do instructional materials and activities meet the following evaluation criteria?

5.13.4.7.1. Are storyboards and supporting documentation detailed enough for reviewers to get a clear picture of precisely what content will be presented, how it will be instructed, and how students will be evaluated for attainment of each learning objective?

5.13.4.7.2. Do storyboards and supporting documentation describe any pre-assessment or bypass tests used to determine which modules, if any, a student is required to take?

5.13.4.7.3. Do storyboards for each module contain:

5.13.4.7.3.1. One or more criterion-referenced learning objectives?

5.13.4.7.3.2. Presentations, demonstrations, and simulations used to support each learning objective?

5.13.4.7.3.3. Relevant examples and non-examples in the form of illustrations, diagrams, demonstrations, scenarios, case studies, etc., used to support each learning objective?

5.13.4.7.3.4. Student projects, exercises, review sessions, quizzes, tests and proficiency evaluations used to support each learning objective?

5.13.4.7.3.5. A clear description of screen design and student interactivity features?

5.13.4.7.3.6. All content supporting each learning objective?

5.13.4.7.3.7. Media used for each instructional and student evaluation activity?

5.13.4.7.3.8. Sequence in which instruction will be delivered to students?

5.13.4.7.4. Does content of the course or learning aid adequately address Gagné’s Events of Instruction?

5.13.4.7.5. Are there sufficient instructional materials and activities to support learning objectives?

5.13.4.7.6. Have instructional concepts been applied when creating examples, non-examples, illustrations, diagrams, demonstrations, scenarios, case studies, etc.?

5.13.4.7.7. Is the sequence of content and learning activities effective?

5.13.4.7.8. Do media and delivery method(s) of instructional materials appear to be appropriate for the situation?

5.13.4.7.9. Is content of the instructional materials directly related to topics, tasks, conditions and standards specified in learning objectives?

5.13.4.7.10. Are examples, non-examples, illustrations, diagrams, demonstrations, scenarios, case studies, and other instructional materials and activities relevant to the target audience?

5.13.4.7.11. Is content of instructional materials accurate and complete?

5.13.4.7.12. Is there enough navigational guidance for a student to easily take the course and any bypass pre-test that may be available?

5.13.4.7.13. Did this activity impact the Instructional Strategy and, if so, have these changes been documented, justified and approved by the customer(s) and SME(s)?

5.13.4.7.14. Was an appropriate mix of SMEs, subject area program managers, system OPRs, instructors, trainers, courseware developers, training managers, etc., involved in the design of this course or learning aid?

5.13.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

5.13.5.1. Nature of perceived issues?

5.13.5.2. Priority of assignment?

5.13.5.3. Appropriate organization to handle tasking request?

5.14. Conduct ISD Evaluation of Design Phase Activities. In this activity, an ISD Evaluation official reviews activities performed during the Design Phase using the ISD Management Plan to determine if they meet the intent of the AFMC Courseware Development and Management Process and its underlying principles of systems engineering, instructional design, and quality improvement. This is a required step in the Design Phase.

5.14.1. Purpose. This activity ensures that the AFMC Courseware Development and Management Process is applied so that instruction has a high probability of being relevant, effective and economical to sustain.

5.14.2. Activity Outcomes:

5.14.2.1. Completion of the applicable section of AFMC ISD Gates Release Points Checklist. The current version of the checklist can be found on the *AFMC ISD Resource* Site. This checklist is used for Command Courses but may also be used for Local Course development. It provides a decision that the appropriate activities in the each gate have

been accomplished to the standards of this manual, or that corrective actions are needed before the project goes into the next gate.

5.14.2.2. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

5.14.2.3. Update ISDMP/PDS (as needed).

5.14.2.4. Decision on where to go next in the courseware process.

5.14.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site and the *AFMC ISD Resource* Site.

5.14.4. Evaluation Criteria.

5.14.4.1. Do the ISD Evaluation Notes for the Design Phase include:

5.14.4.1.1. Basic information about project, date of ISD review, and participants?

5.14.4.1.2. Status of outstanding action items from the Analysis Phase or earlier Design Phase reviews?

5.14.4.1.3. Reasons typical Design Phase activities were skipped, (as applicable)?

5.14.4.1.4. Observations about compliance/non-compliance of Design Phase activities with standards set forth in this manual and the *AFMC Depot Maintenance Training Resource* Site?

5.14.4.1.5. Explanations of waivers and adjustments of standards for Design Phase activities, (as applicable)?

5.14.4.1.6. Documentation and customer approval of changes to the Preliminary Instructional Strategy?

5.14.4.1.7. Corrective actions which need to be accomplished with suspense dates, (as applicable)?

5.14.4.1.8. Design Phase review decision?

5.14.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

5.14.5.1. Nature of perceived issues?

5.14.5.2. Priority of assignment?

5.14.5.3. Appropriate organization to handle tasking request?

Chapter 6

DEVELOPMENT PHASE

6.1. Development Phase Defined. During the Development Phase, instructional and evaluation materials are developed, tested, and revised and the instructional system is prepared for full operational tryout in the field. For additional information and guidance relating to the Development Phase of the Air Force ISD process, refer to DAFH 36-2675.

6.2. Develop Instructional Materials for Review. In this activity, the courseware developer creates an initial draft of presentation materials, learning aids, and instructional guidance that enable a qualified instructor or trainer to deliver consistent and comprehensive instruction repeatedly.

6.2.1. Purpose. This is the activity where non-CBI is developed. (presentations, lesson plans, student handouts, etc.)

6.2.2. Activity Outcomes:

6.2.2.1. Draft instructional materials such as charts, videos, demonstration aids, student handouts, student guides/workbooks, computer technologies, etc.

6.2.2.2. Draft instructor or trainer guidance in the form of POI Part 2, Structured On-the-Job Training (SOJT) Guide or generic lesson plan documentation.

6.2.2.3. Update Course Control Documents (CCDs) such as a POI, CTS, or Course Chart.

6.2.2.4. Modified Instructional Strategy with customer approval (as needed).

6.2.2.5. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

6.2.2.6. Decision on where to go next in the courseware process.

6.2.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

6.2.4. Evaluation Criteria.

6.2.4.1. Are draft instructional materials (e.g., course control documents, presentation, generic lesson plan, student handout, etc.) detailed enough for reviewers to get a clear picture of precisely what content will be presented and how it will be instructed?

6.2.4.2. Does content of the course or learning aid adequately address Gagné's Events of Instruction?

6.2.4.3. Are there sufficient instructional materials and activities to support learning objectives?

6.2.4.4. Have instructional concepts been applied when creating examples, non-examples, illustrations, diagrams, demonstrations, scenarios, case studies, etc.?

6.2.4.5. Is the sequence of content and learning activities effective?

6.2.4.6. Do media and delivery method(s) of instructional materials appear to best appropriate for the situation?

6.2.4.7. Is content of instructional materials directly related to topics, tasks, conditions and standards specified in learning objectives?

6.2.4.8. Are examples, non-examples, illustrations, diagrams, demonstrations, scenarios, case studies, and other instructional materials and activities relevant to the target audience?

6.2.4.9. Is content of instructional materials accurate and complete?

6.2.4.10. Is there enough instructor guidance for a technically qualified substitute instructor/trainer to take over a course and deliver the material consistent with methods used by a lead instructor/trainer?

6.2.4.11. Do learning objectives meet the following evaluation criteria?

6.2.4.11.1. Does each learning objective contain:

6.2.4.11.1.1. One behavior (knowledge or task)?

6.2.4.11.1.2. A condition of performance that is closely related to actual conditions in the work environment?

6.2.4.11.1.3. A standard of behavior that is criterion-referenced and closely related to actual standards required in the work environment?

6.2.4.11.2. Do knowledge and tasks included in learning objectives coincide directly with task statements approved by the customer during knowledge/task analysis in the Design Phase?

6.2.4.11.3. Are proficiency levels for learning objectives appropriate for instructional requirements defined during the Planning, Analysis and Design Phases?

6.2.4.11.4. Do objectives use types of learning appropriate for the situation?

6.2.4.11.5. Did changes to initially planned topics, tasks and proficiency levels occur, and if so, have these changes been approved by the customer and SME?

6.2.4.12. Do student evaluation materials and activities embedded in instructional materials meet the following evaluation criteria?

6.2.4.12.1. Do the amount and types of evaluation provide the customer with a high level of confidence that students have attained the proficiency level of learning specified in the learning objectives?

6.2.4.12.2. Does media and delivery method(s) of student evaluations appear to be appropriate for the situation?

6.2.4.12.3. Is content of student evaluations directly related to topics, tasks, conditions and standards specified in the learning objectives?

6.2.4.12.4. Do evaluation activities emulate conditions and standards of the work environment as closely as possible?

6.2.4.12.5. Is content of the evaluation materials accurate and complete?

6.2.4.12.6. Have widely accepted instructional concepts been applied when selecting student evaluation activities and creating test/measurement materials?

6.2.4.12.7. Is there enough evaluation guidance for a technically qualified substitute instructor/trainer to take over a course and administer exercises, projects and tests consistent with methods used by a lead instructor/trainer?

6.2.4.12.8. Are minimum evaluations of 70% correctable to 100% (as applicable), so that students will be given immediate remediation before completion of the course?

6.2.4.12.9. For pass/fail tests, are there at least two questions/scenarios for each learning objective tested or a question pool to allow for remedial instruction and prevent test compromise situations?

6.2.4.12.10. Is content and minimum passing score for pass/fail tests consistent with training mandates?

6.2.4.12.11. Did this activity impact the Instructional Strategy and, if so, have these changes been documented, justified and approved by the customer(s) and SME(s)?

6.2.4.13. Does development of the course meet the following criteria?

6.2.4.13.1. Presentations, demonstrations, simulations and exhibits used to support each learning objective?

6.2.4.13.2. Student projects, exercises, review sessions, quizzes, tests and proficiency evaluations used to support each learning objective?

6.2.4.13.3. A clear description of facilities, equipment, tools and materials used to deliver the training?

6.2.4.13.4. An overview of content supporting each learning objective?

6.2.4.13.5. Was media used for each instructional and student evaluation activity?

6.2.4.13.5.1. Explain how relevant examples and non-examples in the form of illustrations, diagrams, demonstrations, scenarios, case studies, etc., used to apply instructional concepts?

6.2.4.13.5.2. Ensure media supports the tasks, conditions and standards of the work environment as closely as possible?

6.2.4.13.5.3. Describe any pre-assessment or bypass tests used to determine which modules, if any, a student will be required to take?

6.2.4.13.6. Did this activity impact the Instructional Strategy and, if so, have these changes been documented, justified and approved by the customer and SME?

6.2.4.13.7. Was an appropriate mix of SMEs, subject area program managers, system OPRs, instructors, trainers, courseware developers, training managers, etc., involved in the development of this course or learning aid?

6.2.4.14. If this is a command standard course, does the presentation have placeholder charts for later inclusion of localized information?

6.2.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

6.2.5.1. Nature of perceived issues?

6.2.5.2. Priority of assignment?

6.2.5.3. Appropriate organization to handle tasking request?

6.3. Develop Remaining Distributed Learning (DL). In Development Phase, a courseware development team creates remaining instruction. In this activity, having obtained approval on design, content and functionality of the course or learning aid, a development team moves on to create the remaining segments/modules for review.

6.3.1. Purpose. Where applicable, developing DL courseware in modules or segments, problems are identified and changes are tested one segment/module at a time, greatly reducing the risk of costly rework that occurs when an entire course is reviewed at one time.

6.3.2. Activity Outcomes:

6.3.2.1. Completion of development for segments/modules or full course/learning aid from experts inputs based on complexity level.

6.3.2.2. Modified Instructional Strategy with customer approval, (as needed).

6.3.2.3. Additional tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

6.3.2.4. Decision on where to go next in the courseware process.

6.3.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

6.3.4. Evaluation Criteria.

6.3.4.1. Was an appropriate mix of SMEs, subject area program managers, system OPRs, courseware developers, training managers, etc., involved in the development of this course or learning aid?

6.3.4.2. Are screen and navigational features of courseware/learning aid engaging and easy to use?

6.3.4.2.1. Is there enough navigational guidance for a student to easily take the course and any bypass pre-test?

6.3.4.3. Does presented or interactive media meet objective requirements?

6.3.4.3.1. Does media use types of learning appropriate for the situation?

6.3.4.4. Does development meet Section 508 compliance?

6.3.4.5. Does published/compiled DL load properly on an approved SCORM LMS?

6.3.4.6. Does DL bookmark to allow training to continue from point training was exited?

6.3.4.7. Do student evaluation materials and activities meet the following evaluation criteria?

6.3.4.7.1. Do the amount and types of evaluation provided allow students to attain the proficiency level of learning specified in the learning objectives?

6.3.4.7.2. Do media and delivery method(s) of student evaluations match the outline established during the design phase?

6.3.4.7.3. Is content of student evaluations directly related to topics, tasks, conditions and standards specified in the learning objectives?

6.3.4.7.4. Do evaluation activities emulate conditions and standards of the work environment as closely as possible?

6.3.4.7.5. Is content of evaluation materials accurate and complete?

6.3.4.7.6. Are applicable evaluations correctable to 100% so that students will be given immediate feedback and correct answers before completion of the course?

6.3.4.7.7. For pass/fail tests, are there one or more test pools that allow for remedial instruction and prevent test compromise situations?

6.3.4.7.8. Is the content and minimum passing score for pass/fail tests consistent with training mandates?

6.3.4.8. Do instructional materials and activities meet the following evaluation criteria?

6.3.4.8.1. Do storyboards/outlines match the output of the course/learning aid for each segment or module?

6.3.4.8.1.1. Do they identify media used for instructional and student evaluation activities?

6.3.4.8.1.2. Do they align with the sequence in which instruction is delivered to students?

6.3.4.8.1.3. Is sequence of content and learning activities effective?

6.3.4.8.2. Does content of the course or learning aid adequately address Gagné's Events of Instruction?

6.3.4.8.3. Is content of instructional materials directly related to topics, tasks, conditions and standards specified in learning objectives?

6.3.4.9. Did this activity impact the Instructional Strategy and, if so, have these changes been documented, justified and approved by the customer and SME?

6.3.5. Issues/actions: Are any issues/action items from this activity documented sufficiently to determine:

6.3.5.1. Nature of perceived issues?

6.3.5.2. Priority of assignment?

6.3.5.3. Appropriate organization to handle tasking request?

6.4. Review Content of Entire Distributed Learning Course. In this activity, SMEs, subject area program managers, system OPRs, courseware developers, information technology experts, etc., review Distributed Learning (DL) sequence, appearance, content and functionality before final validation. **Note:** Only minor corrections to content of the course or learning aid are permitted after closeout of Development Phase.

6.4.1. Purpose. This activity verifies relevance, accuracy and effectiveness of content developed before the DL goes to production where changes may become very expensive. It is the equivalent of a technical and instructor review for non-distributed learning courseware.

6.4.2. Activity Outcomes:

6.4.2.1. Course Changes Log input from reviewers on content of the course or learning aid.

6.4.2.2. Documentation on changes to be made to content of instructional activities, evaluation activities, and general presentation of information based on input from experts.

6.4.2.3. Additional tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

6.4.2.4. Decision on where to go next in the courseware process.

6.4.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

6.4.4. Evaluation Criteria.

6.4.4.1. Did appropriate SMEs, training managers, subject area program managers, system OPRs, DL experts, etc., review the course/learning aid compared to storyboard/outline materials?

6.4.4.2. Did the courseware developer document input from reviewers and how any recommended changes are to be accomplished?

6.4.4.3. Did the courseware developer document why any recommended changes will not be accomplished?

6.4.4.4. Are learning objectives clearly identified in the courseware?

6.4.4.4.1. Do knowledge and tasks included in learning objectives coincide directly with task statements approved by the customer during the Design Phase?

6.4.4.4.2. Are proficiency levels for learning objectives appropriate for instructional requirements defined during the Design Phase?

6.4.4.4.3. Do objectives use types of learning appropriate for the situation?

6.4.4.4.4. Did changes to initially planned topics, tasks, and proficiency levels occur? If so, have these changes been approved by the customer and SME?

6.4.4.5. Do student evaluation materials and activities meet the following evaluation criteria?

6.4.4.5.1. Does the amount and types of evaluation provide the customer with a high level of confidence that students can attain the proficiency level of learning specified in the learning objectives?

6.4.4.5.2. Do the media and delivery method(s) of student evaluations appear to be appropriate for the situation?

6.4.4.5.3. Is content of student evaluations directly related to topics, tasks, conditions, and standards specified in the learning objectives?

6.4.4.5.4. Do evaluation activities emulate the conditions and standards of the work environment as closely as possible?

6.4.4.5.5. Is the content of evaluation materials accurate and complete?

6.4.4.5.6. Have accepted instructional concepts been applied when selecting student evaluation activities and creating test/measurement materials?

6.4.4.5.7. Are all applicable evaluations correctable to 100% so that students will be given immediate feedback and correct answers before completion of the course?

6.4.4.5.8. For pass/fail tests, are there one or more question pools/scenarios for each learning objective tested to allow for remedial instruction and to prevent test compromise situations?

6.4.4.5.9. Is content and minimum passing score for pass/fail tests consistent with training mandates?

6.4.4.6. Does the storyboard/outline align with of screen design and student interactivity features in the course/learning aid?

6.4.4.6.1. Identifies content that will support each learning objective?

6.4.4.6.2. Media works successfully for each instructional and student evaluation activity?

6.4.4.6.3. Sequence is in line with instruction delivered to students?

6.4.4.6.4. Confirm that selected delivery method is appropriate for the situation?

6.4.4.6.5. Propose instructional and student evaluation events that are appropriate for the situation and support the learning objectives?

6.4.4.6.6. Did relevant examples and non-examples in the form of illustrations, diagrams, demonstrations, scenarios, case studies, etc., function properly?

6.4.4.6.7. Were conditions and standards of the work environment emulated as closely as possible?

6.4.4.6.8. Did any pre-assessment or bypass tests used to determine which modules, if any, a student will be required to take function properly?

6.4.4.6.9. Did this activity impact the Instructional Strategy and, if so, have these changes been documented, justified and approved by the customer and SME?

6.4.4.7. Do instructional materials and activities meet Section 508 delivery requirements?

6.4.4.8. Does content of the course or learning aid adequately address Gagné’s Events of Instruction?

6.4.4.8.1. Is the content of instructional materials accurate and complete?

6.4.4.8.2. Is there enough navigational guidance for a student to easily take the course and any bypass pre-test?

6.4.4.8.3. Did this activity impact the Instructional Strategy and, if so, have these changes been documented, justified and approved by the customer and SME?

6.4.4.8.4. Was an appropriate mix of SMEs, subject area program managers, system OPRs, courseware developers, training managers, etc., involved in the development of this course or learning aid?

6.4.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

6.4.5.1. Nature of perceived issues?

6.4.5.2. Priority of assignment?

6.4.5.3. Appropriate organization to handle tasking request?

6.5. Perform Technical Review of In-Work Materials. In this activity, technical experts review content of draft instructional and evaluation materials to determine if they are relevant, accurate and complete. This step is usually repeated more than once until technical experts are satisfied with the content and appearance of instruction. Only minor corrections to content and appearance of the course are permitted after technical review final approval in Development Phase. For Distributed Learning (DL) this step has been accomplished.

6.5.1. Purpose. This activity calls on SMEs, subject area program managers, system OPRs, training managers, instructors/trainers, and other customer representatives to verify instruction is relevant to what actually occurs in the workplace of the target population, and content of materials is technically accurate and complete.

6.5.2. Activity Outcomes:

6.5.2.1. Course Changes Log input from reviewers on content of the course or learning aid.

6.5.2.2. Documentation on changes made to instruction based on input from experts.

6.5.2.3. Additional tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

6.5.2.4. Decision on where to go next in courseware process.

6.5.3. Additional Guidance. See *AFMC Depot Maintenance Training Resource* Site or contact Complex CPM.

6.5.4. Evaluation Criteria.

6.5.4.1. Did appropriate SMEs, subject area program managers, system OPRs, training managers, instructors/trainers and other customer representatives review the materials?

6.5.4.2. Do learning objectives meet the following evaluation criteria?

6.5.4.2.1. Does each learning objective contain:

6.5.4.2.1.1. One behavior (knowledge or task)?

6.5.4.2.1.2. A condition of performance that is closely related to actual conditions in the work environment?

6.5.4.2.1.3. A standard of behavior that is criterion-referenced and closely related to actual standards required in the work environment?

6.5.4.2.2. Do knowledge and tasks included in learning objectives coincide directly with task statements approved by the customer in the Design Phase?

6.5.4.2.3. Are proficiency levels for learning objectives appropriate for instructional requirements defined in Design Phase?

6.5.4.2.4. Do objectives use types of learning appropriate for the situation?

6.5.4.2.5. Did changes to initially planned topics, tasks and proficiency levels occur, and if so, have these changes been approved by the customer and SME?

6.5.4.3. Do student evaluation materials and activities meet the following evaluation criteria?

6.5.4.3.1. Do the amount and types of evaluation provide the customer with a high level of confidence that students have attained the proficiency level of learning specified in the learning objectives?

6.5.4.3.2. Do the media and delivery method(s) of student evaluations appear to be appropriate for the situation?

6.5.4.3.3. Is content of student evaluations directly related to topics, tasks, conditions and standards specified in the learning objectives?

6.5.4.3.4. Do evaluation activities simulate conditions and standards of the work environment as closely as possible?

6.5.4.3.5. Is content of evaluation materials accurate and complete?

6.5.4.3.6. Have accepted instructional concepts been applied when selecting student evaluation activities and creating test/measurement materials?

6.5.4.3.7. Is there enough evaluation guidance for a technically qualified substitute instructor/trainer to take over a course and administer exercises, projects and tests consistent with methods used by primary instructor/trainer?

6.5.4.3.8. Are applicable evaluations correctable to 100% so that students will be given immediate feedback and correct answers before completion of the course?

6.5.4.3.9. For pass/fail tests, are there at least two questions/scenarios for each learning objective tested or test pool to allow for remedial instruction and prevent test compromise situations?

6.5.4.3.10. Is content and minimum passing score for pass/fail tests consistent with training mandates?

6.5.4.4. Was an appropriate mix of SMEs, subject area program managers, system OPRs, instructors/trainers, courseware developers, training managers, etc. involved in the development of this course or learning aid?

6.5.4.5. Do instructional materials and activities meet the following evaluation criteria?

6.5.4.5.1. Are draft instructional materials (e.g., course control documents, presentation, generic lesson plan, student handout, etc.) detailed enough for reviewers to get a clear picture of precisely what content will be presented and how it will be instructed?

6.5.4.5.2. Does content of the course or learning aid adequately address Gagné's Events of Instruction?

6.5.4.5.3. Are there sufficient instructional materials and activities to support learning objectives?

6.5.4.5.4. Have accepted instructional concepts been applied when creating examples, non-examples, illustrations, diagrams, demonstrations, scenarios, case studies, etc.?

6.5.4.5.5. Does the sequence of content and learning activities follow the course outline created during the design phase?

6.5.4.5.6. Are the media and delivery method(s) of instructional materials in line with the course outline created during the design phase?

6.5.4.5.7. Is content of instructional materials directly related to topics, tasks, conditions, and standards specified in the learning objectives?

6.5.4.5.8. Are examples, non-examples, illustrations, diagrams, demonstrations, scenarios, case studies, and other instructional materials and activities relevant to the target audience?

6.5.4.5.9. Is content of instructional materials accurate and complete?

6.5.4.5.10. Is there enough instructor guidance for a technically qualified substitute instructor/trainer to take over a course and deliver the material consistent with methods used by a primary instructor/trainer?

6.5.4.5.11. If this is a command standard course, does the presentation have placeholder charts for later inclusion of localized information?

6.5.4.5.12. Did this activity impact the Instructional Strategy and, if so, have these changes been documented, justified and approved by the customer and SME?

6.5.4.6. Did the courseware developer document input from reviewers and how recommended changes will be accomplished?

6.5.4.7. Did the courseware developer document why any recommended changes will not be accomplished?

6.5.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

6.5.5.1. Nature of perceived issues?

6.5.5.2. Priority of assignment?

6.5.5.3. Appropriate organization to handle tasking request?

6.6. Perform Instructor/Trainer Review of In-Work Instructional Materials. In this activity, instructors/trainers and independent courseware developers review draft instruction and evaluation materials to determine if they are effective, efficient, and adequately documented. This step is usually repeated more than once until instructional design and delivery experts are satisfied with the presentation and documentation of instruction. Only minor corrections to presentation and documentation of the course are permitted after final instructor/trainer approval in Development Phase.

6.6.1. Purpose. This activity calls on instructional design and delivery experts to verify that content is effectively presented and adequately documented so that a qualified instructor/trainer can provide consistent and complete instruction time after time.

6.6.2. Activity Outcomes:

6.6.2.1. Course Changes Log input from reviewers on effectiveness of presentations, learning aids, evaluation tools, etc., in helping students attain learning objectives.

6.6.2.2. Documentation on changes that will be made to the instruction based on input from delivery and instructional design experts.

6.6.2.3. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

6.6.2.4. Decision on where to go next in the courseware process.

6.6.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

6.6.4. Evaluation Criteria.

6.6.4.1. Did appropriate instructors/trainers and courseware developers review the instructional materials?

6.6.4.2. Do the learning objectives meet evaluation criteria in Design Phase, Develop/Write Learning Objectives paragraph?

6.6.4.2.1. Have changes to initially planned topics, tasks and proficiency levels been approved by customer and SME?

6.6.4.3. Do student evaluation materials and activities meet the evaluation criteria in Design Phase, Design Test/measurement Materials paragraph?

6.6.4.4. Does the design of the course meet the evaluation criteria in the Design Phase, Design Training paragraph?

6.6.4.5. Do instructional materials and activities meet the evaluation criteria in Development Phase, Develop Instructional Materials for Review paragraph above?

6.6.4.5.1. Did this activity impact the Instructional Strategy and, if so, have these changes been documented, justified and approved by the customer and SME?

6.6.4.6. Did the courseware developer document input from reviewers, and how recommended changes will be accomplished?

6.6.4.7. Did the courseware developer document why any recommended changes will not be accomplished?

6.6.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

6.6.5.1. Nature of perceived issues?

6.6.5.2. Priority of assignment?

6.6.5.3. Appropriate organization to handle tasking request?

6.7. Develop Evaluation Materials (as needed). In this activity, the courseware developer validates the draft version of evaluation materials prepared during Design phase, and makes adjustments.

6.7.1. Purpose. This activity updates instruction materials identified in technical and instructor/trainer review.

6.7.2. Activity Outcomes:

6.7.2.1. Updates to evaluation materials such as pre-tests, bypass tests, review questions, class exercises/projects, quizzes, post-tests, proficiency evaluations, product/process checklists, etc., as identified in the Course Changes Log. Revise design of training (as needed).

6.7.2.2. Update guidance for instructor/trainer on how to administer evaluation tools.

6.7.2.3. Modified Instructional Strategy with customer approval, (as needed).

6.7.2.4. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

6.7.2.5. Decision on where to go next in the courseware process.

6.7.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

6.7.4. Evaluation Criteria. Incorporation of identified updates.

6.8. Assist Instructors/Trainers in Train-the-Trainer Activities. In this activity, a courseware development team helps prepare instructors/trainers to instruct the course by giving them active roles in all review and implementation activities.

6.8.1. Purpose. This activity readies instructors/trainers to deliver instruction when subject matter is new and instructors/trainers do not yet have experience in presenting the material.

6.8.2. Activity Outcomes:

6.8.2.1. Correspondence, meetings, and review sessions that team SMEs with instructors/ trainers for the purpose of discussing and practicing delivery of a new course.

6.8.2.2. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

6.8.2.3. Decision on where to go next in the courseware process.

6.8.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

6.8.4. Evaluation Criteria.

6.8.4.1. Are instructors/trainers kept actively involved in review and tryout activities for this course?

6.8.4.2. Have sufficient tryouts been conducted to effectively allow instructors/trainers to sit through a course presentation at least once, co-teach at least once, and instruct under observation of SMEs at least once?

6.8.4.3. If sufficient tryouts were not feasible, have special arrangements been made for at least one train-the-trainer session with the courseware development team, SMEs, and future instructors/ trainers for the purpose of discussing and practicing delivery of a new course?

6.8.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

6.8.5.1. Nature of perceived issues?

6.8.5.2. Priority of assignment?

6.8.5.3. Appropriate organization to handle tasking request?

6.9. Perform Small Group Tryout of Select Instructional Segments. In this activity, a portion of a course or learning aid is validated under field conditions by selected instructors/trainers, customer representatives and/or students from the target population to verify effectiveness of instruction.

6.9.1. Purpose. This activity is used when a course is lengthy, only part of an existing course has been revised, when a segment of training is particularly risky and requires advance feedback, etc., to determine if changes are needed in content or delivery based on pre-testing of a segment of the course under field conditions.

6.9.2. Activity Outcomes:

6.9.2.1. Average segment length (time needed to provide instruction) should be established.

6.9.2.2. Input from instructors/trainers, content experts and sample students on content, appearance and delivery methods of instruction.

6.9.2.3. Documentation on changes that will be made to content, appearance, and delivery of instruction based on this early field test of materials.

6.9.2.4. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

6.9.2.5. Decision on where to go next in the courseware process.

6.9.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

6.9.4. Evaluation Criteria.

6.9.4.1. Did this early individual/small group tryout include lead SMEs, lead subject area program manager(s), lead system OPRs, instructors/trainers, assigned ISD Evaluation official, lead customer training manager, students from the target population, etc., as appropriate to validate the segment of instruction?

6.9.4.2. Do the learning objectives meet evaluation criteria in Design Phase, Develop/Write Learning Objectives paragraph?

6.9.4.2.1. Did this activity impact the Instructional Strategy and, if so, have these changes been documented, justified and approved by the customer and SME?

6.9.4.3. Do student evaluation materials and activities meet the evaluation criteria in Design Phase, Design Test/measurement Materials paragraph?

6.9.4.3.1. Did this activity impact the Instructional Strategy and, if so, have these changes been documented, justified and approved by the customer and SME?

6.9.4.4. Does the design of the course meet the evaluation criteria in the Design Phase, Design Training paragraph**?**

6.9.4.4.1. Did this activity impact the Instructional Strategy and, if so, have these changes been documented, justified and approved by the customer and SME?

6.9.4.5. Do instructional materials and activities meet the evaluation criteria in Development Phase, Develop Instructional Materials for Review paragraph above?

6.9.4.5.1. Did this activity impact the Instructional Strategy and, if so, have these changes been documented, justified and approved by the customer and SME?

6.9.4.6. Did the courseware developer document input from reviewers, and how recommended changes will be accomplished?

6.9.4.7. Did the courseware developer document why any recommended changes will not be accomplished?

6.9.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

6.9.5.1. Nature of perceived issues?

6.9.5.2. Priority of assignment?

6.9.5.3. Appropriate organization to handle tasking request?

6.10. Modify Course Control Documents (CCDs) (as needed). In this activity, a courseware developer makes needed changes to CCDs based on Design and Development Phase activities.

6.10.1. Purpose. This activity is to update or modify course documents as necessary.

6.10.2. Activity Outcomes:

6.10.2.1. Course Chart, CTS, POI, SOJT guide, and/or generic lesson plan that have been revised to match changes in instruction in course changes log and student evaluation from previous Design and Development Phase activities.

6.10.2.2. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

6.10.2.3. Decision on where to go next in the courseware process.

6.10.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

6.10.4. Evaluation Criteria. N/A.

6.10.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

6.10.5.1. Nature of perceived issues?

6.10.5.2. Priority of assignment?

6.10.5.3. Appropriate organization to handle tasking request?

6.11. Complete Production of Distributive Learning Materials. In this activity, a courseware development team completes final production of the entire computer-based course, learning aid, and associated materials based on input from experts and student representatives in the previous Design and Development Phase reviews.

6.11.1. Purpose. This activity prepares computer-based course or learning aid for use in the field.

6.11.2. Activity Outcomes:

6.11.2.1. Distributive Learning (DL) Test Checklist.

6.11.2.2. Computer-based course or learning aid ready for use in the field.

6.11.2.3. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

6.11.2.4. Decision on where to go next in the courseware process.

6.11.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

6.11.4. Evaluation Criteria.

6.11.4.1. Does the computer-based course or learning aid contain:

6.11.4.1.1. Final content for all modules ready for use in the field?

6.11.4.1.2. Screens and narration for all modules ready for use in the field?

6.11.4.1.3. Navigational buttons and student interactivity ready for use in the field?

6.11.4.2. Has computer-based course/learning aid been tested for SCORM conformance? Tests can be found at: Advanced Distributed Learning (ADL) **www.adlnet.gov/scorm/**

6.11.4.3. Has computer-based course been tested on the primary command approved LMS, courseware database and content library that will host it?

6.11.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

6.11.5.1. Nature of perceived issues?

6.11.5.2. Priority of assignment?

6.11.5.3. Appropriate organization to handle tasking request?

6.12. Arrange Final Validation Activities. In this activity, preparations are made to test the final draft of a course or learning aid using actual methods of delivery. Final validation activities for computer-based courses and learning aids usually involve only an operational tryout in the field. Final validation activities for non-computer based courses and learning aids usually involve a small group tryout and an operational tryout.

6.12.1. Purpose. This activity arranges final validation events, invites participants, and clarifies their role in validation.

6.12.2. Activity Outcomes:

6.12.2.1. Final validation schedule and invitation.

6.12.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

6.12.4. Evaluation Criteria.

6.12.4.1. Do validation plans include key SMEs, lead subject area program managers, system OPRs, instructors/trainers, assigned ISD Evaluation official, customer training manager, students from the target population, etc., as appropriate to validate the training?

6.12.4.2. Is validation plan appropriate for the situation, justified and documented in the ISD Evaluation Review Notes?

6.12.4.3. Are needed resources (e.g., facility, instructor, equipment, tools, etc.) available to support the final validation as planned?

6.12.4.4. Have participants been advised of their role in validation activities?

6.12.4.5. If this is a command course, have participants been given negotiated notice for temporary duty arrangements or electronic conference, for a small group tryout/final SME review of the course?

6.12.4.6. Has validation been scheduled to de-conflict with holidays, inspections, and traditionally heavy workload periods for the target population?

6.12.4.7. Have participants been engaged in the review of training materials before a small group tryout/final SME review of the course?

6.12.4.8. If there are group activities as part of the instruction, have appropriate numbers of students been included in final validation to permit evaluation of these activities during full operational tryout?

6.12.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

6.12.5.1. Nature of perceived issues?

6.12.5.2. Priority of assignment?

6.12.5.3. Appropriate organization to handle tasking request?

6.13. Perform Final Validation of Course. In this activity, the final draft of the full course or learning aid is validated under field conditions by a group of selected instructors/trainers, and customer representatives depending on the situation (as applicable). This is one of the last activities in Development Phase. By conducting a Development Phase ISD Evaluation of courseware and validation plan, risks are minimized, and final validation activities are appropriate. Command standard courses usually involve a final validation of course materials by a group of SMEs, instructors, and training specialists from each Complex using the course. Those who cannot attend are given the opportunity to recommend final changes in writing. Each installation using a command standard course is expected to localize materials to meet their needs and to conduct a full operational tryout of the course at their own complex once the official course is released by the command. In this activity, a final draft of the course or learning aid is tested on average students under normal operating conditions. This activity is usually where a DL project enters the Implementation Phase. This may be in conjunction with the validation process.

6.13.1. Purpose. This activity is the last chance for experts such as SMEs, subject area program managers, system OPRs and instructors to identify changes to a course based on minor issues with content and delivery of instruction not obvious during earlier reviews. This activity also helps determine average length of a course. This activity is the final test for courseware before it is finalized for regular use in the field. It evaluated the effectiveness of instruction on the actual target population.

6.13.2. Activity Outcomes:

6.13.2.1. Establish of average course length (time needed to provide instruction).

6.13.2.2. Input from instructors/trainers and content experts on content, appearance and effectiveness of the instruction.

6.13.2.3. Documentation on changes made to content, appearance, and delivery of instruction based on this field test of materials.

6.13.2.4. Decision on where to go next in the courseware process.

6.13.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

6.13.4. Evaluation Criteria.

6.13.4.1. Did final validation include SMEs, subject area program managers, system OPRs, instructors/trainers, assigned ISD Evaluation official, lead customer training manager, etc., as appropriate to validate training?

6.13.4.2. Did courseware developer document input from reviewers and how recommended changes will be accomplished?

6.13.4.3. Did courseware developer document why any recommended changes from reviewers will not be accomplished?

6.13.4.4. Did operational tryout include students from the target population randomly selected through normal scheduling methods?

6.13.4.5. Did courseware developer document input from the students and instructor/trainer, and explain how recommended changes will be accomplished?

6.13.4.6. Did courseware developer document why any recommend changes from the students and instructor/trainer will not be accomplished?

6.13.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

6.13.5.1. Nature of perceived issues?

6.13.5.2. Priority of assignment?

6.13.5.3. Appropriate organization to handle tasking request?

6.14. Conduct ISD Evaluation of Development Phase Activities. In this activity, at least one ISD Evaluation official reviews activities performed during the Development Phase to determine if they meet the intent of the AFMC Courseware Development and Management Process and its underlying principles of systems engineering, instructional design, and quality improvement. This is a required step in the Development Phase.

6.14.1. Purpose. This activity ensures that the AFMC Courseware Development and Management Process is applied so that instruction has a high probability of being relevant, effective, and economical to sustain.

6.14.2. Activity Outcomes:

6.14.2.1. Completion of the applicable section of AFMC ISD Gates Release Points Checklist. The current version of the checklist can be found on the *AFMC ISD Resource* Site. This checklist is used for Command Courses but may also be used for Local Course development. It provides a decision that the appropriate activities in the each gate have

been accomplished to the standards of this manual, or that corrective actions are needed before the project goes into the next gate.

6.14.2.2. Decision that appropriate activities in the Development Phase have been accomplished to the standards of this manual, or that corrective actions are needed before the project goes into Implementation Phase.

6.14.2.3. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

6.14.2.4. Update ISDMP/PDS (as needed).

6.14.2.5. Decision on where to go next in the courseware process.

6.14.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site and the *AFMC ISD Resource* Site.

6.14.4. Evaluation Criteria.

6.14.4.1. Do ISD Evaluation Notes for Development Phase include:

6.14.4.1.1. Basic information about project, date of ISD review, and participants?

6.14.4.1.2. Status of outstanding action items from Design Phase or earlier Development Phase reviews?

6.14.4.1.3. Reasons any Development Phase activities were skipped?

6.14.4.1.4. Observations about compliance/non-compliance of Development Phase activities with standards in this manual, the *AFMC Depot Maintenance Training Resource* Site and the *AFMC ISD Resource* Site?

6.14.4.1.5. Explanations of waivers and adjustments of standards for Development Phase activities, (as applicable)?

6.14.4.1.6. Documentation and customer approval of changes to the Instructional Strategy.

6.14.4.1.7. Corrective actions which need to be accomplished with suspense dates, (as applicable)?

6.14.4.1.8. Development Phase review decision?

6.14.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

6.14.5.1. Nature of perceived issues?

6.14.5.2. Priority of assignment?

6.14.5.3. Appropriate organization to handle tasking request?

6.14.6. Update ISD Management Plan as required.

Chapter 7

IMPLEMENTATION PHASE

7.1. Implementation Phase Defined. During the Implementation Phase, the instructional system is validated under field conditions, final changes are made to courseware, official courseware master file is posted in the appropriate installation functional courseware library and personalized lesson plans are developed by instructors/trainers. At the end of this phase, the course or learning aid is approved by the government and is ready for full operational implementation with personnel targeted for training. For additional information and guidance relating to the Implementation Phase of the Air Force ISD process, refer to DAFH 36-2675.

7.2. Finalize Materials and Course Control Documents (CCDs). In this activity, the courseware developer makes requested changes and prepares the final version of materials and CCDs.

7.2.1. Purpose. This activity corrects any issues found with content, appearance, delivery methods and documentation, and prepares courseware for the official master file.

7.2.2. Activity Outcomes:

7.2.2.1. Completion of any changes to be made.

7.2.2.2. Minor changes made to content, appearance, and documentation of the instruction based on input during final validation.

7.2.2.3. Decision on where to go next in the courseware process.

7.2.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

7.2.4. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

7.2.4.1. Nature of perceived issues?

7.2.4.2. Priority of assignment?

7.2.4.3. Appropriate organization to handle tasking request?

7.2.5. Follow-on: Has a courseware developer arranged for a follow-on review of revised materials, if warranted?

7.3. Obtain Content Expert, Customer and Training Management Signatures on Courseware (signature page). In this activity a courseware developer obtains signatures from SMEs, subject area program manager(s), system OPRs, instructors/trainers, and other customer representatives who participated in the courseware project. The courseware developer also obtains signatures from training managers, courseware managers, ISD Evaluation official, and others who need to approve the courseware after Implementation Phase activities have been reviewed by the ISD Evaluation official before the course is implemented in the field. A formal signature page is also accomplished to include with course materials. Signatures are course and circumstance dependent.

7.3.1. Purpose. This activity documents final approval of courseware by experts and customer representatives prior to the ISD Evaluation review for Implementation Phase, thus providing evidence that customer representatives are satisfied with the product. This activity also

documents final approval of the courseware by Command and/or Complex Maintenance Training representatives as required prior to creation of an official courseware master file and distribution of materials for regular use in the field.

7.3.2. Activity Outcomes:

7.3.2.1. Electronic Signature Document of applicable representatives on courseware approval.

7.3.2.2. Decision on where to go next in the courseware process.

7.3.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site for templates.

7.3.4. Evaluation Criteria.

7.3.4.1. Were needed digital signatures added on the signature document?

7.4. Conduct ISD Evaluation Review of Implementation Phase Activities. In this activity, at least one ISD Evaluation official reviews activities performed during the Implementation Phase to determine if they meet the intent of the AFMC Courseware Development and Management Process and its underlying principles of systems engineering, instructional design, and quality improvement. This is a required step in the Implementation Phase.

7.4.1. Purpose. This activity ensures that the AFMC Courseware Development and Management Process is applied so that instruction has a high probability of being relevant, effective and economical to sustain.

7.4.2. Activity Outcomes:

7.4.2.1. Completion of the "Implementation" section of the AFMC ISD Gates Release Points Checklist. The current version of the checklist can be found on the *AFMC ISD Resource* Site. This checklist is used for Command Courses but may also be used for Local Course development.

7.4.2.2. Decision that appropriate activities in the Implementation Phase have been accomplished to the standards of this manual, or that corrective actions are needed before the project is closed out.

7.4.2.3. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

7.4.2.4. Finalize ISDMP/PDS (as needed).

7.4.2.5. Decision on where to go next in the courseware process.

7.4.3. Additional Guidance. See the *AFMC ISD Resource* Site.

7.4.4. Evaluation Criteria.

7.4.4.1. Do the ISD Evaluation Notes for Implementation Phase include:

7.4.4.1.1. Basic information about project, date of ISD review, and participants?

7.4.4.1.2. Status of outstanding action items from Development Phase or earlier Implementation Phase reviews?

7.4.4.1.3. Reasons typical Implementation Phase activities were skipped, (as applicable)?

7.4.4.1.4. Observations about compliance/non-compliance of Implementation Phase activities with the standards set forth in this manual, the *AFMC Depot Maintenance Training Resource* Site and the *AFMC ISD Resource* Site?

7.4.4.1.5. Explanations of waivers and adjustments of standards for Implementation Phase activities, (as applicable)?

7.4.4.1.6. Corrective actions which need to be accomplished with suspense dates, (as applicable)?

7.4.4.1.7. Implementation Phase review decision?

7.4.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

7.4.5.1. Nature of perceived issues?

7.4.5.2. Priority of assignment?

7.4.5.3. Appropriate organization to handle tasking request?

7.5. Localize Training Materials. In this activity, a courseware developer tailors a command or other standardized training course to meet local needs. AFMC command standard course learning objective behaviors, conditions of performance, and standards of performance may be increased during localization of training materials, but cannot be lessened in any way. Instructional and evaluation materials, including presentations, computer technologies, class exercises and student handouts, may be altered as long as all learning objectives and proficiency levels of the standard course are attained by the student at the completion of training.

7.5.1. Purpose. This activity allows installations, functional areas, work centers, etc., to make instruction more relevant to the target audience without altering learning objectives or proficiency levels of the course.

7.5.2. Activity Outcomes:

7.5.2.1. Identify and annotate local modifications to instructional and evaluation materials of a standardized course that may add or replace course content which does not affect or change course objectives identified in the POI.

7.5.2.2. Possible modifications to course control documents to reflect changes in instructional and evaluation materials.

7.5.2.3. Decision on where to go next in the courseware process.

7.5.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

7.5.4. Evaluation Criteria. N/A

7.5.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

7.5.5.1. Nature of perceived issues?

7.5.5.2. Priority of assignment?

7.5.5.3. Appropriate organization to handle the tasking request?

Chapter 8

EVALUATION PHASE

8.1. Evaluation Phase Defined. Evaluation is integrated throughout each activity of the instructional development process. It starts in the planning stage with development of an evaluation plan and continues for the life cycle of the training system. The focus of evaluation is continuous improvement in instructional system quality.

8.1.1. Purpose. As previously mentioned, evaluation is an ongoing process. It begins during ISD planning and continues as long as the instructional system is operational. The process includes formative, summative and operational evaluations. These forms of evaluation are described in order to acquaint the user of this manual with the continuous evaluation process.

8.1.2. Activity Outcomes:

8.2. Formative evaluation. Planning for formative evaluation begins in the initial planning stage of ISD. However, formative evaluation activities actually begin during analysis and continue through small-group tryout in development. Within each stage, Analysis, Design and Development formative evaluation seeks to improve the quality of the activities and products of ISD.

8.2.1. Purpose. Formative evaluation is a form of evaluation designed to collect data and information that is used to improve the activities and products of the ISD process while the system is still being developed. Formative evaluation is also used when the design or development phases are re-entered in order to revise or update the system.

8.2.2. Activity Outcomes. Formative evaluation may include any or all of the following activities, depending upon the community. For example, in the Education, Technical Training and Unit Training communities, only the first two activities, process and product evaluations, will likely be performed in the formative evaluation process. However, in the Aircrew Training and Acquisition communities, it may be necessary to perform all of the activities listed below in order to complete the formative evaluation process.

8.2.2.1. Process Evaluation. Process evaluation ensures quality in the Analysis, Design and Development activities. It checks each activity against standards, or metrics, established during ISD project planning, to assure process quality, while continually seeking improvements within each activity. Process evaluation enables instructional developers to form an effective and efficient instructional system based on quality principles.

8.2.2.2. Product Evaluation. Product evaluation is an integral part of each stage of the ISD process. Product evaluation focuses on the products of the Analysis, Design and Development activities such as task lists, objectives, tests, plans of instruction and training materials. During product evaluation the focus is again on quality. Products are measured against standards and metrics established in the planning stage of ISD to ensure quality. Product evaluation also helps form a total quality instructional system. Two activities of product evaluation are:

8.2.2.2.1. Validation. Takes place during training development and is the final activity in the formative evaluation process. Validation identifies improvements that should be

made to the instruction prior to implementing the system which may include conducting technical accuracy reviews, individual try-outs, small group try-outs and operational field try-outs.

8.2.2.2.2. Quality Control. Starts in the initial stages of ISD planning with the strategy of controlling quality and continues throughout training analysis, design and development. This process ensures that each activity such as equipment acquisition, facility construction, etc., is based on quality principles.

8.2.2.3. Developmental Test and Evaluation (DT&E). DT&E is an active part of training system development. As a formative evaluation activity, it is conducted to demonstrate that training system equipment design/development are complete, design risks have been minimized and the system meets performance requirements. It ensures the effectiveness of the manufacturing process, equipment and procedures.

8.2.2.4. Operational Test and Evaluation (OT&E). OT&E completes the formative evaluation process for training system equipment. This formative evaluation activity evaluates the system's operational effectiveness, maintainability, supportability and suitability. It identifies any operational and logistic support deficiencies and the need for modification. In addition, OT&E provides information on organizational structure, personnel requirements, support equipment, doctrine, training and tactics. It should also provide data to verify operating instructions, maintenance procedures, training programs, publications and handbooks.

8.2.2.5. Site Readiness Reviews (SRR). The site readiness review is a formative evaluation activity that focuses on evaluating the readiness of the "bed-down" site for the training system. This evaluation ensures that the site, including training facilities and support equipment, is ready for OT&E of the system. Site readiness reviews help ensure the training system effectiveness.

8.2.3. Evaluation Criteria. ISD Evaluation Checklist and/or ISD Management Plan

8.2.4. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

8.3. Summative Evaluation.

8.3.1. Summative evaluation is the next stage in the continuous evaluation process. This stage of evaluation involves trying out the instruction on the target population in an operational environment. In some organizations, summative evaluations are conducted after the instructional system becomes operational and includes two components: internal and external evaluation. Summative evaluation is focused on the period of operational tryouts. These tryouts begin after the small-group tryouts have been completed and continue until the instructional system is implemented. Normally, the operational tryout period is limited to two or three classes.

8.3.2. Summative evaluation is a form of evaluation designed to collect data and information during the operational (field) tryouts in order to determine the "summed" effect of the instruction under operational conditions and to make any changes or revisions to the system prior to becoming operational. Summative evaluations are also conducted when significant revisions or updates have been made to the instructional system.

8.3.3. Activity Outcomes. The only summative evaluation activity is the operational tryouts. Summative evaluations are conducted on fully integrated instructional systems. This form of evaluation is essential in determining the effectiveness of the system and correcting any deficiencies prior to implementation.

8.3.4. Evaluation Criteria.

8.3.4.1. Does the instructional system work under operational conditions?

8.3.4.2. Does the operational tryout provide feedback from a large sample of target population in which to base revisions prior to implementation of the instructional system?

8.3.4.3. Does the operational tryout identify possible implementation or operational problems?

8.3.4.4. Does the operational tryout determine if instruction is cost-efficient?

8.3.4.5. Does the operational tryout determine if instruction is adequate and needed?

8.3.5. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

8.4. Operational Evaluation.

8.4.1. Period. Once the formative and summative evaluation activities have been completed and the instructional system is implemented, operational evaluation begins. Operational evaluation continues as long as the system is operational.

8.4.2. Purpose. Operational evaluation is a form of evaluation designed to gather and analyze internal and external feedback data to ensure that the system continues to effectively and cost-efficiently produce graduates who meet established training requirements.

8.4.3. Activity Outcomes:

8.4.3.1. Internal evaluation. Focuses on evaluating the instructional system internally. This form of evaluation continuously evaluates feedback data such as instructor comments, student critiques and test results in order to continually improve the system and ensure quality.

8.4.3.2. External evaluation. Focuses on evaluating the instructional system externally. This form of evaluation continually evaluates feedback data from the field such as inspection and evaluation reports to ensure that graduates meet the established job performance requirements. Each operational evaluation activity contributes to the overall quality of the instructional system by ensuring.

8.4.4. Evaluation Criteria.

8.4.4.1. Does instructional system encourage and reward effective teaching practices on basis of student learning outcome?

8.4.4.2. Have outcomes, benefits, or final results most closely linked to the training been identified?

8.4.4.3. Do Course Critiques and Graduate Assessment Surveys provide sufficient feedback to determine strengths and weaknesses of training provided?

8.4.4.4. Does feedback provide sufficient information to assess changes in knowledge, skills, behaviors or attitudes?

8.4.4.5. Do evaluation activities provide sufficient information to reflect performance improvement, i.e., increased production efficiency, reduced waste, speed and quality of performance?

8.4.4.6. Do evaluation activities provide a thorough understanding of how effective training was, and how you can improve reaction, learning, behavior and results?

8.5. Conduct ISD Evaluation Review of Evaluation Phase Activities. In this activity, at least one ISD Evaluation official reviews activities performed during the Evaluation Phase to determine if they meet the intent of the AFMC Courseware Development and Management Process and its underlying principles of systems engineering, instructional design, and quality improvement. This is a required step in the Evaluation Phase.

8.5.1. Purpose. This activity ensures that the AFMC Courseware Development and Management Process is applied so that instruction has a high probability of being relevant, effective and economical to sustain.

8.5.2. Activity Outcomes:

8.5.2.1. Completion of the applicable section of AFMC ISD Gates Release Points Checklist. The current version of the checklist can be found on the *AFMC ISD Resource* Site. This checklist is used for Command Courses but may also be used for Local Course development. It provides a decision that the appropriate activities in the each gate have been accomplished to the standards of this manual, or that corrective actions are needed before the project goes into the next gate.

8.5.2.1.1. Completion of the New Development or Revision will include the completion of the AFMC Release Points Checklist for New Development and Revision found in the *AFMC ISD Resource* Site and will be filed in the ISD Notes folder of the Courseware Master File.

8.5.2.2. Decision that appropriate activities in the Evaluation Phase have been accomplished to the standards of this manual or that corrective actions are needed before the project is closed out.

8.5.2.3. Possible tasking worksheets on training issues, courseware issues, training gaps or non-training issues submitted to appropriate organization for action.

8.5.2.4. Decision on where to go next in the courseware process.

8.5.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

8.5.4. Evaluation Criteria.

8.5.4.1. Do the ISD Evaluation Notes for Evaluation Phase include:

8.5.4.1.1. Basic information about project, date of ISD review, and participants?

8.5.4.1.2. Status of outstanding action items from Development Phase or earlier Evaluation Phase reviews?

8.5.4.1.3. Reasons typical Evaluation Phase activities were skipped (as applicable)?

8.5.4.1.4. Observations about compliance/non-compliance of Evaluation Phase activities with the standards set forth in this manual and the *AFMC Depot Maintenance Training Resource* Site?

8.5.4.1.5. Explanations of waivers and adjustments of standards for Evaluation Phase activities (as applicable)?

8.5.4.1.6. Corrective actions which need to be accomplished with suspense dates (as applicable)?

8.5.4.1.7. Evaluation Phase review decision?

8.5.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

8.5.5.1. Nature of perceived issues?

8.5.5.2. Priority of assignment?

8.5.5.3. Appropriate organization to handle tasking request?

8.5.6. Update ISD Management Plan as required.

Chapter 9

FORMAL COURSEWARE RE-EVALUATION (PERIODIC REVIEW)

9.1. Formal Courseware Re-Evaluation Defined. A "periodic review" of an existing course or learning aid is conducted by a training specialist. Experts are interviewed and courseware is evaluated to determine what revisions are needed to make instruction accurate, effective, and compliant with applicable directives. If only minor corrections are needed, course control documents are updated, any materials needing changes are quickly corrected and courseware is revalidated. If more analysis or major revision is needed, activities of the follow-on revision project are defined during this process.

9.1.1. Differences between Periodic Review, Interim Change and Pen-and-Ink Change:

9.1.1.1. Periodic review is a well-documented evaluation of courseware by a team of experts who determine the overall health of course material. It identifies if a course is still required, if content is still accurate or if changes are needed to update content to make the training more effective.

9.1.1.2. Interim Change is an official correction made to a course not requiring formal revalidations or revisions. An Interim Change is documented in the courseware master file with a courseware issue worksheet and all instructors/trainers are advised if it impacts the delivery of a course.

9.1.1.3. Pen-and-ink change is a minor correction made to courseware by an instructor or trainer without updating the courseware master file.

9.1.1.4. Neither an Interim Change nor a pen-and-ink change fulfills the mandatory requirement for Periodic Review.

9.1.2. Directives Requiring Periodic Courseware Review. AFI 36-2650_AFMCSUP, requires assigned Command training and associated courseware be reviewed at least triennially for effectiveness in meeting the instructional need. Certain safety-related courses require more frequent review in accordance with safety regulations. For standardization establish local policy to align with AFI 36-2650 AFMCSUP for non-Command courseware review. Courses extensions must be approved by HQ AFMC/A4PT.

9.1.2.1. Completion of the Re-Evaluation will include the completion of the Re-Evaluation Release Points Checklist found in the *AFMC ISD Resource* Site and will be filed in the ISD Notes folder of the Courseware Master File. This checklist is used for Command Courses but may also be used for Local Course requirements.

9.2. Conduct Preliminary Research on Courseware Status. In this activity, a training specialist conducts research to determine if the course or learning aid is still needed.

9.2.1. Purpose. This activity saves resources by deciding if courseware is a candidate for archival before resources are spent gathering materials and reviewing the design and content of the instruction.

9.2.2. Activity Outcomes:

9.2.2.1. Decision to archive the courseware or continue with the courseware Re-Evaluation (periodic review) (as applicable).

9.2.2.2. Initiate updated AFMC Form 853 to Complex CPM.

9.2.2.3. Documentation on the status of courseware sufficient to justify decision to archive or continue with periodic review.

9.2.2.4. Possible tasking worksheets on training issues, courseware issues, training gaps or non-training issues submitted to appropriate organization for action.

9.2.2.5. Decision on where to go next in the courseware process.

9.2.3. Additional Guidance. See the *AFMC ISD Resource* Site.

9.2.4. Evaluation Criteria.

9.2.4.1. Did the evaluator:

9.2.4.1.1. Obtain recent student completion data on the course?

9.2.4.1.2. Search TSS and/or other similar functionally-specific system, to verify which organizations still require the course?

9.2.4.2. Was documentation on research activities and status of courseware sufficient to support the recommendation to archive the course or continue with the periodic review?

9.2.5. Conduct ISD Evaluation Review of Preliminary Research on Courseware Activities. In this activity, at least one ISD Evaluation official reviews activities performed during the Preliminary Research to determine if they meet the intent of the AFMC Courseware Development and Management Process and its underlying principles of systems engineering, instructional design and quality improvement. This is a required step in the Preliminary Research on Courseware Activities.

9.2.5.1. Purpose. This activity ensures that the AFMC Courseware Development and Management Process is applied so that instruction has a high probability of being relevant, effective and economical to sustain.

9.2.5.2. Activity Outcomes:

9.2.5.2.1. Completion of the " Re-Evaluation Release Points Checklist.

9.2.5.2.2. Decision that appropriate activities in the Preliminary Research have been accomplished to the standards of this manual or that corrective actions are needed before the project is closed out.

9.2.5.2.3. Possible tasking worksheets on training issues, courseware issues, training gaps or non-training issues submitted to appropriate organization for action.

9.2.5.2.4. Decision on where to go next in the courseware process.

9.2.5.2.5. Creation of Revision Working Folder and DMTL Working Folder for Command Courses.

9.2.5.3. Additional Guidance. See the AFMC ISD Resource Site.

9.2.5.4. Evaluation Criteria.

9.2.5.4.1. Do the ISD Evaluation Notes for Preliminary Research include:

9.2.5.4.2. AFMC Form 853 forwarded to Complex CPM?

9.2.5.4.3. Re-Evaluation Project created?

9.2.5.4.4. Validate the CMD Data?

9.2.5.4.5. Research mandate?

9.2.5.4.6. Archive assessment?

9.2.5.4.6.1. Complete PR process

9.2.5.4.6.2. Refer to AFI 36-2650_AFMCSUP for archive guidance.

9.2.5.4.7. Research of training worksheets?

9.2.5.4.8. Preliminary Research review decision?

9.2.6. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

9.2.6.1. Nature of perceived issues?

9.2.6.2. Priority of assignment?

9.2.6.3. Appropriate organization to handle tasking request?

9.3. Gather Materials and Assemble Review Team. In this activity, a courseware evaluator gathers materials needed to conduct a formal courseware Re-Evaluation and invites experts to participate in the review.

9.3.1. Purpose. This activity sets the stage for an efficient periodic review of a course or learning aid.

9.3.2. Activity Outcomes:

9.3.2.1. All available instructional and evaluation materials and supporting documentation on the course or learning aid.

9.3.2.2. Identification of SMEs who have agreed to participate in the courseware review.

9.3.2.3. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

9.3.2.4. Decision on where to go next in the courseware process.

9.3.3. Additional Guidance. See the *AFMC ISD Resource* Site.

9.3.4. Evaluation Criteria.

9.3.4.1. Did the evaluator gather:

9.3.4.1.1. The official courseware master file from the appropriate installation courseware library?

9.3.4.1.2. Results of Course Critiques?

9.3.4.1.3. Results of Graduate Assessment Surveys?

9.3.4.1.4. Personalized materials, notes, and pen-and-ink changes from instructors/trainers in organizations delivering the course?

9.3.4.1.5. Any recent correspondence related to the course from the functional area and/or Education and Training Office?

9.3.4.2. Did the evaluator contact:

9.3.4.2.1. The functional area home office or quality function to request inspection and staff assistance findings, recent and proposed changes in regulations and policies, process improvement initiatives, performance issues, etc., related to the subject matter of the course?

9.3.4.2.2. All authorized instructors/trainers of the course to request their participation in the review?

9.3.4.2.3. Training managers from organizations where the target population resides to request their participation in the course review?

9.3.4.2.4. System OPRs to request their participation in the course review, if this course involves an automated information system (e.g., Depot Maintenance Accounting Production System, Stock Control System, etc.)?

9.3.4.2.5. Subject area program managers (e.g., safety, security, personnel, acquisition, finance, quality assurance, etc.) for primary topics included in the course to request their participation in the review?

9.3.4.2.6. SMEs from work centers in the target population to request their participation in the course review?

9.3.4.2.7. Experts from all complexes to request their participation in the review if this is a command course?

9.3.4.2.8. An ISD evaluator to request their assistance in reviewing the course or learning aid for ISD compliance?

9.3.5. Conduct ISD Evaluation Review of Gather Materials and Assemble Review Team activities. In this activity, at least one ISD Evaluation official reviews activities performed during Gather Materials and Assemble Review Team activities to determine if they meet the intent of the AFMC Courseware Development and Management Process and its underlying principles of systems engineering, instructional design and quality improvement. This is a required step in the Gather Materials and Assemble Review Team activities.

9.3.5.1. Purpose. This activity ensures that the AFMC Courseware Development and Management Process is applied so that instruction has a high probability of being relevant, effective and economical to sustain.

9.3.5.2. Activity Outcomes:

9.3.5.2.1. Completion of the Re-Evaluation Release Points Checklist.

9.3.5.2.2. Decision that appropriate activities in the Gather Materials and Assemble Review Team activities have been accomplished to the standards of this manual or that corrective actions are needed before the project is closed out.

9.3.5.2.3. Possible tasking worksheets on training issues, courseware issues, training gaps or non-training issues submitted to appropriate organization for action.

9.3.5.2.4. Decision on where to go next in the courseware process.

9.3.5.3. Additional Guidance. *AFMC ISD Resource Site*.

9.3.5.4. Evaluation Criteria.

9.3.5.4.1. Do the ISD Evaluation Notes for Gather Materials and Assemble Review Team activities include:

9.3.5.4.2. Gather Completion Report?

9.3.5.4.3. Gather current material?

9.3.5.4.4. Request SME(s)/Instructor(s)?

9.3.5.4.5. Review worksheets?

9.3.5.4.6. Review personalization, pen-and-ink changes and instructor notes?

9.3.5.4.7. Gather Materials and Assemble Review Team activities review decision?

9.3.6. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

9.3.6.1. Nature of perceived issues?

9.3.6.2. Priority of assignment?

9.3.6.3. Appropriate organization to handle tasking request?

9.4. Perform Re-Evaluation of Courseware. In this activity, a review team will review the content of instructional and evaluation materials to determine if they are still relevant, accurate and complete. The team will also address appearance and design of existing training to determine if it is effective, efficient and adequately documented.

9.4.1. Purpose. This activity calls on SMEs, subject area program managers, system OPRs, training managers, instructors and other customer representatives to verify instruction is relevant to what actually occurs in the workplace of the target population. Ensure content is technically accurate and complete.

9.4.2. Activity Outcomes:

9.4.2.1. Input from reviewers on content, appearance and design of the materials.

9.4.2.2. Documentation on changes required to content, appearance and design of instruction to make it more effective.

9.4.2.3. Possible tasking worksheets on training issues, courseware issues, training gaps or non-training issues submitted to appropriate organization for action.

9.4.2.4. Decision on where to go next in the courseware process.

9.4.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site and the *AFMC ISD Resource* Site.

9.4.4. Evaluation Criteria.

9.4.4.1. Did appropriate SMEs, subject area PMs, system OPRs, TMs and other customer representatives review the materials?

9.4.4.2. Did the review team consider the following when evaluating the design of the course:

9.4.4.2.1. Is the sequence of content and course activities effective?

9.4.4.2.2. Do learning objectives contain conditions of performance and standards of behavior that are closely related to actual conditions and performance standards in the work environment?

9.4.4.2.3. Do media and delivery method(s) of the instruction appear to be appropriate for the situation?

9.4.4.3. Did the review team consider the following when evaluating the instructional materials (e.g., presentation, SOJT Guide, student handouts, etc.):

9.4.4.3.1. Are there sufficient instructional materials and activities (e.g. examples, non-examples, illustrations, diagrams, demonstrations, scenarios, case studies, etc.) to support the learning objectives?

9.4.4.3.2. Are examples, non-examples, illustrations, diagrams, demonstrations, scenarios, case studies and other instructional materials and activities relevant to the target audience?

9.4.4.3.3. Is content of instructional materials accurate and complete?

9.4.4.4. Did the review team consider the following when assessing the student evaluation materials (e.g., progress checks, written tests, proficiency evaluations, etc.):

9.4.4.4.1. Do amount and types of evaluation provide a high level of confidence that students have attained the proficiency level of learning specified in learning objectives?

9.4.4.4.2. Do evaluation activities emulate the conditions and standards of the work environment as much as is feasible?

9.4.4.4.3. Is content of evaluation materials accurate and complete?

9.4.4.4.4. Is content and minimum passing score for pass/fail tests consistent with training mandates?

9.4.4.5. Did a courseware evaluator document recommended changes and other input from reviewers?

9.4.4.6. Did appropriate instructor(s)/trainer(s) and courseware design specialist(s) review the course (as applicable)?

9.4.4.7. During the course design evaluation, did the review team identify any necessary changes to the structure or sequence of the training?

9.4.4.7.1. Does design contain:

9.4.4.7.1.1. One or more criterion-referenced learning objectives?

9.4.4.7.1.2. When evaluating design of a course, did review team determine what changes need to be made to learning objectives so they meet the following evaluation criteria**?**

9.4.4.7.1.3. Does each learning objective contain:

9.4.4.7.1.4. One behavior (knowledge or task)?

9.4.4.7.1.5. A condition of performance that is closely related to actual conditions in the work environment?

9.4.4.7.1.6. A standard of behavior that is criterion-referenced and closely related to actual standards required in the work environment?

9.4.4.7.1.7. Do knowledge and tasks included in learning objectives coincide directly with task statements approved by the customer?

9.4.4.7.1.8. Are proficiency levels for learning objectives appropriate for instructional requirements?

9.4.4.7.1.9. Do objectives use types of learning appropriate for the situation?

9.4.4.7.1.10. Did changes to topics, tasks and proficiency levels occur? If so, have these changes been approved by the customer and documented in the Instructional Strategy?

9.4.4.7.2. Does course material/training aid contain:

9.4.4.7.2.1. Presentations, demonstrations, simulations and exhibits used to support each learning objective?

9.4.4.7.2.2. Student projects, exercises, review sessions, quizzes, tests and proficiency evaluations that will be used to support each learning objective?

9.4.4.7.2.3. A clear description of screen design and student interactivity features?

9.4.4.7.2.4. An overview of the content that will support each learning objective?

9.4.4.7.2.5. Media that will be used for each instructional and student evaluation activity?

9.4.4.7.2.6. Show the sequence in which instruction will be delivered to students?

9.4.4.7.2.7. Confirm that the selected delivery method is appropriate for the situation?

9.4.4.7.2.8. Propose instructional and student evaluation events that are appropriate for the situation and support learning objectives?

9.4.4.7.2.9. Relevant examples and non-examples in the form of illustrations, diagrams, demonstrations, scenarios, case studies, etc. Will be used to apply instructional concepts?

9.4.4.7.2.10. Support tasks, conditions and standards of the work environment as much as is feasible?

9.4.4.7.2.11. A clear description of facilities, equipment, tools and materials used to deliver the training?

9.4.4.7.3. Did this activity impact the Instructional Strategy and, if so, have these changes been documented, justified and approved by the customer and SME?

9.4.4.7.4. Additional areas to consider for DL:

9.4.4.7.4.1. Are storyboards and supporting documentation detailed enough for reviewers to get a clear picture of precisely what content will be presented, how it will be instructed and how students will be evaluated for attainment of each learning objective?

9.4.4.7.4.2. Do storyboards and supporting documentation describe (if any) pre-assessment or bypass tests that could be used to determine which modules will be required?

9.4.4.7.4.3. Is training SCORM compliant with LMS?

9.4.4.7.4.4. Learning development methods employed (job sequence, psychological sequence, logical order sequence, branching scenario, etc.).

9.4.4.7.4.5. Identify Section 508 Compliance requirements for the target audience. **https://www.section508.gov/manage**

9.4.4.7.4.6. Identify LMS SCORM level requirements for DL development.

9.4.4.7.5. Coding methods or Air Force approved software for SCORM publishing to an LMS used in course development?

9.4.4.7.6. A clear description of screen design and student interactivity features?

9.4.4.7.7. Does SCO contain:

9.4.4.7.7.1. SCO level based on LMS SCORM version?

9.4.4.7.7.2. SCO-calls in storyboard/outline/checklist?

9.4.4.8. Is there enough navigational guidance for a student to easily take the course and any bypass pre-test that may be available?

9.4.4.9. Did this activity impact the Instructional Strategy and, if so, have these changes been documented, justified and approved by the customer and SME?

9.4.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

9.4.5.1. Nature of perceived issues?

9.4.5.2. Priority of assignment?

9.4.5.3. Appropriate organization to handle tasking request?

9.4.6. Conduct ISD Evaluation Review of Re-Evaluation of Courseware activities. In this activity, at least one ISD Evaluation official reviews activities performed during Re-Evaluation of Courseware activities to determine if they meet the intent of the AFMC Courseware Development and Management Process and its underlying principles of systems engineering, instructional design and quality improvement. This is a required step in the Re-Evaluation of Courseware activities.

9.4.6.1. Purpose. This activity ensures that the AFMC Courseware Development and Management Process is applied so that instruction has a high probability of being relevant, effective and economical to sustain.

9.4.6.2. Activity Outcomes:

9.4.6.2.1. Completion of the "Re-Evaluation of Courseware Activities" section of the Re-Evaluation Release Points Checklist.

9.4.6.2.2. Decision that appropriate activities in the Re-Evaluation of Courseware activities have been accomplished to the standards of this manual or that corrective actions are needed before the project is closed out.

9.4.6.2.3. Possible tasking worksheets on training issues, courseware issues, training gaps or non-training issues submitted to appropriate organization for action.

9.4.6.2.4. Decision on where to go next in the courseware process.

9.4.6.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site and the *AFMC ISD Resource* Site.

9.4.6.4. Evaluation Criteria.

9.4.6.4.1. Do the ISD Evaluation Notes for Re-Evaluation of Courseware activities include:

9.4.6.4.2. Instructor review?

9.4.6.4.3. SME technical review?

9.4.6.4.4. Review course design?

9.4.6.4.5. Document results if needed?

9.4.6.4.6. Define revision activity if needed?

9.4.6.4.7. Courseware issue worksheet if needed?

9.4.6.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

9.4.6.5.1. Nature of perceived issues?

9.4.6.5.2. Priority of assignment?

9.4.6.5.3. Appropriate organization to handle tasking request?

9.5. Document Courseware Re-Evaluation Results and Recommendations. In this activity, the results of research and review activities performed thus far in support of courseware Re-Evaluation are documented.

9.5.1. Purpose. This activity organizes findings and recommendations of the courseware evaluator and review team so sound decisions can be made to close out periodic review with course control documents update or initiate a revision project.

9.5.2. Activity Outcomes:

9.5.2.1. Closeout periodic review and update course control documents.

9.5.2.2. Closeout periodic review with a decision on where follow-on project should re-enter courseware development and management process.

9.5.2.3. Possible tasking worksheets on training issues, courseware issues, training gaps or non-training issues submitted to appropriate organization for action.

9.5.2.4. Re-Evaluation ISD Checklist documentation on courseware Re-Evaluation activities.

9.5.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site, the *AFMC ISD Resource Site* or contact the Complex CPM.

9.5.4. Evaluation Criteria.

9.5.4.1. Does documentation of courseware Re-Evaluation include:

9.5.4.1.1. Revision of Course Control Documents (as needed)?

9.5.4.1.2. SME review and concurrence of updated Course Control Documents (as needed)?

9.5.4.1.3. Periodic Review Closeout Worksheet?

9.5.4.1.4. Periodic Review Signature Sheet?

9.5.4.1.5. CMD Course and PR Project updated?

9.5.4.1.6. If periodic review process results in a revision of courseware; decide where revision project should re-enter the ISD process.

9.5.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

9.5.5.1. Nature of perceived issues?

9.5.5.2. Priority of assignment?

9.5.5.3. Appropriate organization to handle tasking request?

Chapter 10

COURSEWARE MANAGEMENT FUNCTION

10.1. Courseware Management Function Defined. The Courseware Management Function is an extension of the systems approach that places the ISD process within the functional design of a total instructional system. The basic top-level system functions of the instructional system involves management, support, administration and delivery of courseware through policies, programs, standardized procedures and working aids within the sphere of quality improvement. Quality Improvement (QI) is the continuous, organized creation of beneficial change to the system. The objective of quality improvement is to foster continuous improvement in the products and processes through a constraints based approach. Using these essential functions to design the overall instructional system architecture and then allocating them to the respective instructional system components or people responsible, ensures these functions are operational when the total training system is fielded. ISD products are integrated into the total instructional system and aspects of the system functions are active throughout all phases of the ISD process. **Note:** Courseware development varies between the ALCs based on how the organization is structured. There may be variations of the ISD process as needed.

10.1.1. The system functions of the ISD model are as follows:

10.1.1.1. Management - the function of directing or controlling ISD and operations through AoP principles, practices and techniques.

10.1.1.2. Support - the function of maintaining all parts of the system.

10.1.1.3. Administration - the function of day-to-day processing and record keeping

10.1.1.4. Delivery - the function of bringing instruction to students.

10.1.2. It includes the following types of activities in support of courseware development and sustainment:

10.1.2.1. Establishing courseware policies, programs, procedures and working aids.

10.1.2.2. Planning, obtaining, distributing and managing resources.

10.1.2.3. Promoting standardization and re-use of instructional materials.

10.1.2.4. Monitoring, assessing and reporting on the health of courseware.

10.1.2.5. Implementing quality improvement measures.

10.1.2.6. Controlling Interim Changes.

10.1.2.7. Distribution of instructional materials.

10.2. Establish Local Courseware Policies and Tools (as needed). Maintenance Training Section offices at all levels are responsible for establishing their own policies, procedures and tools, (as needed), to support courseware activities within their span of control.

10.2.1. Purpose. This activity efficiently implements the AFMC Courseware Development and Management Process at each installation so that maximum benefits can be realized.

10.2.2. Activity Outcomes:

10.2.2.1. Local supplements and operating instructions as desired.

10.2.2.2. Local guidance, worksheets, forms, templates, quality checklists, samples and other working aids related to courseware development and management.

10.2.2.3. Possible tasking worksheets on training issues, courseware issues, training gaps or non-training issues submitted to appropriate organization for action.

10.2.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

10.2.3.1. Links to existing directives related to courseware.

10.2.4. Evaluation Criteria.

10.2.4.1. Do policies, procedures or tools established contradict this manual or any other equal or higher level directive?

10.2.4.2. Have policies, procedures, or tools been posted for easy access by courseware process participants?

10.2.4.3. Depending on local procedures, were proposed policies, procedures or tools in support of this activity reviewed by a courseware management supervisor, Complex CPM or ISD evaluation official to verify they meet the requirements of this manual and other applicable directives?

10.3. Promote Standardization of Instruction. Training offices at all levels are responsible for promoting the standardization of instructional requirements and materials, to include use of command and installation standard courseware whenever prescribed or practical, creating and adopting RLOs and reusable learning objects whenever feasible, and sharing information on planned and in-work courseware activities with other organizations and agencies.

10.3.1. Purpose. This activity promotes consistency of instruction across organizations and installations, which saves resources by preventing duplicate courseware and capitalizing on existing materials.

10.3.2. Activity Outcomes:

10.3.2.1. Use of command and installation standard courseware whenever prescribed or practical.

10.3.2.2. Sharing of information on planned and in-work courseware activities with other organizations and agencies.

10.3.2.3. Policies, procedures and guidance on standardization of instruction (as needed).

10.3.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

10.3.4. Evaluation Criteria.

10.3.4.1. Do any established policies, procedures, or tools contradict this document or any other equal or higher level directive?

10.3.4.2. Have policies, procedures, or tools been posted for easy access by courseware process participants?

10.3.4.3. Depending on local procedures, were results of this activity reviewed by a courseware management supervisor, Complex CPM or ISD Evaluation official to verify

that standardization of instruction activities meet requirements in this manual and other AFMC directives?

10.4. Establish Formal Courseware Re-Evaluation Schedule. HQ AFMC/A4PT in conjunction with Complex training offices are responsible for establishing their own schedule to manage periodic reviews and learning aids for which they are responsible.

10.4.1. Purpose. This activity ensures that courseware is re-evaluated as prescribed by applicable directives.

10.4.2. Activity Outcomes:

10.4.2.1. Thirty-Sixty-Ninety Day reports or local management tools and guidance for managing periodic courseware reviews.

10.4.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

10.4.4. Evaluation Criteria.

10.4.4.1. Do established procedures, guidance or tools contradict this manual or any other equal or higher level directive?

10.4.4.2. Have program management procedures, guidance and tools been posted for easy access by courseware process participants?

10.4.4.3. Depending on local procedures, were the results of this activity reviewed by a courseware management supervisor, Complex CPM or ISD Evaluation official to verify proposed courseware Re-Evaluation program management meets the intent of this document?

10.5. Maintain Depot Maintenance Training Library (DMTL). HQ AFMC/A4PT maintains oversight of the command DMTL. Complex offices are responsible for filing and maintaining the DMTL for courseware they are responsible for, which contains (at a minimum) copies of all formal functional courseware and learning aids used at the installation. AFMC command standard courses shall be included. Course data not compatible with library storage mandates will be maintained at the lead complex. Localized courseware may be maintained as locally determined.

10.5.1. Purpose. This activity maintains an inventory of courseware files used at an installation and enables most courseware to be searched, distributed and managed electronically. The Complex CPM can provide additional course information as necessary.

10.5.2. Activity Outcomes:

10.5.2.1. Official courseware master files loaded (as applicable).

10.5.2.2. Command procedures, guidance and working aids for managing the command DMTL.

10.5.3. Additional Guidance. Does the course chart meet the criteria in the quality checklist at the *AFMC Depot Maintenance Training Resource* Site for master file template, instructions and quality checklist? See Attachments.

10.5.4. Evaluation Criteria.

10.5.4.1. Do the official courseware master files posted to the appropriate libraries comply with the structure and content guidance at the *AFMC Depot Maintenance Training Resource* Site?

10.5.4.2. Are there any local procedures, guidance or tools that contradict this manual or any other equal or higher level directive?

10.5.4.3. Were the results of this activity reviewed by a courseware management supervisor or Complex CPM to verify the DMTL or local libraries were updated properly as required?

10.5.5. Prepare Preliminary Courseware Master File For Review. In this activity, instructional materials, evaluation materials, course control documents and all supporting documentation from the courseware project are organized into a master file for review by the Complex CPM.

10.5.5.1. Purpose. This activity allows the CPM to determine if appropriate documents are loaded in the master file before it is distributed to instructors/trainers or loaded in the courseware master library.

10.5.5.2. Activity Outcomes:

10.5.5.2.1. Preliminary courseware master file with all documentation.

10.5.5.2.2. Decision on where to go next in the courseware process.

10.5.5.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

10.5.5.4. Evaluation Criteria.

10.5.5.4.1. Does preliminary courseware master file include all appropriate documentation for the courseware that could be useful for implementation, reference or audit purposes and ISD Evaluation Implementation Phase Notes?

10.5.5.4.2. Does preliminary courseware master file comply with the file structure and content guidance with the *AFMC Depot Maintenance Training Resource* Site?

10.5.5.4.3. Does each document in the courseware master file have a date on it for configuration management, reference and audit purposes? (This is an International Standards Organization 9000, AS9100, etc., requirement.)

10.5.6. Finalize Official Courseware Master File. In this activity, instructional materials, evaluation materials, course control documents and all supporting documentation from the courseware project are reviewed by Complex CPM.

10.5.6.1. Purpose. This activity prepares the official courseware master file for loading into the appropriate installation functional courseware library and distribution for use in the field.

10.5.6.2. Activity Outcomes:

10.5.6.2.1. Official courseware master file which includes all documentation associated with the courseware.

10.5.6.2.2. Decision on where to go next in the courseware process.

10.5.6.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

10.5.6.4. Evaluation Criteria.

10.5.6.4.1. Does the official courseware master file comply with the structure and content guidance at the *AFMC Depot Maintenance Training Resource* Site?

10.5.6.4.2. Does the official courseware master file include all appropriate documentation for the courseware useful for implementation, reference or audit purposes?

10.5.6.4.3. Are documents in the courseware master file dated appropriately for configuration management, reference and audit purposes?

10.5.7. Load Official Courseware Master File in Appropriate Library. In this activity, the official master file, which contains instructional materials, evaluation materials, course control documents and all supporting documentation from the courseware project, is loaded into the appropriate installation functional courseware master library.

10.5.7.1. Purpose. This activity adds the courseware master file to the appropriate electronic repositories for reference, audit and configuration control purposes.

10.5.7.2. Activity Outcomes:

10.5.7.2.1. Official courseware master file loaded in appropriate courseware libraries.

10.5.7.2.2. Decision on where to go next in the courseware process.

10.5.7.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

10.5.7.4. Evaluation Criteria.

10.5.7.4.1. Does the official courseware master file submitted for posting to an official courseware library comply with the structure and content guidance at *the AFMC Depot Maintenance Training Resource* Site?

10.5.7.4.2. Has the official courseware master file been posted in at least two locations, otherwise backed-up to prevent loss of official data?

10.6. Insert Course Into Instructional System.

10.6.1. Purpose. CPM notifies Training Managers, Production Acceptance Certification (PAC) Managers, Civilian Training Plan (CTP) Managers, Course Managers and Instructors that the course has been added to the CMD or equivalent and is ready for utilization.

10.6.2. Activity Outcomes:

10.6.2.1. Command/Complex Notification Email.

10.6.2.2. Contacts are made to show course is active.

10.6.2.3. Course added to survey for scheduling.

10.6.2.4. Training managers identify applicable series for inclusion on work center course catalog.

10.6.2.5. PAC Managers identify applicable PAC records for inclusion.

10.6.2.6. CTP Managers identify applicable CTPs for inclusion.

10.6.2.7. Course Manager verifies course is active in CMD or equivalent.

10.6.2.8. Instructors are made aware course can be scheduled (as applicable).

10.6.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

10.6.4. Evaluation Criteria.

10.6.4.1. Is Command/Complex Notification Report complete?

10.6.4.2. Is course on the scheduling survey or loaded on official LMS?

10.7. Updated Course Information in the Courseware Management Database (CMD) or Equivalent.

10.7.1. Purpose. To update the master courseware database to reflect current information and course status.

10.7.2. Activity Outcomes:

10.7.2.1. Course information is accurately reflected in the Training Scheduling System (TSS)/CMD or equivalent.

10.7.3. Additional Guidance. Contact the Complex CPM.

10.7.4. Evaluation Criteria.

10.7.4.1. Is basic information in database current?

10.7.4.2. Are dates in database fields accurate?

10.7.4.3. Have Complex CPM and Course Manager made updates (as needed)?

10.7.4.4. Have relevant links to courseware documents been added (as necessary)?

10.8. Distribute Official Courseware Master File for Use in Field (as needed). In this activity, the official master file which contains instructional materials, evaluation materials, course control documents and all supporting documentation from the courseware project is distributed to all authorized instructors/trainers. Self-paced, computer based instruction is sent to organizations that will host the instruction on a command approved LMS, courseware database and content library.

10.8.1. Purpose. This activity provides for distribution of the most current version of instruction material that cannot be loaded to an electronic resource.

10.8.2. Activity Outcomes:

10.8.2.1. Official courseware master file distributed to those implementing it in the field.

10.8.2.2. Decision on where to go next in the courseware process.

10.8.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

10.8.4. Evaluation Criteria.

10.8.4.1. Have applicable Group Training and PAC managers been notified of release?

10.9. Personalize Course Materials (as necessary).

10.9.1. Purpose. To provide local information to command level course material to meet local training requirements.

10.9.2. Activity Outcomes:

10.9.2.1. Personalization Document.

10.9.2.2. Personalized entries are identified by document and location.

10.9.2.3. Local directives are included in training material as required.

10.9.3. Additional Guidance. See the *AFMC ISD Resource* Site.

10.9.4. Evaluation Criteria.

10.9.4.1. Does material meet the needs of the local maintenance community?

10.9.4.2. Are all course objectives kept? **Note:** No objectives can be modified.

10.9.4.3. Are new local course objectives added to the appropriate course control documents?

10.10. Budget for Courseware Sustainment Activities. In this activity, Complex CPM and project managers estimate resources needed to sustain a course or learning aid based on historical factors, courseware Re-Evaluation and revision formulas and input from technical experts.

10.10.1. Purpose. This activity allows organizations to plan for manpower/funding needed to accomplish formal courseware Re-Evaluation and revision tasks.

10.10.2. Activity Outcomes:

10.10.2.1. Fiscal Year Budget Plan.

10.10.2.2. Possible tasking worksheets on training issues, courseware issues, training gaps, non- training issues, etc., submitted to appropriate organization for action.

10.10.3. Additional Guidance. Contact Complex CPM for information on formulating budget.

10.10.4. Evaluation Criteria.

10.10.4.1. Does courseware sustainment budget include:

10.10.4.1.1. Estimate of hours needed to accomplish anticipated courseware formal Re-Evaluation and revision tasks in a worst case scenario (i.e., inexperienced organic courseware developer performing the work)?

10.10.4.1.2. Identification of months when formal courseware Re-Evaluation and revision will be due in accordance with directives?

10.10.4.1.3. Explanation of basic formula and adjustment factors used for determining the ROM of resources needed for sustainment of the course or learning aid?

10.10.4.2. Was the resource estimate prepared or validated by a courseware manager or developer experienced in determining rough order of magnitude of resources needed for Air Force courseware activities and projects?

10.10.4.3. Were the adjustment factors for courseware sustainment estimate based on courseware master file references and discussions with instructors/SMEs (as applicable), rather than on assumptions and incomplete data?

10.10.4.4. Were the results of this activity reviewed by the Complex CPM to verify that courseware sustainment budget documentation meets the guidance in this manual and local guidance on budget requirements? If contracted courseware development/revision is considered, the Contracting Officer Representative (COR) must be contacted for further guidance.

10.10.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

10.10.5.1. Nature of perceived issues?

10.10.5.2. Priority of assignment?

10.10.5.3. Appropriate organization to handle tasking request?

10.11. Distribute Courseware to External Customers Upon Request. In this activity, items from the official master file are released to authorized individuals upon request.

10.11.1. Purpose. This activity distributes the most current version of courseware to individuals in the federal government who wish to review or adopt it for use at their installations. Individuals outside of the federal government who are not developing instruction for government use may only be given courseware with approval of HQ AFMC Courseware Manager.

10.11.2. Activity Outcomes:

10.11.2.1. Courseware distributed to authorized individual.

10.11.2.2. Documentation in courseware master file or courseware management system of individuals and organizations receiving a copy of courseware instructional and evaluation materials, not only course control documents.

10.11.3. Additional Guidance. Contact the Complex CPM.

10.11.4. Evaluation Criteria.

10.11.4.1. Has the master file been made available to all authorized individuals, including interim changes and issue worksheets?

10.12. Process Interim Change (IC). In this activity, a significant change to a course or learning aid occurring between official re-evaluation or revision of courseware is distributed to all courseware users for immediate implementation.

10.12.1. Purpose. This activity allows corrections and updates to courseware between formal re-evaluation or revision. It tracks corrections and modifications to instruction to ensure instructors/trainers are using the most recent version of materials.

10.12.2. Activity Outcomes:

10.12.2.1. Interim Change Memo (ICM) distributed to those who use the courseware in the field.

10.12.2.2. Decision on where to go next in the courseware process.

10.12.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

10.12.3.1. Various tasking worksheets, instructions and examples.

10.12.4. Evaluation Criteria.

10.12.4.1. Does ICM content and format comply with the quality checklist in *AFMC Depot Maintenance Training Resource* Site?

10.12.4.2. Has ICM been added to the courseware master file in all appropriate installation courseware master libraries, and in any appropriate command-approved functional training management systems, such as the CMD?

10.12.4.3. Have recommended and approved changes been accomplished?

10.12.5. Issues/actions: Are any issues/action items which emerged during this activity sufficiently documented to determine:

10.12.5.1. Nature of perceived issues?

10.12.5.2. Priority of assignment?

10.12.5.3. Appropriate organization to handle tasking request?

10.13. Deactivate Unteachable Course. In this activity, a course no longer teachable or a learning aid no longer usable is temporarily deactivated pending revision or replacement.

10.13.1. Purpose. This activity removes a course or learning aid from the list of available instruction to ensure students are not scheduled for training which is outdated or awaiting revision.

10.13.2. Activity Outcomes:

10.13.2.1. Indicate “temporarily inactive” status in appropriate command-approved functional training management system (i.e., TSS/CMD)?

10.13.2.2. Possible tasking worksheets on training issues, courseware issues, training gaps, or non-training issues submitted to appropriate organization for action.

10.13.3. Additional Guidance. See CMD User’s Guide:

10.13.4. Evaluation Criteria.

10.13.4.1. Should course be archived rather than deactivated?

10.13.4.2. Has “temporarily inactive” status been entered into any appropriate command-approved functional training management systems (i.e., TSS/CMD)?

10.13.5. Issues/actions: Are any issues/action items that emerged during this activity sufficiently documented to determine:

10.13.5.1. Nature of perceived issues?

10.13.5.2. Priority of assignment?

10.13.5.3. Appropriate organization to handle tasking request.

10.14. Reactivate Course or Learning Aid. In this activity, a course in “Temporarily Inactive” or “Archive” status is reactivated to allow students to be scheduled for instruction.

10.14.1. Purpose. This activity allows “archived” or “Temporarily Inactive” courses to be placed in “Active” status for scheduling and instruction purposes.

10.14.2. Activity Outcomes:

10.14.2.1. Course “active” status entered in appropriate command-approved functional training management systems (i.e., TSS/CMD)?

10.14.2.2. Possible tasking worksheets on training issues, courseware issues, training gaps or non-training issues submitted to appropriate organization for action.

10.14.3. Additional Guidance. Contact the Complex CPM.

10.14.3.1. Various tasking worksheets, instructions and examples.

10.14.4. Evaluation Criteria.

10.14.4.1. Is the course teachable?

10.14.4.2. Has active status been entered in appropriate command-approved functional training management systems (i.e., TSS/CMD)?

10.14.5. Issues/actions: Are any issues/action items that emerged during this activity documented sufficiently to determine:

10.14.5.1. Nature of perceived issues?

10.14.5.2. Priority of assignment?

10.14.5.3. Appropriate organization to handle tasking request?

10.15. Archive Courseware. In this activity, courseware no longer required is officially Archived.

10.15.1. Purpose. This activity documents reasons courseware is no longer needed and is placed in archive status.

10.15.2. Activity Outcomes:

10.15.2.1. Archive Signature Page processed through appropriate using agencies prior to archive action.

10.15.2.2. Course “Archive” status entered into any appropriate command-approved functional training management systems (i.e., TSS/CMD)?

10.15.2.3. Reason for archive added to courseware master file.

10.15.3. Additional Guidance. See the *AFMC Depot Maintenance Training Resource* Site.

10.15.4. Evaluation Criteria.

10.15.4.1. Should courseware be placed in “Temporarily Inactive” status rather than archived?

10.15.4.2. Is courseware a prerequisite to other training?

10.15.4.3. Is this courseware a requirement on any CTP or Formal Training Plan (FTP)?

10.15.4.4. Did appropriate instructors, program managers and training managers from all using agencies approve the archive action?

10.15.4.5. Has Archive Justification and Signature Page been added to the courseware master file?

10.15.4.6. Has "Archive" status been entered into any appropriate command-approved functional training management systems (i.e., TSS/CMD)?

10.15.5. Issues/actions: Are any issues/action items that emerged during this activity sufficiently documented.

10.16. Outsource Courseware Services (as needed). In this activity, a courseware specialist or training manager works with the installation contracting office to contract out courseware development or management services.

10.16.1. Purpose. This activity is a reminder of primary activities involved, and important issues to remember, when outsourcing courseware services.

10.16.2. Activity Outcomes. Outcomes for this activity vary with the outsourcing task being performed and the circumstances of the particular situation. Contact Complex CPM for guidance.

10.16.3. Additional Guidance. Specific guidance on Acquisition topics can be provided by the local contracting functional at each installation. Evaluation Criteria. Evaluation criteria for this activity vary with outsourcing task being performed and circumstances of the particular situation.

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Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

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AFI 36-2650_AFMCSUP, *Maintenance Training,* 03 March 2023

AFSCH 60-101, *Art of the Possible Handbook,* 30 August 2023

DAFH 36-2675, *Information for Designers of Instructional Systems,* 15 April 2022

DAFI 90-160, *Publications and Forms Management*, 14 April 2022

DAFMAN 36-2689, *Training Program*, 31 March 2023

DAFPD 36-26, *Total Force Development and Management,* 18 January 2024

MIL-HDBK-29612-2A, *Instructional Systems Development/Systems Approach to Training and Education,* 31 August 2001

Adopted Forms

AF Form 9, *Request for Purchas*e

AFMC Form 96, *Existing Courseware Evaluation Checklist*

AFMC Form 853, *AFMC Course Chart*

DAF Form 847, *Recommendation for Change of Product*

SF Form 182, *Authorization, Agreement, and Certification of Training*

Abbreviations and Acronyms

ADDIE—Analysis, Design, Development, Implementation, Evaluation

ADL—Advanced Distributed Learning

ADL-R—Advanced Distributed Learning Repository

AFB—Air Force Base

AFI—Air Force Instruction

AFIADL—Air Force Institute for Advanced Distributed Learning

AFMAN—Air Force Manual

AFMC—Air Force Materiel Command

AFMCMAN—Air Force Materiel Command Manual

AFSC—Air Force Sustainment Center

AoP—Art of the Possible

CAM—Content Aggregation Model

CBI—Computer-Based Instruction

CCD—Course Control Document

CMD—Courseware Management Database

COR—Contracting Officer Representative

COTS—Commercial-Off-The-Shelf

CPM—Courseware Program Manager

CTP—Civilian Training Plan

CTS—Course Training Standard

DAF—Department of the Air Force

DAFH—Department of the Air Force Handbook

DAFI—Department of the Air Force Instruction

DAFMAN—Department of the Air Force Manual

DAFPD—Department of the Air Force Policy Directive

DL—Distance Learning or Distributive Learning or Distributed Learning

DMTL—Depot Maintenance Training Library

DoD—Department of Defense

DT&E—Development Test and Evaluation

E&T—Education and Training

ECML—Electronic Courseware Master Library

ETMS—Education and Training Management System

FAR—Federal Acquisition Regulation

FSU—Florida State University

FTP—Formal Training Plan

GAS—Graduate Assessment Survey

HQ AFMC—Headquarters Air Force Materiel Command

IAW—In Accordance With

IC—Interim Change

ICM—Interim Change Memo

ID—Instructional Design

ISD—Instructional System Development

ISDD—Instructional Systems Design & Development

ISDMP—Instructional System Development Management Plan

JPR—Job Performance Requirement

LMS—Learning Management System

MIL-HDBK—Military Handbook

OPR—Office/Officer of Primary Responsibility

OT&E—Operational Test and Evaluation

PAC—Production Acceptance Certification

PADDIE—Planning, Analysis, Design, Development, Implementation, Evaluation

PDS—Project Definition Summary

PMBOK—Project Management Body of Knowledge

POI—Principles of Instruction or Plan of Instruction

PR—Periodic Review/Re-Evaluation

PWS—Performance Work Statement

QI—Quality Improvement

RKO—Reusable Knowledge Object

RLO—Reusable Learning Object

ROM—Rough Order of Magnitude

SAT—Systems Approach to Training

SCO—Sharable Content Object

SCORM—Sharable Content Object Reference Model

SME—Subject Matter Expert

SOJT—Structured On-the-Job Training Guide

SOW—Statement of Work

SRR—Site Readiness Reviews

TMs—Training Managers

TSS—Training Scheduling System

USAF—United States Air Force

WG—Wage Grade

Office Symbols

HQ AFMC/A4P—Headquarters Air Force Material Command, Resource Integration Division

HQ AFMC/A4PT—Headquarters Air Force Material Command, Workforce Development Branch

Terms

Acoustic Memory—Short-Term Memory (STM) uses three processes to store and organize information selectively chosen from the sensory memory: iconic memory, acoustic memory, and working memory processes. Acoustic memory holds sounds, or acoustic representations, from 3-20 seconds before discarding or transferring them to Long-Term Memory (LTM). Also see Memory, Miller's Magic Number and STM.

Acquisition Method—Way of procuring goods and services. For Education and Training there are three methods of acquisition: (1) SF Form 182, *Authorization, Agreement, and Certification of Training*; (2) government credit card; and (3) a contract and AF Form 9, *Request for Purchas*e.

Acquisition Strategy—Activity in which the Education and Training office and contracting office work together to determine the best contract type, extent of competition, and contract vehicle for obtaining quality services at reasonable prices (while at the same time considering efficiency of proposal preparation, proposal evaluation, negotiation, and contract award). See local contracting office for more information.

Active Learning—A method in which the student learns by participating, practicing, and performing. Some common techniques used in active learning are group projects, role playing, case studies, ice-breakers, simulations, games, and in-basket exercises.

Activity Outcome—A product, service, or decision that results from an activity in the AFMC Courseware Development and Management Process.

Activity—For purposes of the AFMC Functional Courseware Development and Management Process, an activity is a set of tasks with a clear performance objective.

ADDIE Instructional Design Model—A flexible and interactive approach to analyzing instructional requirements, designing an instructional system, developing courseware, implementing instruction, and evaluating the effectiveness of an instructional system. Its Analysis, Design, Development, Implementation, and Evaluation Phases can be entered at any point depending on the specific situation. It is the basis of nearly all ID methodologies in use today. AFMC's closely related Planning, Analysis, Design, Development, Implementation, Evaluation (PADDIE) model separates the analysis activities into Planning and Analysis Phases.

Administration Function—This ISD function is responsible for the day-to-day operations of the instructional system throughout its life cycle. Some of the basic administration activities are: preparing documentation such as instructional materials, course control documents, plans and reports; maintaining records such as courseware master files, rosters, test scores, student completions, budgets, and contracts; providing staff support such as processing personnel actions, and maintaining employee development programs; scheduling resources such as personnel, equipment and facilities; implementing and maintaining automated information management systems and other tools to support instructional system activities; administering funds and contracts; etc. Also see Delivery Function, Management Function and Support Function.

Adult Learning Theory—Adults differ from children in how they learn. Adult learning theory contends that mature learners want to know why they are being taught something; want to participate and control their learning experience; prefer active rather than passive learning experiences; have a greater repository of experience to which the new learning can be related, and therefore bring greater diversity to learning situations; prefer to set their own pace for learning;

have a greater need for feedback on their learning; are more task-oriented; prefer a variety of instructional methods; and like to determine the place and time of learning. Also see Andragogy.

Advanced Distributed Learning Initiative—The DoD and the White House Office of Science and Technology Policy launched the ADL Initiative in November 1997. The mission of the ADL Initiative is to provide access to the highest quality education and training that is tailored to individual needs, and can be delivered cost-effectively, anytime and anywhere. This initiative aims to accelerate large-scale development of dynamic and cost-effective learning software and systems and to stimulate the market for these products. This will help meet the expanding E&T needs of government, academia and industry. Also see Asset, Distance Learning, Learning Object Metadata, Sharable Content Object Reference Model (SCORM), Shareable Content Object (SCO), Training Object Repository and Web-Based Instruction.

Advanced Distributed Learning Repository—The DoD registry for SCOs, assets, and repositories. It is the "Yellow Pages" of what DoD repositories are out there and what is in each repository. See Asset, Learning Object Metadata, SCORM, SCO, Training Object Repository and Web-Based Instruction.

Affective Domain—It is widely believed that our brains learn in three distinct ways, by using three mental processors (cognitive, psychomotor and affective) to encode any information that we receive into stored mental images (memory). In the training world these three domains (types) of learning are usually called knowledge, skills and attitude. The affective domain processes incoming information in terms of feelings and emotions, which impacts our attitude in dealing with something. Bloom's Taxonomy lists five progressive levels of learning using the affective domain: receiving phenomena, responding to phenomena, valuing, organizing and internalizing values.

Air Force Institute for Advanced Distributed Learning—The office of primary responsibility for implementation of Distance Learning (DL) policy and emerging DL technologies within the Air Force.

Air Force Specialty—A grouping of jobs in the Air Force military service that have similar duties and require common qualifications. Each group of jobs is assigned an Air Force Specialty Code and title (i.e., 2A7X3, Aircraft Structural Maintenance). See Occupational Series.

Algo-Heuristic Theory—This learning theory model identifies mental processes, especially unconscious processes, which underlie expert learning, thinking and performance. It includes a system of techniques for asking the right questions of experts, which allows analysts to uncover the processes involved in their behavior, and to capture their expert knowledge. It contends students be taught not only knowledge but the algorithms and heuristics of experts as well.

Analysis Phase—ISD phase in which training requirements are analyzed to determine precisely what should be included in instruction, types of learning involved, proficiency level of learning needed for the target audience, and any prerequisite or follow-on instruction required.

Analysis—Bloom's Taxonomy lists six progressive levels of learning using the cognitive domain: knowledge, comprehension, application, analysis, synthesis, and evaluation. Analysis is dividing information into simpler objects and ideas and then seeing how these parts are organized and relate to one another.

Anchored Instruction—This instructional design theory contends that instruction should be designed around an anchor, or focal point, which should be a case study or problem situation, and that instructional materials should allow exploration by the learner, such as interactive Web- based instruction or other information repository. Also See Instructional Design Theory.

Andragogy—This Greek term is widely used by adult educators to describe the theory of adult learning. It is an alternative to pedagogy (educating children).

Application—Bloom's Taxonomy lists six progressive levels of learning using the cognitive domain: knowledge, comprehension, application, analysis, synthesis, and evaluation. Application is adapting knowledge to actual situations.

ARCS—See Attention, Relevance, Confidence and Satisfaction (ARCS) Model of Motivation. Articulation. The psychomotor domain processes incoming information in terms of physical movement and coordination, which result in a physical skill. The R.H. Dave model lists five progressive levels of learning using the psychomotor domain: imitation, manipulation, precision, articulation, and naturalization. Articulation is arranging a sequence of actions to achieve harmony and consistency.

Articulation—The psychomotor domain processes incoming information in terms of physical movement and coordination, which result in a physical skill. The R.H. Dave Model lists five progressive levels of learning using the psychomotor domain: imitation, manipulation, precision, articulation and naturalization. Articulation is arranging a sequence of actions to achieve harmony and consistency.

Assessment—Measures a student's transfer of knowledge and attainment of learning objectives. Assessment is a major element of blended and electronic learning because it allows students to test out of content that they already know and take only the parts of the instruction that they need.

Asynchronous Learning Event—Instruction in which a subject matter expert or facilitator responds to questions and comments from students at remote locations, but with a delay. This means that the facilitator does not have to be on-line when students are taking Computer-Based Instruction (CBI), or at a help desk when students are taking correspondence courses, videotaped courses, etc. Asynchronous means not real-time.

Attention, Relevance, Confidence and Satisfaction Model of Motivation—John Keller's ARCS model of motivation is a proven approach to developing interesting and effective instruction. It focuses on keeping the learner engaged in the instruction so that optimal learning can take place.

Attention—Getting and holding the learner's interest. This is one of the four tenants of John Keller's ARCS Model of Motivation, which uses Attention, Relevance, Confidence and Satisfaction to create interesting and effective instruction so that optimal learning will take place. Starting the class with an icebreaker activity such as telling a joke or story, polling students with a thought-provoking question, etc., engages the students and prepares them for learning.

Authoring System—Computer program designed specifically to assist in creating Computer-Based Instruction.

Behavioral Learning Theory—A theory based on the idea that learning takes place when students make an association between a stimulus (cue) and the desired response (behavior). In behavioral learning, students are taught how to recognize cues and respond with actions, and their learning is

reinforced through instructor feedback on successful performance. This stimulus-response-reinforcement method is an effective means of training personnel to perform procedural tasks. However, the mental processes and hidden behaviors that occur within the learner are not addressed as part of the learning process with behavioral learning theories, so it has only limited application to the instruction of intellectual skills and attitudes. Also see Behaviorism and Learning Theory.

Behaviorism—The leading school of thought in the early 1900s, behaviorism declares that the only subjects worth psychological study are observable behaviors.

Blended Learning Theory—An instructional design (ID) approach that combines several different instructional delivery methods such as live events, self-paced learning, collaboration, assessment and performance support materials. Blended Learning Theory combines the traditional theories and models of Benjamin Bloom, Robert Gagné, Walter Dick, Lou Carey, David Merrill, John Keller, Tom Gilbert, Gloria Gery, etc., to create situational ID that effectively meets the needs of a particular target audience.

Bloom's Taxonomy—In the 1950s, Benjamin Bloom chaired a committee that created an easy to understand taxonomy of learning behavior that is still the most widely used today. The Bloom Taxonomy consists of three domains of learning behavior: cognitive (mental skills or knowledge), affective (attitude, or growth in feelings or emotional areas), and psychomotor (manual or physical skills). Within those domains, there are typical levels of understanding that the learner experiences:

Body—Major section of a lesson in which learning is developed through support material and various teaching exercises to achieve learning objectives, preceded by an introduction and followed by a conclusion. Also see Introduction, Conclusion.

Bypass Test—A proficiency test that, when passed, gives students equivalency credit for a course. Bypass tests are assigned separate course numbers and are tracked in employee education and training records as course completions.

Case Study—A teaching method in which students encounter a real-life or fictional situation under the guidance of an instructor in order to achieve an instructional objective.

Characterization—Bloom's Taxonomy lists five progressive levels of learning using the affective domain: receiving phenomena, responding to phenomena, valuing, organizing, and internalizing values (characterization). Characterization is applying an internal, personal value system which results in consistent, predictable behavior that is "characteristic" of the learner.

Charting—This is an instructional technique that strengthens comprehension and increases learning transfer by having student fill in the blanks on a skeletal version of a flow chart or similar diagram.

Civilian Training Plan—Comprehensive core training document that identifies life-cycle education and training requirements, training support resources, and minimum core task requirements for an occupational series. It serves as a training roadmap for civilian employees. Also see Occupational Training Template, Career Field Education and Training Plan.

Clarification Support—Type of instructional material used in the body of a lesson to develop learning and clarify ideas. It includes definitions, examples, non-examples, comparisons, statistics, testimony from experts, etc. Also see Body, Proof Support.

Closure—The final segment of a lesson conclusion during which instruction is appropriately ended. Also see Conclusion.

Coaching—Providing one-to-one support on a particular project or task. When this is accomplished, primarily via the internet, it is called e-coaching. Also see Mentoring and Tutoring.

Cognitive Domain—It is widely believed that our brains learn in three distinct ways, by using three mental processors (cognitive, psychomotor and affective) to encode any information that we receive into stored mental images (memory). In the training world these three domains (types) of learning are usually called knowledge, skills and attitude. The cognitive domain processes incoming information in terms of facts, procedural patterns, and concepts. Bloom's Taxonomy lists six progressive levels of learning using the cognitive domain: knowledge, comprehension, application, analysis, synthesis and evaluation.

Cognitive Flexibility Theory—A constructivist learning theory based on the premise that effective learning is dependent on context, so instruction needs to be very specific and learners must be allowed to develop their own personal interpretations of information in order to learn. It is flexible in that the learner is assumed to restructure knowledge and adapt responses (ideas, decisions, judgments, behaviors, etc.) to changing situations.

Cognitive Learning Theory—A theory based on cognitive psychology, which focuses on mental processes that occur in the student's mind during learning. It assumes that learning is an active, constructive process and that learning can be designed based on the cognitive processes, hierarchies, and representations which occur within students' minds to create organized and retrievable knowledge, skills, and attitudes that can later be transferred to real world situations. This acquisition-storage-retrieval approach is an effective means of instructing personnel to retain and transfer knowledge. Also see Constructivism, Learning Theory.

Cognitive Load Theory—A learning theory popularized by John Sweller that describes learning structures in terms of an information processing system involving long term memory and working memory. It is based on the premise that instruction is more effective if the role and limitations of working memory are considered during instructional design.

Cognitive Strategies—(1) The capability of people to control their own learning, remembering, and thinking behavior. (2) The ability of people to create strategic and tactical decisions and judgments based on previous knowledge and incoming situational information.

Cognitive System—Robert Marzano's theory describes how our brains learn by passing incoming information through four thought operating systems: self-system, meta-cognitive system, cognitive system, and the knowledge domain. The cognitive system is the third operating system, which processes the information needed to learn tasks and solve problems. It enters and retrieves knowledge in the permanent memory repository of the brain. The cognitive system is used to adapt knowledge to solve problems, accomplish tasks and communicate with others.

Collaboration—This is a major element of blended learning. Communities of practice, on-line chat rooms, threaded discussions and e-mail are examples of collaborative environments in which learners can communicate with subject matter experts, coaches, mentors, and one another to reinforce learning.

Command Standard Course—A course used at more than one installation in AFMC.

Component Display Theory—David Merrill established this situational instructional design theory that assumes for every learning situation, (e.g., facts, concepts, procedures, principals and processes), there are corresponding instructional treatments (rules, examples, recall exercises, practice, prerequisites, mnemonics, feedback, etc.) that should be used.

Comprehension—Bloom's Taxonomy lists six progressive levels of learning using the cognitive domain: knowledge, comprehension, application, analysis, synthesis and evaluation. Comprehension is grasping the meaning of information.

Computer Technologies—A type of instructional material that uses technologies using computer systems and accessories that connect to the computer to teach the student (i.e., Virtual Reality, Augmented Reality, etc.).

Computer-Assisted Instruction—The use of computer learning aids to assist in the delivery of classroom instruction. Computer-based exercises, training simulations of live systems and computer-hosted tutorials are examples of using computers to aid in instructor-led training events.

Computer-Based Instruction Complexity Levels—Categories which describe the difficulty of content, sophistication of the presentation and student interactivity (control over the level, order, pace and content of instruction) of CBI materials.

CBI Complexity Level 1—Often called the "page turner," this is the most basic level of CBI development. It is typically an awareness or familiarization course with a linear sequence of instruction, and student interactivity is limited to moving to the next or previous screen.

CBI Complexity Level 2—At this level, the presentation allows the student to navigate through the instruction in any order. It includes multimedia inserts (animation, narration, video clips, graphics, sounds, etc.) that the user can activate in addition to the usual CBI navigation buttons. It typically incorporates single path scripted activities (user is requested to select the correct position for a dial or switch, identify the correct sequence of steps in a procedure, etc.) and appropriate feedback messages in response to student input.

CBI Complexity Level 3—At this level, the presentation involves extensive multimedia and student interactivity. Sophisticated graphics and constrained (not full free play) simulations of systems, equipment, processes, tasks, etc., are typical. The instruction involves complicated content and higher-than-familiarization proficiency levels, which require more frequent use of input devices (e.g., mouse, keyboard, track ball, light pen, touch screen, etc.) to enhance the recall, retention and transfer of learning. Students may be required to alternate between numerous screens to accomplish learning activities and to control the sequence and pace of their instruction. 2-3 software branches and rapid response are utilized for remediation.

CBI Complexity Level 4—A highly sophisticated presentation that includes extensive interactivity, multimedia, and full free play simulations. The content is complex and proficiency levels of learning required go far beyond familiarization. Every possible performance step and decision point is analyzed and presented with full, on-screen interaction, similar to the approach used when developing aircraft simulators. Extensive software branching (four or more levels) and responses which fall just short of artificial intelligence are typical.

Computer-Based Instruction Segment—A representative model of the instruction that shows how the screens will look, how the navigational buttons will operate, how the narrator will sound, how the interactivity will work, etc. Creating a prototype module or segment of the actual

instruction reduces the risk of complete rework by producing at least one representative example of the proposed CBI for review by typical students and experts before the whole course or learning aid is produced. See Rapid Prototyping Design.

Computer-Based Instruction—Self-paced instruction controlled and monitored by a computer. *(Also refer to Distributed Learning (DL).*

Computer-Based Training—See CBI and Distributive Learning.

Conclusion—A major section of a lesson that follows an introduction and body. It contains a summary, re-motivation and closure.

Condition—Element of a learning objective that describes the situation (e.g., equipment, tools, materials, assistance, etc.) under which a student is expected to demonstrate a behavior.

Confidence—Expectation that one will attain an objective. This is one of the four tenants of John Keller's ARCS Model of Motivation, which uses Attention, Relevance, Confidence and Satisfaction to create interesting and effective instruction so that optimal learning will take place. When learners have confidence in their ability to attain the instructed skills and knowledge, they stay motivated and want to learn more. Clearly explaining learning objectives and expectations, providing immediate feedback on student responses, providing distributed practice sessions for rehearsing new skills, etc., allows students to experience incremental success during a learning event, which builds confidence.

Configuration Control—Managing changes to a document so that everyone is using the latest version.

Constraints—Limiting conditions or factors, such as regulatory mandates and policy, funding, deadlines, facilities, equipment and tools, materials, skilled personnel, etc.

Constructivism—A doctrine based on the premise that learners construct their own knowledge by using their prior knowledge and experience to create possible approaches and solutions to the unknown, applying their prior knowledge and experience to the new situation, and then integrating their newly learned information into their prior knowledge and experience. Constructivism relies on learners' active participation in analyzing and resolving problems presented to them as learning activities which they find relevant and engaging.

Content Aggregation Model—How to put learning content together so it can be moved and reused. See Sharable Content Object Reference Model.

Content Packaging—In electronic learning, provides a standardized way to exchange digital learning resources between different systems or tools. Content packaging can also define the structure (content organization) and the intended behavior of a collection of learning resources. See Sharable Content Object Reference Model.

Contract Requirements Document—Document such as a Statement of Work, PWS, task order, etc., which identifies specific requirements for products and/or services and defines the criteria by which the contractor's performance will be evaluated. The contracts requirements document is prepared by the functional area. See Contract Requirements Package.

Contract Requirements Package—Package that is submitted by the functional area to the contracting office in order to acquire commercial products or services. It contains a contracts

requirements document, government cost estimate, certification of available funds, approvals, and other agency or installation required documentation.

Cooperative Learning—Use of small groups to facilitate instruction by placing students in groups whose primary goal is to provide each member of the group the support, encouragement and assistance needed to attain learning objectives. It results in higher achievement, more positive relationships among students, and stronger retention of subject matter.

Course Chart—A course control document that contains basic information such as course title, number, length, description, target audience, prerequisites, follow-ons, training locations, lead complex, and other basic catalog data. When developing instruction for AFMC audiences, AFMC Form 853 is used.

Course Control Documents—Set of documents used to organize, conduct, and control the quality of a formal course or learning aid. The course chart; course training standard; plan of instruction, or structured-on-the-job-training guide; and course signature page are the typical course documents in AFMC.

Course Critique—Evaluation of a course by the student immediately after completion of the instruction. Also see Courseware Assessment Program, Graduate Assessment Survey.

Course Training Standard—A course control document that identifies each behavior and proficiency level of learning that the student is required to attain by the completion of the instruction.

Courseware Acquisition and Contract Oversight Function—ISD function in which appropriate acquisition method and funding are determined; contract type and vehicle are chosen, contract requirements packages are prepared and processed; best value contractors are selected; post-award contractor activities are monitored; corrective actions on contracts are taken (as needed); and contract deliverables are accepted.

Courseware Assessment Program—Installation-level Education and Training offices are responsible for implementing a comprehensive courseware assessment program for their complex that includes, at a minimum, end-of-course student critiques, graduate assessment surveys, and periodic evaluation of local processes and programs which impact courseware development and management. See Course Critique, Graduate Assessment Survey.

Courseware Developer—Individual responsible for creating relevant, engaging, and effective instruction in an efficient and low-risk manner. The developer performs most of the tasks in the Analysis, Design, and Development Phases. Also called instructional designer, instructional system designer.

Courseware Development Resource Site—A website that hosts the most current version of checklists, worksheets, templates, examples and other working aids that support the activities of the AFMC Courseware Development and Management Process. It also contains references, points of contact, and links to information about ISD, educational technology, and other topics related to the development and management of instructional systems.

Courseware Documentation Manager—Individual responsible for maintaining configuration control of courseware master files. See Courseware Program Manager.

Courseware Evaluator—Individual responsible for performing formal courseware re- evaluation on a course or learning aid.

Courseware Issue Analysis—Activity in which a courseware project manager or courseware developer works with the customer to better define and analyze a perceived courseware deficiency and to determine the nature and scope of the fix.

Courseware Issue Worksheet—Used when an individual, group, or organization determines that existing courseware is not adequate in resolving a performance deficiency. It documents basic information about the instructional requirements and the weaknesses in the courseware sufficient to determine the scope and priority of the assignment, and to decide which training function should receive the tasking. Also see Non-Training Issue Worksheet, Training Gap Worksheet and Training Issue Worksheet.

Courseware Issue—Perceived deficiency in existing courseware.

Courseware Management Function—ISD function that involves the management, support, and administration of courseware through policies, programs, standardized procedures and working aids. It includes establishing courseware policies, programs, procedures and working aids; planning, obtaining, distributing and managing resources; promoting standardization and re-use of instructional materials; monitoring, assessing and reporting on the health of courseware; implementing quality improvement measures; and controlling Interim Changes and distribution of instructional materials.

Courseware Master File—Place where all instructional materials, evaluation materials, course control documents and supporting documentation from the courseware project are organized and maintained during the life cycle of the course or learning aid.

Courseware Program Manager—Individual responsible for applying the ISD methodology and the AFMC Courseware Development and Management Process for a functional area or installation. The primary role is to serve as a liaison between their area of responsibility and higher training offices on issues related to courseware development and management.

Courseware Project Manager—Individual responsible for monitoring the progress of assigned courseware projects, assisting the courseware development team in removing obstacles that threaten the timeline or budget of the project, and reporting on project status to Education and Training office and customer reps. The courseware project manager performs most of the tasks in the Planning process to identify and prioritize training requirements, and estimate the resources needed to accomplish development and revision of courseware.

Courseware Re-Evaluation—Activity in which technical experts, instructors, and instructional design experts review the content of instructional and evaluation materials to determine if they are still relevant, accurate, complete and effective in meting defined objectives.

Courseware Re-Validation—Activity in which the courseware developer updates course control documents and makes any cosmetic changes and minor corrections to instructional and student evaluation materials based on reviews by technical experts and instructors/ trainers. It is performed at least every three years.

Courseware Specialist—A Courseware Program Manager, Courseware Project Manager, Courseware Developer, or other individual knowledgeable in Instructional System Development and skilled in applying the AFMC Courseware Development and Management Process.

Courseware Sustainment—The maintenance of a course or learning aid after development to keep it current, relevant, and effective. It includes periodic Re-Evaluation, revision and revalidation activities.

Courseware—Course control documents, instructional materials, evaluation materials, and supporting documentation which make up an instructional system. Examples of courseware items are course charts, course training standards, plans of instruction, instructional presentations, training manuals, training simulation programs, student exercises, product/process checklists, written tests, training requirements analysis reports, source code and authoring files, correspondence, decision documentation, etc.

Criterion-Referenced Test—A test to evaluate, as objectively as possible, a student's achievement in relation to standards specified in criterion-referenced learning objectives. Also see Norm-Referenced Test.

Criterion-Referenced—Based on a measurable (objective) standard. Also see Norm- Referenced Test.

Criterion—The standard by which something is measured.

Critical Attributes—Necessary characteristics for determining class membership in a concept.

Cuing—An instructional technique that prepares a student mentally for what they are about to learn by capturing attention with a headline or a preview of what is to come. Presentation slides, video clips, exhibits, demonstrations, instructor questions, etc., are commonly used to "cue" students on what they are about to learn by helping them to reference an abstract topic to something they already know.

Curriculum Developer—Also known as an instructional coordinator, is an educational professional who focuses on the topics that students learn and the materials that teachers use in the classroom.

Deficiency—The difference between "what is" and "what should be." In education and training, the gap between desired knowledge/performance and the actual knowledge/performance. Also called a need or a discrepancy.

Deliverables Format Analysis—Activity in which a courseware project manager or courseware developer decides if any of the deliverables need to be in a particular electronic format for easy maintenance and reusability. It ensures deliverables will be in a government-approved format, so the government has organic capability to deliver and maintain courseware without buying additional software applications, licenses, training, etc.

Delivery Function—This ISD function provides the instruction to students throughout its life cycle. Some of the basic delivery activities are providing an infrastructure for Distance Learning including an on-line course management system, equipment, and facilities; supplying skilled personnel to instruct courses; and maintaining current, effective and appealing courseware. Also see Administration Function, Management Function and Support Function.

Delivery Resources Analysis—Activity in which a courseware developer identifies the resources (e.g., equipment, automated systems, training regions, instructor skills, facilities, tools, etc.) that will be needed to implement the instruction. It provides the lead time necessary to purchase tools and equipment, arrange for instructors and facilities, and obtain other necessary resources.

Demographics—Statistical data about a population, such as average age, education level, occupation, gender, etc.

Demonstration-Performance—A teaching method in which students observe and then practice a sequence of events designed to teach a procedure, technique, or operation. It combines oral explanation with the operation or handling of systems, equipment, or materials.

Depot Maintenance Training Library—A central repository used to contain courseware that has been developed organically or by contractor to meet identified training requirements for AFMC personnel.

Design Phase—ISD phase in which the courseware developer uses the results of Planning and Analysis Phase activities to create a blueprint of the course or learning aid that shows what instruction, evaluation, and media will be used in each module to meet the training requirements.

Desktop Conferencing—A means of two-way electronic communication accomplished with a personal computer. This one-on-one communication allows instructors, coaches and tutors to focus on the student's individual issues without the embarrassment that might be experienced in a group setting. This form of communication is often used remedially to help a student understand particular material with which they are having difficulty.

Desktop Reference Guide—Self-paced, stand-alone reference and learning aid that meets a defined training objective.

Development Phase—ISD phase in which instructional and evaluation materials are developed, tested, and revised and the instructional system is prepared for full operational tryout in the field.

Diagnostic Test—Tool used to evaluate attainment of supporting skills and knowledge necessary to attain an objective. Diagnostic tests are designed to search for the specific source of learning deficiencies and errors so that weaknesses in instruction can be identified and corrected.

Dick and Cary Model for Instructional Design—In their authoritative book The Systematic Design of Instruction, Walter Dick and Lou Carey popularized their systems approach to courseware development that builds on Robert Gagné's learning theory and the ADDIE model. The Dick and Cary model is the basis of the current Systems Approach to Training and ISD. It adapts the systems engineering process to provide a means for sound decision making and integrates art and technology with the scientific principles of psychology, sociology and anthropology to ensure quality of instruction. It is a practical and flexible approach for developing and delivering instruction that promotes transfer of learning from the instructional setting to the workplace. Also see Gagné's Learning Theory and ISD.

Directive—Publication, regulation, official policy, technical data specification, operating instruction, process order, etc., that prescribes standards, activities or procedures.

Discrimination—The process of perceiving incoming information, discerning the differences between varying situations and then making decisions on how to respond appropriately.

Distance Learning—Instruction that is exported to students in geographically separate locations, either electronically or by standard mail. Also called distributed learning or ADL. See CBI and Web-Based Instruction.

Distributed Learning—A more encompassing term for Computer Based Training, Computer Based Instruction, Electronic Learning or other training instruction that includes electronic media

courseware or learning aids that can be provided through a standalone computer; web based or internal network LMS, internet source like "Mil Suite" or "YouTube"; electronic conference systems like Defense Collaboration Services (DCS); Distance Learning locations or by satellite audio/video transmissions.

Distributed Practice—An instructional technique that increases the student's ability to perform newly learned skills, including intellectual skills such as problem-solving. According to studies performed by C. L. Hull, when practice periods are spaced apart, performance is superior to what it is when practice periods are close together. Also, when learners are given a rest break or a diversion from the topic to be practiced in between practice periods, they will attain a higher level of learning than those who practice without an intervening diversion.

Duty—Arrangement of related work tasks assigned to an individual.

Education and Training Management System—ETMS implements AFMC's training management processes; requirements identification, resource identification, resource management and class management.

Education and Training Office—The organization (at installation level or higher) which is responsible for military and civilian E&T as described in DAFMAN 36-2689, *Training Program*. An installation may have more than one organization performing E&T office duties, based on local needs, competitive outsourcing studies, etc.

Educational Technology—Using technological means (e.g., computers, simulators, teleconferencing, satellite, etc.) to support learning. Also called instructional technology.

Education—Instruction that helps a learner acquire new knowledge disciplines, skill sets, and viewpoints that are not based on the performance requirements of the learner's current job. Education prepares the learner for non-specific future requirements, while functional training instructs the learner how to perform based on known (current or upcoming) job requirements. The effectiveness of education cannot be fully evaluated until the employee is on a job that requires performance related to the educational learning objectives.

Electronic Courseware Master Library—The ECML is a complete inventory in digital format of formal training used at an installation which can be searched, distributed and managed electronically. Complex-level Education and Training offices are responsible for maintaining an ECML which contains (at a minimum) copies of all formal functional courses and learning aids used at the installation. Both local and AFMC command standard courses are included.

Embedded Training—A training capability which is designed into or added on to operational systems and equipment.

Employee Development Instruction—Instruction that focuses on the growth of the individual rather than the performance of a job. While it may indirectly help the mission by increasing the promotion potential of employees or improving their perspective or morale, employee development instruction does not directly relate to specific job tasks. The effectiveness of employee development learning events cannot be fully evaluated because they do not have performance-based learning objectives.

Enterprise Theory—An enterprise is a complex process or project that involves numerous decisions and tasks. Gagné and Merrill established a method that develops learning activities which consider all of the learning objectives of a course and the comprehensive set of knowledge,

tasks and attitudes needed to attain them. By incorporating techniques such as "denoting" (identifying an object and explaining its physical and functional characteristics), "manifesting" (showing others the steps in a task) and "discovering" (creative problem solving), an instructor or trainer can integrate multiple objectives into each learning activity. This approach has the added advantage of providing the learner with meta-skills (understanding based on multiple learning hierarchies) which increases retention and learning transfer.

Environment—The physical conditions and surroundings in which a job is performed, or in which learning takes place, including tools, equipment and job aids.

Equivalent Course—Any bypass test or instruction that covers the same material at the same or higher proficiency level of learning and can be considered a substitute for the subject course.

Evaluation Materials—Materials used to measure a student's attainment of learning objectives, such as class exercises, progress checks, written tests, process and product checklists, etc.

Evaluation—(1) One of the primary functions of the ISD methodology, evaluation is the bridge between the quality improvement infrastructure and the system phases of the ISD process. It is a continual measurement of activity outcomes against defined performance requirements, and is part of every step in the ISD process. It gathers feedback to assess instructional system and student performance. Activities include formative, summative and operational evaluation. (2) Bloom's Taxonomy lists six progressive levels of learning using the cognitive domain: knowledge, comprehension, application, analysis, synthesis and evaluation. Evaluation is making judgments based on incoming information and previously-learned definitions, concepts, principles, formulas, etc.

Example—An instructional tool that strengthens comprehension and increases learning transfer and is essential to effective learning. Stories, short videos, pictures, demonstrations, etc., that provide both positive and negative examples create strong mental images for learners. Examples enable students to transfer their learning by showing them how they can adapt new information to make it work for them in real-world situations.

Existing Content Research and Evaluation—Activity in which a courseware specialist searches for any available materials (e.g., directives, technical data, working aids, courseware, vendor instruction, etc.) that might meet all or part of the defined requirement and evaluates its suitability. A decision is made as to what can be easily adopted and what remaining courseware development (if any) is needed. This saves resources by adopting as much existing courseware and vendor instruction as is practical.

Expectancy—An individual's perception or confidence that they can perform or succeed at a given task.

Facilitator—Individual who does not have technical expertise in the content of the training, but assists students in progressing through the instructional material so that they can attain learning objectives more efficiently. Learning centers often employ a facilitator who assists novices with the electronic technology of self-paced training and Electronic Performance Support Systems.

Federal Acquisition Regulation System—Directives which provide standard policies and procedures for acquisition of products and services. FAR Part 11, Describing Agency Needs; FAR Part 37, Service Contracting; and FAR Part 46, Quality Assurance are the sections that directly relate to the outsourcing of courseware development and management services.

Feedback—Information provided back in response to an action. According to B.F. Skinner, lack of immediate feedback on a student's action, especially when that action is wrong, invites the student to learn a wrong response. These wrong responses then have to be unlearned before the correct behavior can be learned, which wastes valuable time. Immediate feedback is critical to efficient learning.

Field Tryout—See Operational Evaluation.

Field Unit—See Functional Area.

Field Validation—See Operational Evaluation.

Final Validation—See Small Group Tryout and Operational Evaluation.

Follow-on Course—Any course proficiency evaluation for which subject course is a prerequisite. Follow-on data prevents the accidental archival or temporary deactivation of a course that is a prerequisite for some other course. A substitute prerequisite course must be established, or an Interim Change removing the prerequisite from the follow-on course must be implemented, before a course with follow-ons can be deactivated or archived.

Formal Courseware Re-Evaluation Phase—ISD phase in which experts are interviewed and courseware is evaluated to determine what changes are needed to make the instruction accurate, effective, and compliant with applicable directives. If only minor corrections are needed, the course control documents are updated, any materials needing changes are quickly fixed, and the courseware is revalidated. Also see Courseware Re-Evaluation, Courseware Revalidation.

Formal Training—Formal training is instruction based on established learning objectives that enables delivery of training to be accurate, complete and consistent information time after time. Formal training is assigned a course number to document student completions in employee records.

Formative Evaluation—Gathers feedback from subject matter experts, instructors, customer reps, etc., about the effectiveness of products and processes as they are being developed, or formed. It is performed periodically from Planning through Development Phase and includes technical accuracy reviews and individual or small-group tryouts of instructional modules or features. The objective of formative evaluation is to identify deficiencies early, when revision is least expensive. Also see Evaluation.

Full Instructional System Analysis—See Instructional System Analysis.

Full Operational Tryout—See Operational Tryout, Summative Evaluation.

Function—A category of activities (e.g., management, evaluation, technical support, etc.) that support various phases or segments of a process. Phases are usually sequential in nature, while function activities are performed during the entire process.

Functional Area Supervisor—Individual who is ultimately responsible for training, qualification and certification of assigned personnel. From a courseware standpoint, the supervisor is responsible to work with the appropriate Education and Training office to identify work center training requirements based on regulatory guidance and work center tasks, provide subject matter experts and students as requested in support of courseware development and management activities, and provide graduate assessment survey feedback to evaluate the relevancy and effectiveness of courses and learning aids.

Functional Area—Organization or work center where the target population of the training resides (e.g., logistics, depot maintenance training, acquisition, civil engineering, etc.).

Functional Training Manager—Position responsible for identifying and arranging training for employees assigned to a particular functional organization or mission area, such as logistics, depot maintenance training, acquisition, civil engineering, etc.

Functional Training—Instruction that helps employees to learn how to perform specific work processes so that they can efficiently and economically accomplish the mission. It differs from education and employee development instruction which focus on learning for the growth of the individual rather than immediate support of the mission. Soft skills such as teambuilding, change management, communication, etc., are considered functional training for purposes of this manual if they are implemented to support performance in the workplace. See Training.

Gagné Conditions of Learning Model—Robert Gagné established a fundamental model on the conditions of learning that combines five types of learning and nine events of instruction. His theories on learning are a basic element in nearly all instructional design models in use today. Gagné's Types of Learning; Intellectual skills, Cognitive strategies, Verbal information, Motor skills, and Attitudes. Gagné's Events of Instruction; Gaining attention, Informing learner of objectives, Stimulating recall of prior learning, Presenting new material, Providing learning guidance, Eliciting practice/performance, Providing feedback about correctness, Assessing performance, and Enhancing retention and transfer.

Generalization—Responding to new information that is similar, not identical, to that presented during original learning, (i.e., during learning a student calls an F-15 or C-130 an "aircraft"). A student who has generalized would later respond "aircraft" when presented with a B-2.

Group Lock-Step Instruction—Students are given the instruction and pass through the sequence of the course at a predetermined pace, completing the instructional sequence on schedule, regardless of the ability and need of the group. It is useful when facility, equipment, instructor and student conflicts require the instruction to remain on schedule to avoid conflicts.

Group-Paced Instruction—Students are given the instruction and pass through the sequence of the course as a group, at the same rate. Depending on the ability and need of the group, an instructional event may progress faster or slower than the predetermined course length.

Guided Discussion—A training method in which the instructor leads a focused discussion that involves participation by all students through a variety of exercises such as case study and role-playing. It is highly effective for analyzing, debating, exploring a topic, value, or attitude.

Human Resource Development—An organized approach to learning that integrates training, development and education.

Iconic Memory—STM uses three processes to store and organize information selectively chosen from the sensory memory: iconic memory, acoustic memory and working memory processes. Iconic memory holds visual images or icons from 3-20 seconds before discarding or transferring them to LTM. Also see Memory, Miller's Magic Number and STM.

Imitation—The psychomotor domain processes incoming information in terms of physical movement and coordination, which result in a physical skill. The R.H. Dave Model lists five progressive levels of learning using the psychomotor domain: imitation, manipulation, precision, articulation and naturalization. Imitation is observing and copying behavior after someone else.

Implementation Phase—ISD Phase in which the instructional system is validated under field conditions, final changes are made to the courseware, the official courseware master file is posted in electronic libraries, and personalized lesson plans are developed by instructors/trainers. At the end of this phase, the course or learning aid is fully operational and ready for use in the field.

Informal Training—A learning experience that is not documented in employee training records upon completion. It does not have defined learning objectives or evaluation methods. It includes coaching, mentoring, job shadowing, tutoring, informal on-the-job training, etc., that is based entirely on the variables of the situation such as the day-to-day workload, background of the employee, background of the trainer, etc., and cannot be predicted or repeated. Also see Formal Training.

Instructing Organization—Organization authorized to teach a particular course. Instructing organizations are responsible to keep the owning organization informed of possible problems with accuracy and completeness of course materials and to participate in any courseware activities associated with courses they teach.

Instructional Briefing—A regularly presented briefing which has an established learning objective and is developed using the Air Force ISD process. It is often presented by a facilitator who may not have technical expertise in the subject matter. An instructional briefing is considered formal training and is tracked in employee records.

Instructional Design Theory—A set of principles that describe how to facilitate learning of knowledge, skills and attitudes. See ADDIE Instructional Design Model, Anchored Instruction, ARCS Model of Motivation, Component Display Theory, Dick and Carey Model for Instructional Design, Enterprise Theory, Gagné Conditions of Learning Model, Human Performance Technology, Human Resource Development, ISD, Minimalism Instructional Design Theory, Planning, Analysis, Design, Development, Implementation and Evaluation Instructional Design Model, Rapid Prototyping Design and System Approach to Training.

Instructional Design—A systematic approach to planning, creating and implementing relevant, efficient and effective instructional materials. See ISD.

Instructional Guidance—Information for the instructor or trainer designed to assist them in teaching the content of the course in an efficient and effective manner. While there is no standardized format for instructional guidance, it must be sufficient for a technically qualified substitute instructor to step in halfway through a course and teach the material that supports the learning objectives in a manner consistent with that of the primary instructor. Instructional guidance includes lessons learned, teaching steps, points to emphasize, tips on use of instructional materials, etc.

Instructional Materials—Items used to instruct, such as a course control documents, lesson plans, slide presentations, learning aids, computer-based simulations, student handouts, class exercises, instructional guidance, etc.

Instructional Method—Manner in which training is delivered to the student, such as instructor-led classroom, computer assisted instruction, Structured On-the-Job Training, self-paced Web-based training, etc.

Instructional Strategy—Activity in which a courseware project manager or courseware developer analyzes data from previous Planning activities and tentatively selects the primary

delivery method (e.g., instructor-led, computer-based, etc.), instructional/ evaluation methods and tools (e.g., presentations, student handouts, proficiency evaluations, etc.) and media (e.g., video, graphics, text, lecture, etc.) that will likely be included in the course or learning aid. In AFMC, it also includes a Delivery Resources Analysis to plan for the instruction. Also see Delivery Resources Analysis, ISD Management Plan and Project Definition Summary.

Instructional System Analysis—Activity in which a courseware project manager or courseware developer analyzes all courses and learning aids within an entire instructional system for a particular work process, program, system, or subject area. It is performed in order to identify training gaps and duplication, and to decide if existing courses and learning aids within the system are effective in meeting the defined training requirements (from initial awareness through refresher) for all target populations.

Instructional System Designer/Developer—See Courseware Developer.

Instructional System Development—A practical and flexible approach for developing and delivering training that promotes transfer of learning from the instructional setting to the workplace. It adapts the systems engineering process to provide a means for sound decision making and integrates art and technology with the scientific principles of psychology, sociology and anthropology to ensure quality of instruction. ISD produces relevant, effective and economical instructional systems.

Instructional System—An organized combination of procedures, techniques, and resources (e.g., instructors, students, courseware, equipment, tools, facilities, etc.) used to attain specified learning objectives.

Instructional Systems Development Evaluator—An expert on ISD and the AFMC Courseware Development and Management process evaluates services, products and decisions for compliance with prescribed AFMC standards and formats. At a minimum, the ISD Evaluation reviews courseware projects at the end of Planning, Analysis, Design, Development, Implementation and Formal Courseware Re-evaluation Phases.

Instructional Systems Development Management Plan—Commonly referred to as Project Definition Summary. It is the plan guiding development or revision processes. Defines the activities, deliverables and resources needed to accomplish a courseware project. It prepares a courseware project for assignment or outsourcing and allows organizations to plan for the manpower/funding needed to accomplish the required tasks.

Instructional Technology—See Educational Technology.

Instruction—Instruction is information provided in a structured manner for education, employee development and training purposes. See Education, Employee Development, Functional Training, Training.

Instructor/Trainer Review of Courseware—Activity in which courseware developers and instructors or trainers review the draft instruction and evaluation materials to determine if they are effective, efficient and adequately documented so that consistent and complete instruction can be provided time after time. This step is usually repeated more than once until instructional design and delivery experts are satisfied with the presentation and documentation of instruction.

Instructor—Individual who applies the principles of learning and instruction by delivering knowledge to students in a systematic manner. In the courseware process, the instructor serves as

an expert on the effectiveness of instructional materials, learning activities and evaluation methods in meeting their intended purpose.

Instructor-Led Lab Training—Hands-on, skills-focused instruction that is group-paced and led by an instructor who is technically qualified in the subject matter. It takes place in a laboratory setting with workstations set up with tools, equipment, materials, technical data, etc., that simulate the actual work environment.

Instructor-Led Training—A delivery method in which an instructor provides knowledge to learners in a systematic manner in a classroom or laboratory setting. It is used when a large group must be taught the same thing, or when instructor skills, equipment, tools and facilities needed to teach the objectives would significantly deter production if training were conducted at the worksite. Also called classroom instruction/training, lecture and stand-up instruction.

Intellectual Skills—Cognitive skills that involve identifying, classifying, categorizing, using rules, solving problems, thinking, reasoning, analyzing, discriminating, evaluating, and judging. Also called cognitive processes, mental skills.

Intelligent Computer-Assisted Instruction—A component in interactive courseware that serves diagnoses student performance and individualizes instruction.

Interactive Courseware—Computer-controlled training designed to allow the student to control the learning environment through input devices such as a mouse, keyboard, joystick or light pen. The student's decisions and inputs to the computer determine the level, order, pace and content of instruction delivered. See CBI Complexity Level.

Interim Change Memo (ICM)—Explains any significant change (i.e., impacts content or delivery method) made to a course or learning aid in between formal revalidations or revisions. It tracks corrections and modifications to the instruction so that all instructors/trainers are using the most recent version of materials. A course number change, correction to content, change in prerequisites, etc., are examples of changes that require an ICM.

Interim Change Number—Shows the official tracking number of an IC. Example: CHPMAS0001900SU_IC-03. It consists of: the 15-character AFMC course number of the item being changed, followed by an underscore "_ ", followed by "IC" for Interim Change, followed by a two digit sequential number that is one number higher than the previous IC number for the subject courseware.

Internal Evaluation—See Evaluation, Formative Evaluation, Summative Evaluation, Operational Evaluation.

Introduction—Major section of a lesson designed to establish a common ground between the instructor and students, designed to capture and hold attention, to outline the lesson and relate it to the overall course, to point out benefits to the students, and to lead the students into the body of the lesson. It usually contains attention, motivation, and overview steps. Also see Body, Conclusion.

Issue Analysis, Performance Root Cause Analysis, Training Gap Analysis, Training Issue Analysis and Training Mandate Analysis.

Job Analysis—See Knowledge/Task Analysis.

Job Performance Order—Method of sequencing instruction to match the order in which the tasks and subtasks are performed on the job.

Job Performance Requirement—The tasks required of employees, the conditions under which these tasks may be performed and the quality standards for acceptable performance. JPRs describe what people should do to perform their jobs.

Knowledge Domain—Robert Marzano's theory describes how our brains learn by passing incoming information through four thought operating systems: self-system, meta-cognitive system, cognitive system and the knowledge domain. The knowledge domain consists of information, mental processes and psychomotor processes. It is the mental library in our brains that organizes vocabulary, facts, time sequences, cause/effect sequences, episodes, generalizations, principles and concepts.

Knowledge Test—Evaluation which determines if the student has learned information that supports tasks, such as safety, security, or basic knowledge needed to perform the task.

Knowledge Transfer—A process of decision making in which individuals decide whether learned knowledge should be applied to a particular situation. It involves understanding how and when to apply learned information in the real world.

Knowledge—(1) Ability to recall facts, identify concepts, apply rules or principles, solve problems and think creatively using cognitive (mental) processes. (2) Bloom's Taxonomy lists six progressive levels of learning using the cognitive domain: knowledge, comprehension, application, analysis, synthesis and evaluation. Knowledge is remembering previously learned information. (3) Information required to develop the skills and attitudes for effective accomplishment of a step, task or job.

Knowledge/Task Analysis—Traditional task analysis concentrates only on observable behavior; while knowledge and task analysis goes on to examine the critical decisions and mental processes that separate the expert from the novice, thus identifying all of the elements of successful performance. In this activity, the courseware developer identifies precisely those tasks and subtasks which require instruction, the conditions under which they are performed, the performance standard that must be achieved in the workplace, and the strategies used by experts at critical decision points. Also called job analysis, cognitive task analysis. See Objectives, Job Performance Requirements.

Lead Complex—Installation appointed by HQ AFMC to provide technical guidance to complexes, HQ AFMC and other organizations on issues related to the assigned subject area. The lead complex serves as liaison between complexes, ensuring that representatives from appropriate installations and organizations are involved in decisions related to the assigned training. The lead complex is responsible for development, review, revision and revalidation of command courseware, and for providing train-the-trainer assistance when applicable.

Learning Activity—A means of instructing through direct experience rather than through text study or lecture. It actively engages the student learning.

Learning Aid—Device that makes it easier for a student to learn. A formal learning aid is a stand-alone video, computer-based simulation aid, desktop reference guide, training manual, etc., that is used for instruction but which is not linked to one particular course. It is stored in a separate master file in the courseware library for rapid retrieval and reference.

Learning Analysis—Activity in which a courseware developer analyzes the tasks to be taught and the demographics of the target audience in order to describe the types of learning involved, the proficiency levels of learning needed, and any student prerequisites that will be required. This ensures that training will be effective in resolving the performance deficiency.

Learning Center—Facility in which learners can take self-paced instruction such as computer-based courses away from the distractions of the work environment. Learning centers are popular in functional areas such as depot maintenance, where not all employees have ready access to a computer workstation.

Learning Management System—An automated interface for administering training activities of an organization, such as managing curriculum and courses, scheduling/registering of instructors/students for training events, launching computer-based and Web-based instruction, tracking and reporting student progress and completions, evaluating generating metrics and reports, etc. Also see Internet Course Management System.

Learning Object Metadata—A metadata schema established by the Institute for Electrical and Electronics Engineers for implementing SCORM that contains categories that make it valuable to the education and training community. It is the metadata coding primer when applying SCORM to learning objects. See Learning Object, Metadata, SCORM.

Learning Objective—See Objective.

Learning Object—Per Robert Reiser, a learning object is any piece of information (e.g., text, sound, video clip, etc.) that is used for instructional purposes. Per David Wiley, The main idea of learning objects is to break educational content down into small chunks that can be reused in various learning environments.

Learning Theory—A set of principles that describe how students learn knowledge, skills and attitudes. Learning theories themselves do not offer guidance on how to instruct, they only describe what is going on inside the learners' heads as they process incoming information. However, it is important for courseware developers, instructors and trainers to fully understand the learning theories that are the basis of instructional design theories and models used to facilitate and accelerate learning. The two major categories of learning theories are "behavioral" and "cognitive." See Adult Learning Theory, Algo-Heuristic Theory, Blended Learning Theory, Behavioral Learning Theories, Behaviorism, Bloom's Taxonomy, Cognitive Flexibility Theory, Cognitive Learning Theory, Cognitive Load Theory, Constructivism, Gagné's Conditions of Learning Model and Meta-Cognitive System.

Learning—Knowledge, attitude or skilled acquired which changes the way the learner behaves in some way. The new behavior can be a different way of thinking, reacting or doing, and may be overt (visible) or covert (hidden).

Lesson Plan—An approved plan for instruction that provides specific definition and direction to the instructor on terminal and enabling learning objectives, equipment, training devices, instructional media requirements, and conduct of an education or training component of the instructional system. Lesson plans are a principal component of curriculum materials in that they sequence the presentation of learning experiences and program the use of supporting instructional materials, devices, and equipment.

Level of Learning—The degree to which a student is expected to internalize (master) a mental subject, values, or ability to perform psychomotor skills.

Live Events—This is a major element of blended learning. A live event is led by an instructor or facilitator and all students participate at the same time, either in a traditional or virtual classroom setting.

Live Simulation—A simulation involving real people operating real systems.

Localized Training Materials—A command or other standardized training course tailored to meet local needs. AFMC command standard course learning objective behaviors, conditions of performance, and standards of performance may be increased during localization of training materials, but cannot be lessened in any way. Instructional and evaluation materials may be altered as long as all learning objectives and proficiency levels of the standard course are attained by the student at the completion of training.

Lock-Step Instruction—Everyone proceeds at the same pace, such as occurs with typical classroom training events. It is also called group-paced instruction.

Logical Order—Combination of job-performance order and psychological order using the whole-part-whole concept: showing the whole, breaking it down into parts, then back to the whole. Also see Cognitive Load Theory.

Long—Term Memory—Memory is usually divided into three progressive levels: sensory, short-term, and long-term. In order to transfer information from STM to LTM the brain must consolidate, organize and encode the incoming data for future retrieval. LTM is our relatively permanent and unlimited repository of knowledge. Information is stored here on the basis of relevance and meaning, which is why it is so important to make instruction relevant and meaningful to the learner. If knowledge does not make it to the LTM repository it cannot be transferred to future real world situations. Also see Acoustic Memory, Iconic Memory, Memory, Miller's Magic Number and Working Memory.

Management Function—This ISD function directs, monitors and controls all activities associated with the instructional system throughout its life cycle. Various participants (supervisors, instructors, courseware developers, etc.) can perform management roles. Some of the basic management activities are planning for instructional system activities arranging resources such as skills, facilities, equipment, funding, etc., needed to accomplish planned activities in support of the instructional system; and reporting on the status of instructional system elements and activities. Also see Administration Function, Delivery Function and Support Function.

Manipulation—The psychomotor domain processes incoming information in terms of physical movement and coordination, which result in a physical skill. The R.H. Dave Model lists five progressive levels of learning using the psychomotor domain: imitation, manipulation, precision, articulation, and naturalization. Manipulation is performing certain actions by following instructions and practicing.

Market Research—Activity in which the government seeks information on the availability of commercial products and services and current market practices before making decisions about acquisition strategy. Advanced market research includes a sources sought synopsis, meetings with industry/academia, or other information exchanges appropriate to the circumstances.

Mastery—A learner's proficiency plus retention equate to mastery of a behavior.

Media—Media is the plural of medium, which is the means of communicating information to another. Media are not the messages themselves, but are only the methods in which they are conveyed. However, because messages must be received and understood before learning can take place, media are just as critical to instruction as the information itself. In the blended learning approach, several types of media (e.g., verbal lectures and discussions, visual presentations and illustrations, video and photographs, written materials, on-line information searchable repositories and chat rooms, demonstration and performance exercises, etc.) are used to effectively deliver the message to individuals with different learning styles and motivations.

Memory—Ability to retain and recall mental images and impressions. Memory is usually divided into three progressive levels: sensory, short-term, and long-term. Also see LTM, Miller's Magic Number, Sensory Memory and STM.

Mentoring—A supervisor, co-worker, subject matter expert or consultant monitors employee performance and provides guidance, feedback, and direction to assure successful job performance. A mentor is a one-on-one trainer who uses real tasks, issues and variables to help the learner become more proficient on the job. When this is accomplished primarily via the internet, it is called e-mentoring. Also see Coaching and Tutoring.

Meta—Cognitive System—Robert Marzano's theory describes how our brains learn by passing incoming information through four thought operating systems: self-system, meta-cognitive system, cognitive system, and the knowledge domain. The meta-cognitive system is the second operating system, which we use to plan strategies for accomplishing goals. It is also the system we use to apply insight into our own learning strengths and weaknesses, evaluate the demands of our current learning task, and self-regulate our progress in attaining our learning goal.

Metadata Tags—Coding at the front of a SCO that contains descriptive data about the content of the object independent of the larger structure (presentation, module, course, etc.) in which the SCO resides. See SCORM, SCO, SCO Metadata.

Metadata—Metadata is data about data. Description of how, when and by whom a particular set of data was collected, and how the data is formatted. Metadata is essential for organizing information stored in electronic repositories. Learning resources that are described with metadata can be systematically sought and retrieved for use and reuse. See Content Packaging, Sharable Content Object Reference Model.

Meta-skill—Knowledge or skill based on multiple learning hierarchies, which increases retention and learning transfer. Meta-skills also include mechanisms for error detection and avoidance. Attaining a meta-skill results in the ability to perform cognitive and psychomotor skills associated with a subject in a variety of situations.

Methodology—A system of procedures, principles and rules. In the scientific field of learning, it is the analysis of the subjects and students to be taught and the study of the methods for instructing them.

Metrics—Measurement tools used for assessing the qualitative and quantitative progress of instructional development with respect to the development standards specified.

Microlearning—Deals with relatively small learning units and short-term learning activities. Generally, the term "microlearning" refers to micro-perspectives in the context of learning, education and training.

Miller's Magic Number—George Miller discovered that the amount of information which can be remembered during one exposure is between five and nine items, depending on the information. The number seven is the average number of items which can be held in STM at any one time, so "7" became known as Miller's Magic Number. Also see Memory, STM.

Minimalism Instructional Design Theory—J.M. Carroll established this theory which is often used when designing CBI. It contends that (1) all learning tasks should be meaningful, stand-alone activities, (2) learners should be given relevant, realistic projects as quickly as possible, (3) instruction should allow self-directed reasoning and adaptation with numerous learning activities, (4) instructional materials and activities should provide immediate feedback and remediation, and (5) there should be a close relationship between the training and actual performance.

Mnemonics—Techniques used to assist in retaining memory.

Model—A graphic representation of information that is often used as an instructional tool to increase learning transfer by presenting information in a visual manner rather than through speech or the written word. A model can be an illustration (e.g., diagram, chart, drawing, map, etc.) or a three-dimensional exhibit.

Module—A module of training is one or more objectives that present a logically divided portion, or lesson, of instruction.

Motivation Step—The segment of a lesson introduction in which the instructor provides specific reasons why students need to learn whatever they are about to learn. Also see Introduction.

Motivation—A belief or value about something that prompts an individual to act.

Motor Skill—Physical actions required to perform a specific task.

Multimedia—Programs that use a combination of data, graphics, video, and sound to disseminate information. Multimedia facilitates transfer of knowledge and skills because it stimulates more than one type learning (see Bloom's Taxonomy). Multimedia is considered interactive when the individual in the audience has some control over the presentation.

Naturalization—The psychomotor domain processes incoming information in terms of physical movement and coordination, which result in a physical skill. The R.H. Dave Model lists five progressive levels of learning using the psychomotor domain: imitation, manipulation, precision, articulation and naturalization. Naturalization is when high level performance becomes automatic or natural.

Non-Example—People, objects, events, ideas, symbols, or actions that lack one or more critical attributes of a particular concept and which should not be called by that concept name. A close- in non-example is a non-example that is missing only one critical attribute. For instance, a close-in non-example of a chair is a stool.

Non-Training Issue Worksheet—Worksheet used to capture preliminary information when an individual, group, or organization is performing a training activity and discovers that there is a significant issue that impacts the effectiveness of training, but that issue falls outside the scope of the training function. Examples are requirement for a desk audit on a position, a position management study, repair to a non-training facility, streamlining of a process, etc.

Norm-Referenced Test—Process of determining a student's achievement in relation to other students. Grading on a curve is an example of norm-referenced testing. Criterion-referenced testing is used in most Air Force instruction, as norm-referenced testing is seldom appropriate.

Object Repository—Storage area for assets, SCOs and content packages. Also see Advanced Distributed Learning Repository (ADL-R) and SCO.

Objective—A statement of what a student will be able to do at the completion of instruction. In functional training, learning objectives are criterion-referenced and are expressed in terms of the behavior to be attained, the conditions under which it is to be exhibited, and the specific standards to which it will be demonstrated. Also called behavioral objective, criterion- referenced objective, enabling objective and training objective. Also see Terminal Training Objective.

Occupational Series—A grouping of jobs in the federal civilian service that have similar duties and require common qualifications. Each group of jobs is assigned the same pay plan and occupational series code, (e.g., Wage Grade (WG)-3703, Welder; GS-1670, Equipment Specialist, etc.) although titles and grade levels may vary. Also see Air Force Specialty.

Occupational Training Analysis—Activity which consolidates several other courseware activities to identify the training needed for an occupation from entry level through advanced journeyman. Occupational training analysis is performed in support of civilian Career Field Education and Training Plans or occupational template development. It helps supervisors and training specialists develop individual and occupational training plans for employees and assists developers by identifying training requirements and target populations.

Occupational Training Template—Comprehensive core training document that identifies life-cycle education and training requirements, training support resources, and minimum core task requirements for an occupational series. It serves as a training roadmap for civilian employees. Also see Civilian Training Plan, Career Field Education and Training Plan.

Online Collaborative Learning—Methods of electronic communication with others for the purpose of learning. Involves a community of learners using asynchronous (not real-time) media such as e-mail and electronic bulletin boards, and synchronous media such as chat rooms, audio/video conferencing, application sharing, virtual classrooms, etc., to communicate with one another.

On-the-Job Training—Over-the-shoulder, practical instruction on tasks required for job performance. It differs from structured-on-the-job training in that it is based entirely on the variables of the situation such as the day-to-day workload, background of the employee, background of the trainer, etc., does not have defined learning objectives, and cannot be predicted or repeated. Also see Formal Training, Informal Training and Structured-On-the-Job Training.

Operational Evaluation—Evaluates effectiveness of the instructional system during full-scale operations. Gathers feedback from students, supervisors, customer reps, etc., about whether graduates are meeting established job performance requirements (external evaluation). Gathers feedback from instructors, subject matter experts, customer reps, etc., about the currency, relevance, efficiency and effectiveness of the courseware (internal evaluation). Operational evaluation continues throughout the life cycle of the instructional system. Also called Field Evaluation. See Evaluation.

Operational Tryout—A review in which instructional materials are presented under normal operating conditions to the actual target audience (average students selected using typical scheduling procedures) to identify deficiencies. Also called a field tryout or full operational tryout. See Evaluation and Summative Evaluation.

Organizing—Bloom's Taxonomy lists five progressive levels of learning using the affective domain: receiving phenomena, responding to phenomena, valuing, organizing, and internalizing values (characterization). Organizing is comparing different values resolving conflicts between them, arranging values into priorities, and then structuring a personal, internal value system.

Other Prerequisite—Any skill, qualification, experience, grade level, etc., which cannot be expressed in terms of a simple course number, that students are required to have before they can be scheduled for a course or proficiency evaluation.

Overview—Segment of a lesson introduction in which the instructor provides clear and concise explanation of the lesson objective, subject matter and teaching method to be used.

Owning Organization—The organization responsible for the sustainment (i.e., periodic review and update) of the courseware and the delivery of the course. Other organizations may teach the course only with the owning organization's permission.

Performance Issue—A gap or deficiency between desired knowledge/performance and the actual knowledge/performance.

Performance Plan—Government plan for evaluating contractor performance to ensure that the terms and conditions of the contract are met. Formerly called a Quality Assurance Surveillance Plan.

Performance Root Cause Analysis—Activity in which a training function or customer organization examines a performance deficiency, analyzes the underlying causes, and determines the solution. It replaces the "training needs assessment" or "training needs analysis," which do not fully analyze the shortcomings of the customer's entire work environment. Performance root cause analysis looks not only at the lack of skills or knowledge, but also considers other factors that might contribute to the performance deficiency. It then proposes holistic solutions for improving performance which may or may not include a training intervention.

Performance Support Materials—This is a major element of blended learning. It includes any materials provided to reinforce retention of learning and enhance job performance, such as desktop reference guides, copies of instructional materials from live events, flow charts, diagrams, checklists, templates, examples, procedural guides, decision tables, worksheets, algorithms, etc. Also called job aids. See Electronic Performance Support System.

Performance Test—Evaluation in which the student actually performs the skill required by the learning objective. Tasks are performed to show proficiency of the skill. When a skill is to recall information or solve a problem, a written performance test is appropriate. When a skill is to perform a physical task, a physical performance test is appropriate.

Performance Work Statement—A contract requirements document that defines services needed in terms of outcomes and the standard of acceptable performance for those outcomes. The emphasis is on performance - what is to be done and not how it is to be done. It is also called a Performance-based Work Statement. See Contract Requirements Document, Contract Requirements Package and Statement of Work.

Performance—Part of a criterion objective that describes the observable student behavior (or the product of that behavior) that is acceptable to the instructor as proof that learning has occurred. Also see Behavior.

Periodic Review—AFMC requires that courses and learning aids be reviewed every two years for accuracy, completeness and effectiveness. This periodic Re-Evaluation of courseware is sometimes called periodic review. Also see Courseware Re-Evaluation, Courseware Revalidation and Formal Courseware Re-Evaluation Phase.

Personalized System of Instruction—Fred Keller developed this type of instruction in which there are occasional motivational lectures, a text-based course divided into small units with a study guide, review questions, and unit tests. A proctor scores the test, provides feedback, and asks probing questions to see if the learner really understands the material. The learner must score at least a 90% before moving on to the next unit, and there is no penalty for failing a unit test. Those that do are coached, given remedial assignments, and then retested until they pass. Once all units have been passed, the learner is granted a completion.

Phase—A group of activities that support a major segment of a process. Phases are usually sequential in nature, while functions are types of activities that are performed during the entire process.

Placeholder Charts—Reminder charts included in standardized training to cue installations or organizations to enter their localized information. The placeholder charts are hidden or replaced by localized charts before the presentation is used in the field.

Plan of Instruction—A course control document used for planning, organizing and conducting instruction. The POI is organized by units or modules of instruction with each module containing such information as instructional method, evaluation method, time allocations, learning objectives, description of student instructional materials, list of training support equipment, etc.

Planning, Analysis, Design, Development, Implementation and Evaluation Instructional Design Model—PADDIE model for courseware development. See ADDIE Design Model.

Post-Award Phase of Acquisition—This phase of the acquisition process ensures that the contractor meets the work statements and product specifications of the contract requirements document (e.g., Statement of Work, performance work statement, task order, etc.). The ISD Evaluation Board is the primary method of overseeing contractor performance of courseware development and management tasks. This phase ends with the government's acceptance of deliverables and final payment for the contractor's performance.

Post-Test—A criterion-referenced test designed to measure student performance on objectives taught during a module or course of instruction.

Pre-Assessment Test—A criterion-referenced test designed to measure student attainment of learning objectives prior to the start of a course. It is often used in CBI to tailor the learning so that students only take the modules that they need and not the entire course.

Precision—The psychomotor domain processes incoming information in terms of physical movement and coordination, which result in a physical skill. The R.H. Dave Model lists five progressive levels of learning using the psychomotor domain: imitation, manipulation, precision, articulation and naturalization. Precision is refining actions to make them more exact and correct.

Predictive Test—If a learning objective behavior cannot be demonstrated during the instructional event because it is too costly, too dangerous, impractical, etc., a predictive test is used to test critical parts of the learning objective behavior. If a student can define the steps in a procedure, or perform certain key tasks, it is predictable that the student will be able to perform the actual learning objective skill.

Prerequisite Course—Any course that students are required to have before they can be scheduled for subject course or proficiency evaluation.

Prerequisite—Prior knowledge or experience.

Pre-Solicitation Phase of Acquisition—This phase of the acquisition process includes identification of the need by the functional area; preparation and processing of a contracts requirements package, development of contractor performance evaluation criteria, and planning of an acquisition strategy. See Contract Requirements Package.

Primary Customer Organization—Organization in which primary segments of the target audience reside. Primary customer organizations coordinate on any courseware reviews, revisions, de-activations, archives, or new courseware developments associated with courses they use.

Priority—The priority for an activity or project is determined with input from the primary customer organizations, primary instructors, appropriate program managers, and Education and Training office representatives as appropriate. A priority ranking set for a particular activity is relative to all of the other activities competing for attention at the same time.

Process Checklist—Performance tests, which require the student to perform a task, typically use a checklist to document student proficiency. When the series in which steps are performed is more critical than the end product, a process checklist that corresponds to the steps or activities of the task being performed is used to annotate the evaluator's observations as the student performs the task.

Product Checklist—Performance tests, which require the student to perform a task, typically use a checklist to document student proficiency. When the end product is more important than the series in which steps are performed, a product checklist that corresponds to criteria required for the end product is used to annotate the evaluator's observations as the student performs the task.

Production Acceptance Certification—Program used in the AFMC depot maintenance community to document employees' training, qualifications, and certifications to perform and certify assigned work.

Proficiency Code—Codes used to describe the level of knowledge or performance that the student is required to attain at the completion of instruction.

Proficiency Evaluation—Performance test that requires the student to perform the task while being observed by an evaluator (typically an instructor or trainer).

Proficiency Level of Learning—Level of knowledge or performance that the student must attain in order to accomplish a learning objective.

Program Manager—See Courseware Program Manager.

Program Object Memorandum—DoD program proposal used to submit budgets constraints and priorities for military forces, modernization, readiness and sustainability and supporting business processes and infrastructure activities. It is the link between strategic planning and programming.

Programmed Learning—Sidney Pressey developed and B.F. Skinner modernized this approach to learning which is commonly used today when developing computer-based and Web-based instruction. Learners are given small amounts of information and proceed at their own pace from one slide/screen to the next. They then answer questions about the information presented and receive immediate feedback on correct and incorrect responses. If they are incorrect, they are directed to remedial information, depending on the mistake they made.

Project Definition Summary—Document which defines the activities, deliverables and resources needed to accomplish a courseware Project. It prepares a courseware project for assignment or outsourcing and allows organizations to plan for the manpower/funding needed to accomplish the required tasks.

Project Kickoff Meeting—Activity in which the courseware project manager hosts a meeting with primary participants in the project to verify roles and responsibilities and review the proposed project plan. It clarifies for all key participants in the project what role each of them will have and the planned project activities and timelines. This reduces misunderstandings and allows the government to emphasize the importance of Subject Matter Expert, subject area Program Manager, System OPR, Instructor/Trainer, Complex CPM, ISD Evaluator, customer Training Manager, etc., involvement in the Project.

Proof Support—A type of instructional material used during the body of a lesson that provides hard data or expert testimony in support of an assertion. Also see Body and Clarification Support.

Prototype—A representative model. See CBI Prototype Module, Rapid Prototyping Design.

Psychological Order—Method of sequencing instruction based on ease of learning. Students are taught the easiest tasks first, then progress to the more complex tasks.

Psychomotor Domain—It is widely believed that our brains learn in three distinct ways, by using three mental processors (cognitive, psychomotor, and affective) to encode any information that we receive into stored mental images (memory). In the training world these three domains (types) of learning are usually called knowledge, skills and attitude. The psychomotor domain processes incoming information in terms of physical movement and coordination, which result in a physical skill. The R.H. Dave Model lists five progressive levels of learning using the psychomotor domain: imitation, manipulation, precision, articulation and naturalization. Also see Bloom's Taxonomy.

Qualitative Data—Information that is subjective in nature because it is based on opinions or concepts rather than measurable facts.

Quality Improvement—Organized creation of beneficial change; improvements made in products procedures, learning, etc. The phases and functions of the ISD model are embedded in, and held together by, an overarching QI process. It continually generates refinements to the instructional system based on the evaluation of activity outcomes against defined performance requirements.

Quantitative Data—Information that is objective in nature because it is based on observable behavior that can be translated into numbers suitable for analysis and measurement.

Rapid Prototyping Design—Instructional design model which deviates somewhat from ADDIE in that a sample module is quickly developed during Design Phase that can be tested with the target population. This rapid prototyping allows students to try out the instruction early in the process,

when changes in media, instructional methods, and design are easier and less expensive to implement. It is appropriate when the subject or technology is new and few similar training programs have been built. It has been used most extensively with computer-based and Web- based instruction, when emerging technologies are initially incorporated into instructional systems.

Ratification—Act of approving an unauthorized commitment, by an official who has the authority to do so. Even when a ratification action is approved, the commander of the individual who entered into the unauthorized commitment must prepare a statement about disciplinary action taken, if any, and steps taken to prevent any further occurrence. If ratification action is not approved, the contracting officer notifies the contractor of the government's decision and the contractor is free to seek reimbursement from the person who committed the act.

Receiving Phenomena—Bloom's Taxonomy lists five progressive levels of learning using the affective domain: receiving phenomena, responding to phenomena, valuing, organizing, and internalizing values (characterization). Receiving phenomena is accepting input of information.

Rehabilitation Act Section 508—Section 508 of the Rehabilitation Act, as amended by the Workforce Investment Act of 1998, requires that when federal agencies develop, procure, maintain, or use electronic and information technology, they shall ensure that the electronic and information technology allows federal employees with disabilities to have access to and use of information and data that is comparable to the access to and use of information and data by federal employees who are not individuals with disabilities, unless an undue burden would be imposed on the agency. See the *AFMC ISD Resource* Site for guidance on how this applies to education and training products and services.

Relevance—This is one of the four tenants of John Keller's ARCS Model of Motivation, which uses Attention, Relevance, Confidence and Satisfaction to create interesting and effective instruction so that optimal learning will take place. When learners believe that the instruction is relevant to their personal situation they stay focused and interested. Examples, discussions, stories, and other means of helping the student to relate to the material and transfer their learning to the real world is incorporating relevance into the instruction.

Reliability—(1) The degree to which a test instrument can be expected to yield the same result upon repeated administration to the same population. (2) The capability of a device, equipment item, or system to operate effectively for a period of time without a failure or breakdown.

Re-motivation—Segment of a lesson conclusion during which the instructor explains how students can use the information presented and challenges the students to use what they have learned. Also see Conclusion.

Repository—A collection of large quantities of digitized data that are kept and maintained in a database. Content and the relationships among its components are stored in the repository for retrieval. Also called a digital library.

Responding to Phenomena—Bloom's Taxonomy lists five progressive levels of learning using the affective domain: receiving phenomena, responding to phenomena, valuing, organizing and internalizing values (characterization). Responding to phenomena is reacting to incoming information. Also called active participation.

Retention—The act of remembering things. Also see Memory, Sensory Memory, LTM and STM.

Reusable Knowledge Object Assessment—Activity in which a courseware project manager or courseware developer determines the reusability of courseware components. This activity estimates the amount of RKOs that will likely be of value in later performance support activities (e.g., instruction, quality, process improvement, etc.) so that the labor involved in coding the objects with metadata tags can be calculated. A knowledge object that can be re- used saves the time and effort of creating a new item, promotes consistency in information provided to the workforce, and reinforces the retention of knowledge

Reusable Knowledge Object—A slide, video clip, photograph, chart, text or other piece of information that has high potential for being useful in future instruction or management activities and is coded as metadata and stored in an electronic repository for rapid search, retrieval and re-use.

Reusable Learning Object—See Reusable Knowledge Object. In industry this term is more frequently used to identify specific type's media, graphics, or text that can be reused to insert in training that is being developed.

Revalidation—See Courseware Revalidation.

Revision Date—Date that courseware was last formally revalidated (i.e., reviewed, updated and signed again). For newly developed courseware, it is the date of original development. It is NOT the date of the most recent Interim Change (IC), as an IC does not require full review of courseware effectiveness or a new signature page. The revision date information is taken directly from the "Revision Date" block on the course chart.

Root Cause Analysis—See Performance Root Cause Analysis.

Run Time Environment—How content is launched and the learner's progress is tracked and reported back. See Sharable Content Object Reference Model.

Satellite Instruction—Video-based instruction over satellite broadcast television networks. Composed of video teleconferencing and interactive television.

Satisfaction—This is one of the four tenants of John Keller's ARCS Model of Motivation, which uses Attention, Relevance, Confidence and Satisfaction to create interesting and effective instruction so that optimal learning will take place. Learners are more likely to retain and apply their knowledge and skills to real world situations if they are satisfied with the results of their learning experience. Preparing learners to transfer their knowledge by simulating work environments, explaining how the knowledge and skills learned in the training event will integrate with follow-on learning events, providing the learner with post-event support such as on-line reference and job aids, etc., makes the instructional experience rewarding and complete.

Schema—Intellectual skills that are integrated into existing knowledge to be remembered and recalled. They contain well-understood features of an object or event. When encountering new information, the student refines the features or fills in the blanks of their schema.

Section 508—See Rehabilitation Act Section 508.

Segment—A logical division of instruction which contains one or more modules. Most courses do not need tracking of blocked modules of instruction, and so the segment number is the same as the learning module number. However, when a course is especially lengthy, learning sessions are separated over several days/weeks, more than one instructor is needed to present material, etc., the segment number can be a useful tool for organizing the course.

Self-Paced Instruction—This is a major element of blended learning. It is learning that is attained independently by the student, who controls the timing, pace and (in the case of interactive courseware) the content of instruction within certain parameters. Computer-based and Web-based instruction and correspondence self-study are examples of self-paced learning.

Self-Paced, Computer-Hosted Course—A self-paced course that is taken via a computer by the student, but is not designed to connect to a command approved LMS, courseware database and content library, which tracks employee progress and automatically documents student completions. This category of delivery is used for PowerPoint presentations, videos, etc., that are not designed to be hosted on the Web and taken via an automated. Computer-Based Training that can either be hosted on a stand-alone computer or can be taken via a command approved LMS on the Web is categorized as Web-based training.

Self-Paced, Web-Based Course—A self-paced course that is taken via a computer and a Web-based command approved LMS, which tracks employee progress and automatically documents student completions. The training is usually taken at the employee's workstation or in a learning center environment away from the work center.

Self—System—Robert Marzano's theory describes how our brains learn by passing incoming information through four thought operating systems: self-system, meta-cognitive system, cognitive system and the knowledge domain. Self-system is the first filter in which we decide if the incoming information has value to us. If it does, our brain will put effort into further processing it.

Sensory Memory—Memory is usually divided into three progressive levels: sensory, short- term, and long-term. Sensory memory retains an exact copy of what is seen or heard. It lasts for only a second or two. It has unlimited capacity. Also see Memory.

Service Contract—Federal Acquisition Regulation (FAR) 37.101 define a service contract as one which directly engages the time and effort of a contractor whose primary purpose is to perform an identifiable task rather than to furnish an end item of supply. Courseware development and management are considered services because the deliverables are not furnished off-the-shelf, but are a result of numerous tasks, or activities, that must be performed to the standards of this manual and other specifications.

Services Summary—Section of the PWS which states the criteria for determining compliance with the standards for required services. It is also called a Performance Requirements Summary.

Sharable Content Object Assessment—Activity in which a courseware project manager or courseware developer determines the level of SCO packaging needed to support transportability of training to other courses, and to track student progress, test scores and completions for self-paced, CBI. It predicts the level of SCO granularity needed so that the labor involved in coding the objects with metadata tags and launch assets can be roughly estimated.

Sharable Content Object Reference Model Content Aggregation Model—Provides a common means for composing learning content from searchable, reusable, sharable and interoperable sources. Defines what metadata SCORM requires and how to package content. Refer to the *AFMC ISD Resource* Site for a link to the most recent version of SCORM requirements.

Sharable Content Object Reference Model Run-Time Environment Data Model—A standard set of data elements used to define the information being communicated, such as the status of the

learning resource. In its simplest form, the data model defines elements that both the command approved LMS, courseware database, content library and Sharable Content Object are expected to "know about." The command approved LMS, courseware database and content library must maintain the state of required data elements across sessions, and the learning content must utilize only these predefined data elements if reuse across multiple systems is to occur. Refer to the *AFMC ISD Resource* Site for a link to the most recent version of SCORM requirements.

Sharable Content Object Reference Model Sequencing and Navigation—Rules that a command approved LMS, courseware database and content library. Must follow in order to present a specific learning experience. The content developer is responsible for defining the rules to which a command approved LMS, courseware database and content library must adhere. These rules are expressed within content structure and encoded in the organization section of content packaging. Through this means, the intended behavior of a collection of learning resources may be moved with a package from one command approved LMS, courseware database and content library environment to another. Refer to the *AFMC ISD Resource* Site for a link to the most recent version of SCORM requirements.

Sharable Content Object Reference Model—The ADL Initiative established the SCORM to define the technical foundations of a Web-based learning environment. It is a model that references a set of interrelated technical standards, specifications and guidelines designed to meet high-level requirements for learning content and systems. SCORM defines a Web-based learning "Content Aggregation Model (CAM)" and "Run-time Environment" for learning objects. Refer to the *AFMC ISD Resource* Site for a link to the most recent version of SCORM requirements. See ADL Initiative, Sharable Content Object, SCORM CAM, SCORM Run-Time Environment Data Model and Web-Based Training.

Sharable Content Object—A self-contained package of knowledge objects (i.e., images, shapes, text, etc.) that can be tracked electronically by a command approved LMS. It includes a specific launch able asset that utilizes the SCORM Run-Time Environment to communicate with a command approved LMS, courseware database and content library. A SCO represents the lowest level of granularity of learning resources that can be tracked by a command approved LMS using the SCORM Run-Time Environment.

Short—Term Memory—Memory is usually divided into three progressive levels: sensory, short-term, and long-term. Short-Term Memory (STM) uses three processes to store and organize information selectively chosen from the sensory memory: iconic memory (which holds visual images); acoustic memory (which holds sounds) and working memory (which holds a thought for immediate use). STM last from 3-20 seconds. It is limited in capacity to about seven items and is quite vulnerable to interruption or interference. Also see Acoustic Memory, Iconic Memory, Memory, Miller's Magic Number and Working Memory.

Signature Page—A course control document that records acceptance of new, revised, and revalidated courseware by authorized officials. For courses used at more than one AFMC installation, a standardized format has been designed to document command-wide approval of the instructional system.

Simple graphics, video and audio clips may be incorporated into the presentation.

Simulation—A technique whereby job environment phenomena are mimicked for learning purposes. Simulation can save time, reduce costs, eliminate the need to interfere with production

while instructing and eliminate potential dangers of the workplace. The simulation usually focuses on a small subset of the features of the actual job environment which are complex, risky or expensive to perform in the workplace.

Skill Transfer—A process of decision making in which individuals decide whether learned skills should be applied to a particular situation. It involves understanding how and when to apply learned skills in the real world.

Skill—The ability to perform a job-related activity that contributes to the effective performance of a task. Skills involve physical or manipulative activities, often requiring knowledge for their execution. All skills are actions having specific requirements for speed, accuracy, or coordination. Also see Attitude and Knowledge.

Small Group Tryout—(1) A portion of a course or learning aid is validated under field conditions by selected instructors/trainers, customer representatives and/or students from the target population to verify effectiveness of instruction. This type of small group tryout is used when a course is lengthy, only part of an existing course has been revised, when a segment of the training is particularly risky and requires advance feedback, etc., to determine if changes are needed in content or delivery based on pre-testing of a segment of the course under field conditions. (2) The final draft of the course or learning aid is validated under field conditions by instructors/trainers, customer representatives and students from the target population. This type of small group/individual tryout often reveals minor issues with content and delivery of the instruction that were not obvious during earlier technical and instructor/trainer reviews, and helps to determine the average length of the course. It is often called final validation. See Evaluation, Formative Evaluation, Operational Tryout.

Soft Skills—Soft skills address feelings, emotions and attitudes, which are part of the affective domain of learning. Instruction on teambuilding, change management and customer service are examples of soft skills training. Soft skills courses are considered functional training for purposes of this manual if they are implemented to support performance in the workplace.

Solicitation and Award Phase of Acquisition—This phase of the acquisition process begins when the contract requirements document is submitted to the contracting office for processing. A contractor is then selected by the contracting officer with the assistance of the functional area representative. This phase ends with the award of the contract.

Source Selection—Activity in Solicitation and Award Phase of acquisition in which the proposals (bids) of prospective contractors (sources) are evaluated to determine best value for the government. The contract is awarded to the bidder who ranks highest against the source selection criteria.

Standard—Defines the criteria for acceptable performance by the student. It is stated in terms such as accuracy requirements, technical specifications, time constraints, performance rates, completeness and qualitative requirements. It identifies the proficiency that the students are expected to achieve when they perform the behavior under the specified conditions. See Behavior, Objective and Condition.

Statement of Work (SOW)—A contract requirements document that describes services and products needed in terms of how they are to be performed or produced and criteria for determining whether these requirements have been met. It differs from the PWS in which the emphasis is on

performance - what is to be done and not how it is to be done. See Contract Requirements Document, Contract Requirements Package and Performance Work Statement.

Storyboard—Panels on which a sequence of sketches depicts the significant changes in action, appearance, or student interactivity in planned CBI and videos.

Structured On-the-Job Training (SOJT) Guide—A course control document that replaces the plan of instruction for SOJT courses. It includes learning objectives, evaluation methods, instructional guidance, required materials and equipment, etc., that assist the trainer in providing quality instruction at the worksite.

Structured On-the-Job Training (SOJT)—Over-the-shoulder, practical instruction on tasks required for job performance. It differs from informal on-the-job training in that it has documented procedures for delivering accurate and complete instruction in a consistent manner.

Student Frequency—How often an employee is required to take a particular course or proficiency evaluation in accordance with training mandates.

Subject Area Program Manager—Individual responsible for implementing a program that crosses functional areas, such as environmental management, safety, occupational health, security, personnel, acquisition, finance, quality assurance, etc. In the courseware process, the subject area program manager serves as an expert on program training requirements.

Subject Matter Expert (SME)—An individual who has high level knowledge and skill in a particular topic, task, system or process. The SME is responsible for the accuracy and completeness of course content.

Subtask—Performance steps that, when combined, make up a task.

Summary—Segment of a lesson conclusion during which the instructor reiterates key points of lesson content (knowledge level) or reviews and expands on key material and develops relationships that lead to generalizations (comprehension level). Also see Conclusion.

Summative Evaluation—Gathers feedback from students, instructors, customer reps, etc., about the effectiveness of instruction at the summation of development activities. It is used to try out the instruction on the target audience (average students) in an operational environment. Summative evaluation is the final step in the validation process. The objective is to ensure that the instructional system is fully integrated and achieves desired outcomes. Also called operational tryout or field validation. See Evaluation, Operational Tryout.

Superseded Course—Course or learning aid that has been replaced.

Support Function—This ISD function involves long-range and day-to-day tasks that are performed in order to implement and maintain the instructional system throughout its life cycle. Some of the basic support activities are providing funding, facilities, manpower authorizations, and services in support of learning activities; supplying equipment, skilled personnel, courseware and instructional aids; and maintaining facilities, equipment, skilled personnel, courseware and instructional aids. Also see Administration Function, Delivery Function, and Management Function.

Supporting Documentation—Items such as ISD Evaluation Board review notes, correspondence, monthly status reports, authoring files for computer-based courses and sophisticated learning aids, Interim Change Memos, graduate assessment surveys, etc., that could be useful when revalidating

or revising the course next time or which justify decisions made about existing course format, content or delivery.

Synchronous Training—Training in which a facilitator responds real-time to questions and comments from students. See Asynchronous Training.

Synthesis—Bloom's Taxonomy lists six progressive levels of learning using the cognitive domain: knowledge, comprehension, application, analysis, synthesis, and evaluation. Synthesis is building a structure or pattern from diverse elements, putting parts together to form a whole which has a new meaning to the learner.

System Approach to Training (SAT)—Procedures used by instructional system developers to develop instruction. Each phase requires input from the prior phase and provides input to the next phase. Evaluation provides feedback which is used to revise instruction. Also see ISD.

System Architecture—Design and interaction of hardware and/or software components in a computer system. Developing a common architecture ensures interoperability of models and simulations in collective training. Open architecture refers to the ability of devices to be easily connected to programs made by different manufacturers. Closed or proprietary architecture makes connecting to the system difficult and is in direct conflict with ADL Initiative goals. Open architectures use Commercial-Off-The-Shelf (COTS) hardware and software and conform to approved standards. See Sharable Content Object Reference Model.

System OPR (Officer of Primary Responsibility)—Individual responsible for implementing an AFMC-approved automated management system that supports one or more functional areas. In the courseware process, the system Officer of Primary Responsibility OPR serves as an expert on system training requirements.

Target Audience Analysis—Activity in which a courseware developer analyzes the characteristics (e.g., age, experience, education, location, attitudes, etc.) of the target audience. It helps the developer decide what types of instructional methods, media, examples, exercises and evaluation techniques will be most effective in gaining and holding the students' attention.

Target Audience/Population—Students for whom an instructional system is intended.

Task Analysis—See Knowledge/Task Analysis.

Task—An observable and measurable unit of work activity that forms a significant part of a job. It constitutes a necessary step in performance, and has a logical beginning and end.

Tasking Worksheet—Used when an individual, group, or organization determines that there is a significant issue that requires the attention of an Education and Training office. The tasking worksheet defines basic information about the issue sufficient to determine the scope and priority of the assignment, and to decide which function should receive the tasking.

Tasking—A formal request to address an issue.

Technical Accuracy Review—Activity in which subject matter experts, system Officers of Primary Responsibilities (OPRs), subject area program managers, and other customer representatives review the content of draft instructional and evaluation materials to determine if they are relevant, accurate and complete. This step is usually repeated more than once until technical experts are satisfied with the content and appearance of instruction. Also called SME review. See Formative Evaluation, Periodic Courseware Review.

Technical Re-Evaluation of Courseware—Activity in which Subject Matter Experts (SMEs), system Officer of Primary Responsibilities (OPRs), subject area program managers, and other customer representatives review the content of the instructional and evaluation materials to determine if they are still relevant, accurate and complete. It is also called Periodic Review and is performed at least every two years.

Terminal Training Objective—The overall objective that the student is expected to attain upon completion of instruction. It is made up of subordinate, enabling, or learning objectives within the modules of the course. Also see Objective.

Test Validity—The degree to which a criterion test actually measures what it is intended to measure.

Trainer—Individual with subject matter expertise who provides task training to students, often at the worksite using actual equipment, tools, automated systems and technical data. In the courseware process the trainer serves as an expert on the effectiveness of instructional materials, learning activities and evaluation methods in meeting their intended Purpose.

Training Category—Shows the locally designated subject matter category of the courseware item.

Training Function—Organization with training administration responsibilities. It does not need to be a stand-alone training office. Most functional areas (logistics, depot maintenance training, acquisition, civil engineering, etc.) have a training function. The installation Education and Training Office prescribed by DAFMAN 36-2689 is also considered a training function.

Training Gap Analysis—Activity in which a courseware project manager or courseware developer works with the customer to better define and analyze a perceived training need and determine the nature and scope of the fix.

Training Gap Worksheet—Used when an individual, group, or organization determines that a lack of knowledge or proficiency requires formal training (i.e., documented instruction with defined learning objectives), but no such training is currently in place. It defines basic information about the training gap sufficient to determine the scope and priority of the assignment, and to decide which training function should receive the tasking. Also see Courseware Issue Worksheet, Non-Training Issue Worksheet, and Training Issue Worksheet.

Training Gap—See Training Need.

Training Issue Worksheet—Used when an individual, group, or organization determines that there is a training issue that needs to be addressed, but the Training Gap Identification and Courseware Issue Identification Worksheets do not seem to apply. This worksheet defines basic information about the training issue sufficient to determine the scope and priority of the assignment, and to decide which training function should receive the tasking. Also see Courseware Issue Worksheet, Non-Training Issue Worksheet, and Training Gap Worksheet.

Training Lead Complex—Organization or installation designated as the manager of training issues associated with a particular subject matter area, system, or occupation. An AFMC Training Lead Complex serves as liaison between complexes, HQ AFMC and other organizations on issues related to assigned command training; ensures that representatives from all complexes are involved in decisions related to the assigned training; periodically reviews assigned command training and associated courseware; develops, distributes and maintains command courseware to support

assigned training; and provides train-the-trainer assistance when applicable. Also called lead complex or training lead.

Training Manager—Individual with a full or part-time training manager role who is responsible for ensuring that the AFMC Courseware Development and Management Process is used to develop relevant, effective and economical instructional systems within their span of control. A training manager may be a supervisor in a training function, or a designated training specialist in any organization.

Training Mandate—Directive (e.g., regulation, instruction, supplemental instruction, operating instruction, etc.) that specifically mandates the existence of the course. Technical data and regulations that govern the material of the course do not necessarily require that the course exist. Only a directive that requires training to exist is a training mandate for courseware management purposes.

Training Manual—Self-paced, stand-alone instructional system that meets a defined learning objective using text and graphics media.

Training Method—See Instructional Method, Media.

Training Needs Assessment (TNA)—Systematic process of identifying a deficiency or gap between what "is being taught" and what "should be taught." See Deficiency, Courseware Issue Analysis, Performance Root Cause Analysis, Training Gap Analysis, Training Issue Analysis, Training Mandate Analysis, Training Requirements Analysis.

Training Need—The difference between what "is being taught" and what "should be taught." See Deficiency.

Training Objective—See Objective.

Training Planning Team (TPT)—An action group composed of representatives from all pertinent functional areas, disciplines, and interests involved in the life cycle design, development, acquisition, support, modification, funding, and management of a specific occupation or subject matter area.

Training Region—Isolated section of an automated system that simulates the characteristics and functionality of the live system, but which is used only for training purposes. Also called a training instance.

Training Requirements Analysis (TRA)—Systematic process of identifying a deficiency or gap between what "is being taught" and what "should be taught." In AFMC, the term is commonly used to describe Planning activities, in which a performance issue, training gap or courseware deficiency is analyzed to define a courseware project. See Deficiency and Courseware.

Training Requirements Matrix—Outcome from Training Mandate Requirements Analysis activity.

Training Requirement—The skills, knowledge, and attitudes that are needed to satisfy the job performance requirements, and that are not already in the trainee's incoming repertoire.

Training Scheduling System (TSS)—A comprehensive relational computer database system used in the AFMC depot maintenance community that documents training, proficiency demonstration and other qualification actions.

Training Specialist—An individual who understands the training process and is assigned either training administration, instruction or courseware development/ management responsibilities.

Training Standardization Assessment—Activity in which a training specialist determines if proposed training development or revision can be expanded to include other organizations, installations or agencies. It promotes consistency of instruction across organizations and saves resources by preventing duplicate courseware.

Training Strategy—See Instructional Strategy.

Training System—See Instructional System.

Training—A set of events or activities presented in a structured or planned manner, through one or more media, for the attainment and retention of skills, knowledge, and attitudes required to meet job performance requirements. See Functional Training and Instruction.

Train-the-Trainer Activities to Qualify Instructors—The courseware development team helps prepare instructors/trainers to instruct the course by giving them active roles in all review and implementation activities. This readies future instructors/trainers to deliver the instruction when subject matter is new and instructors/trainers do not yet have experience in presenting the material.

Transfer of Learning—Extent to which a student successfully applies what is learned to real world situations. It is the result of genuine understanding, not just memorization. Barriers to learning transfer include: instruction is not relevant to actual job performance; instruction is inadequate for the real world situation; students are not given adequate time to incorporate what they have learned into their job tasks; organization contradicts or does not support what was instructed.

Transition—Statements used by the instructor to move from the introduction of a lesson to the body, between main points, between sub-points within each main point, and from the body to the conclusion of the lesson. These statements show a logical relationship between the lesson segments they connect.

Tutorial—A session of intensive one-on-one instruction. It can be an on-line program or provided face-to-face by a qualified subject matter expert.

Tutoring—Providing one-to-one subject matter instruction to a learner. When this is accomplished primarily via the internet, it is called e-tutoring. Also see Coaching and Mentoring.

Unauthorized Commitment—An agreement that is not binding solely because the government representative who made it lacked the authority to enter into that agreement on behalf of the government. Only certain individuals in the installation Education and Training (E&T) office and the installation contracting function have the authority to enter into agreements with contractors for education, employee development and training support. Functional area E&T offices may obtain quotes from contractors, but may NEVER negotiate any deals. This would be an unauthorized commitment that would result in the need for ratification action.

Unfair Competitive Advantage—Information that might give one contractor a better chance over another contractor of being awarded a future contract, such as information about spend plans or budgets, plans for outsourcing services, clarification of a contract requirements document during Solicitation and Award Phase, etc. These are unacceptable communications that could lead to protests from other contractors.

Unsolicited Proposal—Document submitted by a contractor to suggest a solution to a perceived deficiency or to recommend an improvement to government products and services. It is unsolicited because it is not in response to an advertised request for proposals. In the education and training community, a training manager or other qualified individual considers the unsolicited proposal and decides how to proceed.

Validation—The process of conducting technical accuracy reviews, small group tryouts and operational tryouts to evaluate the quality and effectiveness of courseware before it is approved and fielded under normal operating conditions. It involves a repetitive cycle of development, tryouts, and revision until evidence shows that the instructional intent has been achieved. See Evaluation, Operational Tryout, Small Group Tryout, Summative Evaluation and Technical Accuracy Review.

Validity—The degree to which an activity, tool, program, etc., achieves its desired result.

Value—The worth assigned by an individual to an object or outcome. See Attitude, Bloom's Taxonomy.

Valuing—Bloom's Taxonomy lists five progressive levels of learning using the affective domain: receiving phenomena, responding to phenomena, valuing, organizing and internalizing values (characterization). Valuing is attaching worth or value to a particular object, phenomenon, or behavior. Values range from acceptance to a state of commitment.

Vendor Training—Training purchased from a commercial source, or vendor. The vendor company uses its own off-the-shelf courseware and instructs the course either on-site or off-site.

Virtual Classroom—Classroom where students and instructors are not physically collocated. See Learning Center.

Virtual Reality—An artificial environment which is experienced through sensory stimuli (such as sights and sounds) provided by a computer and in which one's actions partially determine what happens in the environment.

Web-Based Training (WBT)—An approach to ADL in which Computer-Based Training is delivered using internet technologies. This self-paced instruction is also called electronic learning, e-learning, on-line learning, networked learning, enterprise learning, Internet Distance Learning (IDL) and ADL. See Computer-Based Training and ADL Initiative.

Workbook—Student handout that includes activities and exercises to reinforce the learning concepts.

Working Memory—STM uses three processes to store and organize information selectively chosen from the sensory memory: iconic memory, acoustic memory, and working memory processes. Working memory holds a process (dialing a particular phone number, etc.) for 3-20 seconds until it is put to use (think of a phone number you'll repeat to yourself until you can dial it on the phone). The purpose of working memory is only to hold, quickly use, and discard, and is not to transfer the information to LTM. Also see Memory, Miller's Magic Number, and STM.

Attachment 2

AFMC COURSEWARE MASTER FILE STRUCTURE

A2.1. Purpose of Standardized Master File Structure. A standardized master file structure facilitates the searching, posting and reviewing of courseware. It prevents important documents from being overlooked because materials are filed in the same folders every time.

A2.2. When Standardized Master File Structure is Required. The master file structure shown below is mandatory on the effective date of this manual for AFMC command courseware. Installations shall implement a standardized master file structure for local courses and learning aids which complements the standard AFMC master file structure.

A2.3. AFMC Versus Local Folders in Master File. The AFMC Courseware Master File contains an AFMC folder and installation folders with the same subfolder categories. The AFMC folder contains the generic AFMC courseware materials and documentation. Each installation folder contains localized materials and documentation for the course or learning aid, as applicable.

A2.4. Dropping Unnecessary Subfolders. When no materials or documentation exist for a particular subfolder, (e.g., Historical CCDs, etc.) there is no need to leave it in the master file. The empty folder can be deleted from the courseware master file.

A2.5. Configuration Management of AFMC Courseware Master Files. AFMC courseware master files are official documents that are used for instruction, reference and audit purposes. They require careful configuration control to prevent discrepancies in the versions used and to rapidly disseminate Interim Changes to material.

Figure A2.1. Master File Structure for Active AFMC Command Course.

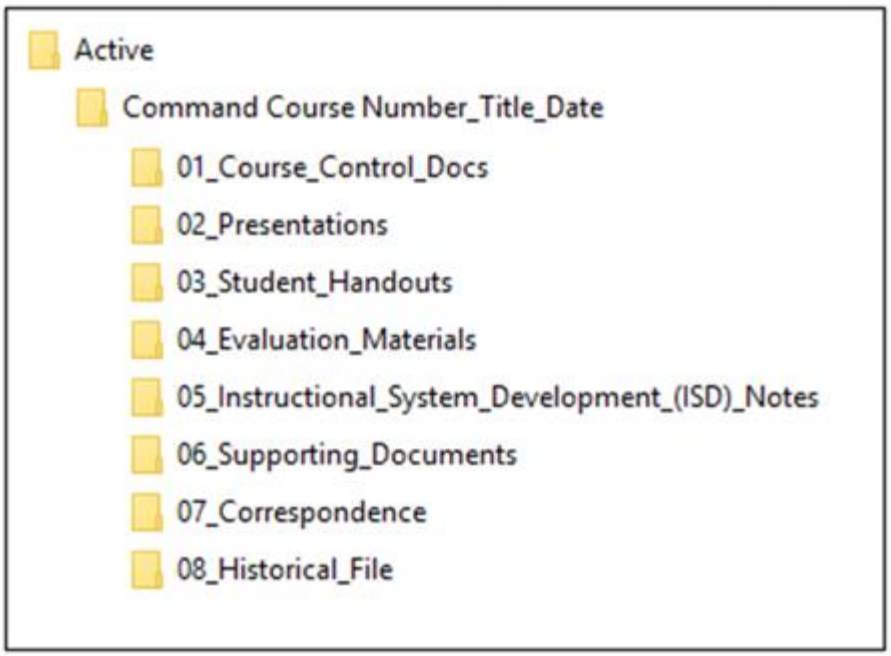
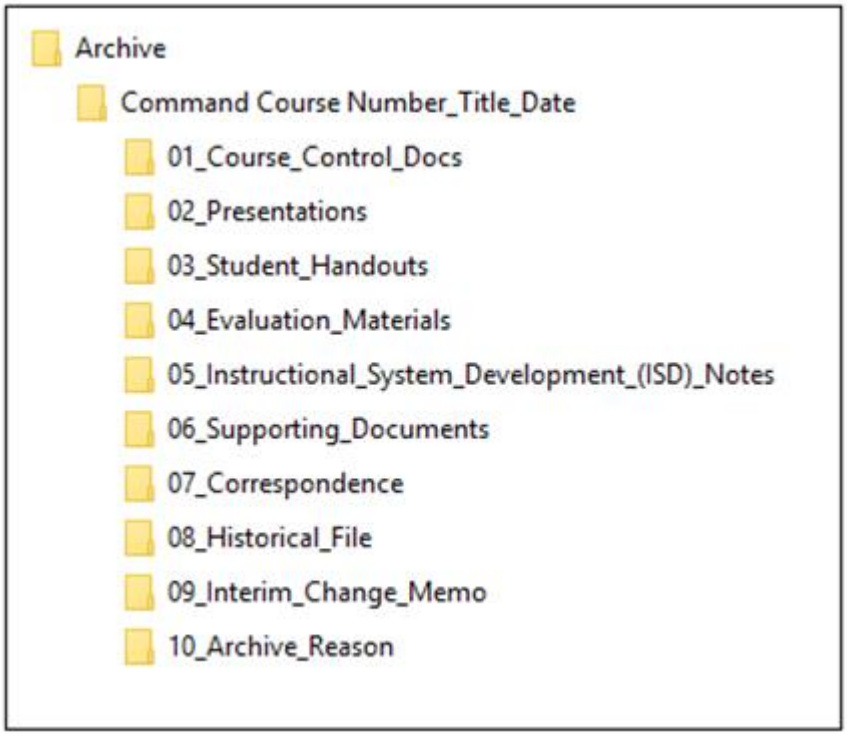


Figure A2.2. Master File Structure for Archived AFMC Command Course.



A2.6. Guidance on Creating a Master File. Below are the listed Folders (bold) and what files will be stored in the folders.

A2.6.1. **Course Control Documents (CCDs)** – This master file contains the documents that organize and manage the instruction. It typically includes documents such as:

A2.6.1.1. AFMC Course Chart (AFMC Form 853).

A2.6.1.2. Course Training Standard (CTS).

A2.6.1.3. Plan of Instruction.

A2.6.1.4. Structured On-the-Job Training Guide.

A2.6.1.5. Coordination Sheets and Signature Pages.

A2.6.1.6. Course Data Sheet (CDS).

A2.6.1.7. Existing Courseware Evaluation Checklist (AFMC Form 96).

A2.6.2. **Presentation** – This master file contains the instructional presentation materials. It typically includes documents such as:

A2.6.2.1. Presentation Charts.

A2.6.2.2. Computer-Based Presentations.

A2.6.2.3. Learning Aids.

A2.6.2.4. Exhibits.

A2.6.2.5. Videos.

A2.6.2.6. Other Audio-Visual Aids.

A2.6.2.7. Formal Lesson Plan or Instructor Guide.

A2.6.3. **Student Handouts** – This master file contains information the student can use and would take with them after leaving the class such as:

A2.6.3.1. Student Guides and Workbooks.

A2.6.3.2. Student Handouts.

A2.6.3.3. Reference Guides.

A2.6.4. **Evaluation Materials** – This master file contains the materials that are used to evaluate student progress and attainment of learning objectives. It is carefully controlled to prevent test compromise, if the situation requires such measures. It typically includes documents such as:

A2.6.4.1. Pre-Assessment Test.

A2.6.4.2. Bypass Test.

A2.6.4.3. Progress Checks.

A2.6.4.4. Class Exercises or Projects.

A2.6.4.5. Standardized Review or Discussion Questions.

A2.6.4.6. Proficiency Evaluations.

A2.6.4.7. Structured On-the-Job Training (SOJT) Worksheets.

A2.6.4.8. Process or Product Checklists.

A2.6.4.9. Written Tests.

A2.6.5. **Instructional System Development Notes** – This master file contains ISD Notes which are not purged regardless of dates; used for reference and audit purposes. ISD Evaluation notes are not purged from the master file regardless of date because they are the audit trail of decisions made about the courseware.

A2.6.5.1. ISD Evaluation Board Reports.

A2.6.5.2. ISD Evaluation Board correspondence on action items, etc.

A2.6.5.3. AFMC Release Point Checklist.

A2.6.6. **Supporting Documentation** – This master file contains documents that support decisions made about the course. The files contain any documentation that could be useful when revalidating or revising the course in the future, or which justify decisions made about existing course format, content or delivery. It typically includes documents such as:

A2.6.6.1. Tasking Worksheets.

A2.6.6.2. Analysis Reports. It typically includes, but is not limited to, documents such as: Occupational Training Analysis Report, Performance Root Cause Analysis Report, Instructional System Analysis Report, Training Gap Analysis Report, Courseware Issue Analysis Report, Knowledge/Task Analysis Report, Target Audience Analysis Report, Learning Analysis Report and Delivery Resources Analysis Report.

A2.6.6.3. Training Requirements Matrix.

A2.6.6.4. Existing Content Suitability Report.

A2.6.6.5. Training Standardization Assessment.

A2.6.6.6. Final Reusable Knowledge Object (RKO) Assessment.

A2.6.6.7. Final Sharable Content Object Assessment.

A2.6.6.8. Deliverables Format Analysis documentation.

A2.6.6.9. Courseware Project Definition Summary.

A2.6.6.10. Existing Content Integration Plan.

A2.6.6.11. RKOs ready for reuse.

A2.6.6.12. Graduate Assessment Survey questions.

A2.6.6.13. Customer Satisfaction Survey.

A2.6.6.14. Courseware deactivation or reactivation documentation.

A2.6.6.15. Source code and authoring files for computer based instruction. This info is carefully controlled to prevent compromise of proprietary information, if the situation requires such measures.

A2.6.7. **Correspondence** – This folder contains any supporting correspondences. It typically includes, but is not limited to, documents such as: Emails, Memos for Record, Monthly Status Reports and Requests for copies from organizations not on the courseware signature sheet.

A2.6.8. **Historical File** – the most recently superseded version of documents or files for reference and audit purposes. It typically includes, but is not limited to, documents such as:

A2.6.8.1. Course Chart.

A2.6.8.2. CTS.

A2.6.8.3. Plan of Instruction.

A2.6.8.4. SOJT guide.

A2.6.8.5. Formal lesson plan or instructor guide.

A2.6.8.6. Coordination Sheets and Signature Pages.

A2.6.9. **Archive Files Only**

A2.6.9.1. **Interim Change Memo.** This folder of the master file contains any Interim Change Memos and supporting documentation related to the course or learning aid. The Interim Change Memo folder is not embedded within any other folders because it is important for courseware users to see this information immediately when they open the master file. It typically includes documents such as:

A2.6.9.1.1. Interim Change Memo.

A2.6.9.1.2. Correspondence on obtaining approval for Interim Change.

A2.6.9.1.3. Correspondence on advising customers of Interim Change.

A2.6.9.2. **Archive Reason.** This folder of the master file contains a memo, e-mail, ISD Evaluation Board report, or other documentation that explains the reason that the course is no longer needed and/or teachable.